

# Why Rotating Fluids Make Polygons

## Abstract

On the flat ( $K = 0$ ) plane, the regular  $N$ -gon of point vortices is an energy minimum on the surface of conserved angular impulse  $L = \sum |z_k|^2$  and linear impulse  $\mathbf{P} = \sum z_k = 0$  for all  $N \leq 7$ , and unstable for  $N \geq 8$  without a central vortex. This stability boundary is an invariant of the two-dimensional Laplacian: the logarithmic Green’s function  $G \propto -\ln r$  is the unique radially symmetric planar interaction for which the Havelock eigenvalue  $\lambda_m = (N-1) - m(N-m)/2$  is scale-independent, and  $N = 7$  is the threshold where the leading eigenvalue  $\lambda_3 = 0$  (the linear restoring force vanishes). A quartic calculation on the constraint surface confirms that the  $N = 7$  ring is nonlinearly stable: the quartic coefficient  $\alpha_0 = 45/14 \approx 3.21 > 0$  (in the normal form  $\mathcal{H} = (\alpha_0/4)\varepsilon^4$ ) supplies a restoring force where the linear one is absent.

For  $N \geq 8$  the stability boundary requires a central vortex of strength  $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$ . The sharpest prediction is at  $N = 8$ :  $\kappa_{\text{crit}}(8) = 1/2$ . In a Bose–Einstein condensate, where circulation is quantized, this is an integer-valued threshold: a ring of eight unit-winding vortices is stable if and only if the central vortex carries winding number  $q_0 \geq 1$ .

The central unifying result is a *constraint-intersection principle*: three selection mechanisms constrain the polygon wavenumber  $N$  from above, and the observed  $N$  is determined by whichever is tightest. Two are fully derived from the 2D Laplacian  $\nabla^2$ : *Thomson stability* (eigenvalues of the Green’s function  $G(\nabla^2)$ ) and *Rossby stationarity* (stationary solutions of  $(\nabla^2 + \beta \partial_y)\psi = 0$ ). The third—*geometric packing*—has a Laplacian-derived skeleton (the domain admissibility condition  $\sin(\pi/N) \geq \varepsilon/R$  for the Green’s function on the punctured domain) but requires one empirical input: the  $\beta$ -drift enlargement  $r_{\text{excl}} \approx 1.3 r_{\text{core}}$  from nonlinear vortex dynamics on a rotating sphere [Gavriel and Kaspi, 2021], not derivable from the linear operator alone. Saturn’s hexagon ( $N = 6$ ) is Rossby-limited (fully derived). Jupiter’s north-pole octagon ( $N = 8$ ) is Thomson-limited (fully derived; the prediction  $N_{\text{max}} = 8$  relies on  $\kappa_0/\kappa \lesssim 1$ , testable with improved Juno wind measurements). Jupiter’s south-pole pentagon ( $N = 5$ ) is packing-limited (Laplacian skeleton plus the empirical 1.3 factor).

On the hyperbolic plane  $\mathbf{H}^2$  with curvature  $K = -1/a^2$ , the stability boundary rises monotonically:  $N_{\text{crit}} = 12$  at ring-to-curvature ratio  $R/a = 1$ —a falsifiable prediction for circuit-QED hyperbolic lattices [Kollár et al., 2019]. The  $N \rightarrow (N+1)$  transition threshold  $\xi^*(N)$  satisfies a *palindromic* quadratic  $A\xi^2 + B\xi + A = 0$ , whose equal leading and trailing coefficients force  $\xi \cdot \xi' = 1$ —the roots are multiplicative inverses. This is not a computational coincidence: it encodes the conformal inversion  $\xi \leftrightarrow 1/\xi$  of the Poincaré disk (Proposition 2.1), connecting stability thresholds to the Pell equation  $a^2 - Db^2 = 1$  of algebraic number theory.

The threshold  $\xi^*(N)$  is an algebraic integer—living in the maximal order of  $\mathbb{Q}(\sqrt{D})$ —for exactly five values:  $N \in \{8, 9, 11, 15, 23\}$  (characterized by  $(N-7) \mid 16$ ;

differences  $\{1, 2, 4, 8\}$ , powers of 2). At each,  $\xi^*$  is a power of the inverse fundamental unit:  $\xi^*(8) = (8+3\sqrt{7})^{-1}$  in  $\mathbb{Z}[\sqrt{7}]$ ,  $\xi^*(23) = \varphi^{-2}$  where  $\varphi = (1+\sqrt{5})/2$  is the golden ratio. The monic polynomial traces  $\{16, 10, 6, 4, 3\}$  follow a binary subdivision  $c(k) = 2 + 2^{4-k}$  at  $N = 7 + 2^k$ , converging to the golden ratio ( $c = 3$ ) as the algebraically simplest palindromic unit. Each threshold has continued fraction  $1/\xi^* = [c-1; \overline{1, c-2}]$ , degenerating to the all-ones expansion of  $\varphi$  at  $N = 23$ .

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# Preface

This project began years ago when I first encountered the Voyager photographs of Saturn’s polar hexagonal storm. Why a polygonal ring? Why the persistence? I researched the relevant evidence and arguments, and while informative, the underlying framework did not adequately satiate my desire for a clear answer. I catalogued it as one of those interesting intersections between the physical world and geometry—crystals, viruses, snowflakes, coral reefs, beehives, nautilus shells—many of which have been rigorously explained, yet the explanatory framework for polygonal formation in the atmospheres of the two largest planets in our Solar System remained disconnected and incomplete.

My educational background is primarily philosophy with a mathematical timbre. Formally, I am more familiar with Peano, Frege, Russell, and Gödel than the applied disciplines. A hobbyist’s obsession with number theory is an apt description for my post-educational years. Professionally, I am a full-stack developer working in healthcare.

Fast-forwarding: I was watching an astronomy video with my children and the polar polygons on Jupiter and Saturn immediately astonished them and once again piqued my curiosity. I dove back into the most current research. The Cassini and Juno images hinted at a deeper mathematical connection between the multiple  $N$ -gons on the gas giants. The hook was set. I ordered a copy of *Geophysical Fluid Dynamics* by Joseph Pedlosky and started writing code.

Observationally, the storms show central vortices and logarithmic spirals eventually mutating into polygonal rings. From my wanderings in the mathematical literature, I knew there was a formal connection between the first two via Möbius transformations. This was my mathematical starting point and eventual foundation for the resulting paper.

I became determined to pull the connecting string until the knot unravelled. This project has crossed many mathematical, physical, and astrophysical domains as a result. I am by no means a subject matter expert in all of these fields; my research directions were motivated by the singular goal of finding an answer to the original question, wherever that led. Institutional research correctly forces disciplinary depth; the cost is that cross-domain connections can go unnoticed. I am unaffiliated, which meant I could follow the mathematics wherever it led, without the professional incentive to stay within a single field. Whether that freedom produced something worth the lack of expert supervision is for the audience to assess.

My process was iterative: at each step I identify a potential connecting point, honestly assess what I do and do not understand about the underlying structures, research the applicable domain, study that specific subset until I have an intuitive understanding in the context of my project, then develop and test software that confirms my understanding and verifies against existing mathematics, physics, or astrophysics. At that point I architect code that will support, prove, or disprove my connective hypotheses. Rinse and repeat.

My background gave me a solid logical foundation of what constitutes a “proof” in the strictest sense. This mapped well onto the sections of the paper that required such reasoning. The software allowed me to quickly assess feasibility, routes to proving, potential logical errors or contradictions, and blind spots.

Full transparency: the review process drew on AI-assisted critique. With Claude Code, I built structured review agents—a professional mathematician, a physicist, and a technical writer. I needed my subjective understanding, writing, and results to be in a language conversant with the modern research community. Once I had my ideas fully fleshed out in writing, I used the agents to scrutinise, review, and find logical errors. The

philosophical ideas, logic, proof scaffolding, development, and insights from the paper are my own. This process repeated until I was confident that the paper accurately reflected the quality, tone, and accuracy expected of a published article in academic journals.

I have made a serious attempt to cite and attribute assiduously, as well as to honestly assess any weakness in argumentation. Acknowledging vulnerabilities, although in a sense “knee-capping” portions of the paper, is intellectually honest, and since my goal was to answer *why*, I do not think the acknowledged shortcomings detract from the overall aim.

The paper is admittedly dense in places and broad in scope. No part requires fluency in all parts. The mathematician will be less inclined to devote attention to planetary atmospheric dynamics. The planetary scientist might find the algebraic number field portions less interesting than the mechanisms at play in polar storms. Experimentalists in circuit QED will be drawn to the BEC predictions. I have attempted to build the paper logically, starting with the mathematics first and progressing to the applied planetary physics. The table of contents will direct the reader to the areas most relevant to their interests.

Feedback, comments, and suggestions are welcome.

*Gordon Speagle*

## 1 Introduction

These observations raise a question with two parts. The first—*how* does a specific polygon form?—has received considerable attention. The second—*why* should different physical mechanisms on different planets all produce the same geometric outcome?—has not.

In 1988, David Godfrey examined Voyager images of Saturn’s north pole and found a hexagon [Godfrey, 1988]—a persistent, regular, six-sided polygon inscribed in the jet stream at  $76^\circ\text{N}$ , stable across decades. Its most striking property is that it co-rotates with Saturn’s System III period ( $10^{\text{h}}39^{\text{m}}22.4^{\text{s}}$ ), the rotation rate of the deep interior inferred from the magnetic field, rather than being advected by the  $\sim 120\text{ m/s}$  eastward jet in which it sits [Godfrey, 1988, Sánchez-Lavega et al., 2014]. This phase-locking to the deep interior has been confirmed from Voyager through Cassini across more than 30 years and is the primary observational constraint on any proposed mechanism. Thirty years later, the Juno spacecraft revealed that Jupiter’s poles harbor their own polygonal arrangements: eight cyclones in an octagonal ring at the north pole, five in a pentagonal ring at the south [Adriani et al., 2018]. Numerical simulations can reproduce these structures [Morales-Juberías et al., 2011, Gavriel and Kaspi, 2021], and vortex crystals emerge spontaneously in 2D turbulence simulations on polar caps [Siegelman et al., 2022].

We answer the second question, subject to one physical assumption (Onsager negative-temperature statistics at the polar scale; see Section 16 for a detailed assessment). The mechanism operates at two scales:

1. **Macro scale: cluster formation.** The Kraichnan (1967) inverse energy cascade drives 2D turbulence toward large-scale coherent structures. In the Onsager (1949) framework, this produces a negative-temperature state where same-sign vorticity concentrates into  $N$  coherent clusters [Bouchet and Venaille, 2012, Siegelman et al., 2022]. This step explains why discrete vortex clusters exist; it does not determine their spatial arrangement.

2. **Meso scale: polygon arrangement.** Given  $N$  vortex clusters on a ring of fixed angular momentum, the regular  $N$ -gon *minimizes* the interaction energy on the conserved-quantity surface (Theorem 8.2). By the energy-Casimir method, this makes the  $N$ -gon a nonlinearly stable equilibrium for  $N \leq 7$  without a central vortex, or for any  $N$  with sufficient central vortex strength. The polygon is selected by dynamical stability, not by statistical mechanics.

The meso-scale proof (Theorem 8.2) applies to discrete point vortices and is directly relevant to Jupiter’s cyclone rings. Saturn’s hexagon is a continuous jet meander, not a discrete vortex ring; its wavenumber  $N = 6$  is selected instead by Rossby wave stationarity of the QGPV equation (Proposition 17.1). Both mechanisms share the logarithmic Green’s function as a structural foundation, but the mathematical arguments differ.

Which polygon forms is determined by planet-specific selection criteria:  $N = 6$  by Rossby stationarity (Saturn),  $N = 8$  by the Thomson–Onsager maximum-stability bound (Jupiter north). The observed  $N = 5$  at Jupiter’s south pole is consistent with Thomson stability ( $\kappa_{\text{crit}}(5) = 0$ , so no central vortex is required) but is not uniquely predicted by the present framework; the south-polar packing geometry provides the additional constraint (Section 17).

Beyond the solar system, the framework makes its most physically distinctive prediction on the hyperbolic plane: a ring of  $N = 12$  vortices at  $R/a = 1$  is stable *without* a central vortex—an impossibility in flat geometry, where any  $N \geq 8$  ring requires central stabilization. Recent implementations of hyperbolic lattice Hamiltonians in circuit QED [Kollár et al., 2019] and topoelectrical circuits [Lenggenhager et al., 2022] make this prediction experimentally accessible (Section 5).

## Part I

# Mathematical Foundations

## 2 The logarithmic interaction: uniqueness and symmetry

### 2.1 Dilation invariance and the Cauchy functional equation

**Remark 2.1** (The logarithm as the unique dilation-invariant interaction). *Among all pairwise interactions  $h : (0, \infty) \rightarrow \mathbb{R}$  with  $h' < 0$ , the logarithmic interaction  $h(r) = -\ln r$  is the unique interaction invariant under spatial dilation up to an additive constant. Define the scale map  $T_b[h](r) = h(br)$  ( $b > 0$ ). Dilation-invariance means  $h(br) = h(r) + c(b)$  for all  $r$ —that is, rescaling changes  $h$  only by an additive constant (irrelevant to the force  $-h'$ ). This is a Cauchy functional equation whose continuous solutions with  $h' < 0$  are exactly  $h(r) = -a \ln r + b$ ,  $a > 0$ , confirming class (ii) as the unique dilation-invariant class. The  $N$ -body lift of this per-pair statement is Corollary 2.2: the Möbius covariance of  $H$  (Proposition 2.1), restricted to the dilation subgroup, gives  $H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b$ , which is position-independent—the same fixed-point condition at the  $N$ -body level. This symmetry is not merely a geometric observation about the logarithm: it is the structural reason for the scale-independence of  $N_{\text{crit}} = 7$ . Because scaling changes  $H$  by an additive constant (Corollary 2.2), the Havelock eigen-*

value  $\lambda_m = (N-1) - m(N-m)/2$ —which determines whether the  $N$ -gon is a stable constrained minimum—is scale-independent, and the threshold  $N = 7$  cannot drift under spatial rescaling. Prior treatments of Möbius covariance in vortex dynamics (e.g. Newton 2001) identify it as a conformal symmetry of the stream function; Corollary 2.2 shows it additionally implies that the stability threshold is dilation-invariant.

**Remark 2.2** (RG classification of perturbations). Write  $h(r) = -\ln r + f(r)$  where  $f$  is a perturbation. Consider the rescaling  $h \mapsto h_b(r) := h(br)$  at scale  $b > 1$  (coarse-graining). Since  $-\ln(br) = -\ln r - \ln b$  differs from  $-\ln r$  only by an additive constant, the log part is scale-invariant modulo constants (this is Corollary 2.2). The perturbation  $f$  transforms as  $f_b(r) = f(br)$ . Classify perturbations by their large-scale behavior:

1. Irrelevant:  $|f(br)| \rightarrow 0$  as  $b \rightarrow \infty$  for fixed  $r$ . Example:  $f(r) = r^\alpha$  with  $\alpha < 0$  (power law with negative exponent,  $r^\alpha \rightarrow 0$  as  $r \rightarrow \infty$  i.e.  $b \rightarrow \infty$  at fixed  $r$ —wait, that’s  $b^\alpha r^\alpha$  which  $\rightarrow 0$  iff  $\alpha < 0$  for  $b \rightarrow \infty$ ). The perturbation is suppressed at large scales.
2. Relevant:  $|f(br)| \rightarrow \infty$  as  $b \rightarrow \infty$ . Example:  $f(r) = r^\alpha$  with  $\alpha > 0$ . The perturbation dominates at large scales.
3. Marginal:  $|f(br)| \rightarrow \text{const.}$  Example:  $f(r) = \text{const}$  or  $f(r) = \ln r$ .

This is a standard physics-level classification of scaling behavior under coarse-graining, not a theorem. It captures the large-scale asymptotics of the perturbed Hamiltonian and its stability consequences (see connection to the sign rule below).

Connection to the sign rule. For  $h = -\ln r + \varepsilon r^\alpha$ :  $h' = -1/r + \varepsilon \alpha r^{\alpha-1}$ ; the sign rule condition  $h' < 0$  holds at all  $r > 0$  iff  $\alpha \leq 0$  (irrelevant perturbations). For  $\alpha > 0$  the sign rule is violated at sufficiently large  $r$ : relevant perturbations destroy polygon stability. Thus the sign rule condition (i) is equivalent to the requirement that  $h$  lies in the basin of attraction of the logarithmic fixed point.

The stable manifold from multipole expansion. The basin of attraction arises naturally from compact vorticity. For  $|z| < \varepsilon \ll |w| = d$ , expanding  $\ln |w - z| = \ln d - \text{Re}(z/w) + O(\varepsilon^2/d^2)$  and integrating against  $\omega(z)$  with centroid at the origin gives the effective interaction  $H_{\text{eff}}(d) = -K^2 \ln d + O(\varepsilon^2/d^2)$  for  $d \gg \varepsilon$ . The corrections are irrelevant ( $\alpha = -2$ ), so any compact vortex distribution flows to the logarithm at large separation (Proposition 2.2).

Unifying the blob and  $K_0$  bridges. This RG perspective unifies two results proved later in the paper:

- The blob bridge (Section 12):  $h_\varepsilon = -\frac{1}{2} \ln(r^2 + 2\varepsilon^2) = -\ln r + O(\varepsilon^2/r^2)$ . The blob correction is an irrelevant perturbation ( $\alpha = -2$ ); the  $O(\varepsilon^2)$  convergence rate of Proposition 12.1 is the approach rate to the fixed point.
- The  $K_0$  interaction (Section 13):  $K_0(r/R_d) = -\ln r + O(e^{-r/R_d})$  for  $r \ll R_d$ . The exponential tail is irrelevant (faster than any power); the deformation-radius limit  $R_d \rightarrow \infty$  is the RG flow toward the fixed point.

In the Kraichnan inverse cascade [Kraichnan, 1967], vorticity concentrates into compact structures of growing scale  $L(t)$ , driving the inter-blob interaction toward the logarithm

with corrections suppressed as  $(L/d)^2 \rightarrow 0$ . The cascade is the physical RG flow; the sign rule demarcates its basin of attraction.

The Gaussian curvature  $K$  is a destabilizing perturbation when  $K > 0$  and a stabilizing one when  $K < 0$ . On the sphere ( $K > 0$ ) the deviation from the logarithmic Green's function lowers  $N_{\text{crit}}$  below 7; on the hyperbolic plane ( $K < 0$ ) it adds a positive  $O(|K|)$  correction  $12|K|$  to the formerly-marginal  $\lambda_3$  eigenvalue (5.4). The flat-plane case  $K = 0$  is therefore the least stable point of the constant-curvature family with respect to polygon formation:

## 2.2 Möbius covariance of the logarithmic energy

**Proposition 2.1** (Möbius covariance). *Under a Möbius transformation  $f: \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$  (i.e.,  $f(z) = (az + b)/(cz + d)$  with  $ad - bc \neq 0$ ), the logarithmic energy transforms as*

$$H(f(z_1), \dots, f(z_N)) = H(z_1, \dots, z_N) - \sum_{k=1}^N \frac{N-1}{2} \ln |f'(z_k)|.$$

(For general conformal maps the identity  $|f(z) - f(w)| = |f'(z)|^{1/2} |f'(w)|^{1/2} |z - w|$  holds only to leading order in  $|z - w|$ ; for Möbius  $f$  it is exact.) Consequently, the point vortex equations of motion are Möbius-invariant: the single-site correction  $\frac{N-1}{2} \ln |f'(z_k)|$  depends only on  $z_k$ , so  $\partial/\partial \bar{z}_j$  for  $j \neq k$  annihilates it, and the Hamiltonian gradient  $\partial H/\partial \bar{z}_j$  is unchanged.

*Proof.* The identity  $|f(z) - f(w)| = |f'(z)|^{1/2} |f'(w)|^{1/2} |z - w| \cdot (1 + O(|z - w|))$  is exact when  $f$  is Möbius (the error term vanishes), so  $\ln |f(z_j) - f(z_k)| = \ln |z_j - z_k| + \frac{1}{2} \ln |f'(z_j)| + \frac{1}{2} \ln |f'(z_k)|$ . Summing over all  $\binom{N}{2}$  pairs, each  $\ln |f'(z_k)|$  appears  $N-1$  times. Since  $\frac{1}{2} \ln |f'(z_k)|$  depends on  $z_k$  alone,  $\partial[\sum_k \frac{N-1}{2} \ln |f'(z_k)|]/\partial \bar{z}_j = \frac{N-1}{2} \partial \ln |f'(z_j)|/\partial \bar{z}_j$ , which is the single-site term in the  $j$ -th equation of motion. For Möbius  $f$ ,  $f'$  is rational and  $\ln |f'|$  is harmonic away from poles, so this term contributes a gauge-like shift to the Hamiltonian that does not couple distinct vortices.  $\square$

**Corollary 2.2** (Dilation subgroup = scale-invariant Hamiltonian). *Among all Möbius transformations, the dilation subgroup  $\{f(z) = bz : b > 0\}$  is uniquely characterized by having constant derivative:  $f'(z) = b$  for all  $z$ . Substituting into Proposition 2.1 gives a correction that is position-independent:*

$$H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b. \quad (2.1)$$

This is the  $N$ -body form of the dilation-invariance condition of Remark 2.1: scaling changes  $H$  only by an additive constant, so the equations of motion  $\dot{\bar{z}}_j = \partial H/\partial \bar{z}_j$  are dilation-invariant. For any other Möbius map  $f$ , the derivative  $|f'(z_k)|$  varies between sites, producing a configuration-dependent correction that is not a pure constant; the Hamiltonian is not scale-invariant under general conformal maps, only under the one-parameter dilation subgroup.

Equation (2.1) is the  $N$ -body lift of the per-pair uniqueness result in Remark 2.1: the logarithmic energy is the unique  $N$ -body Hamiltonian that is dilation-invariant.

**Remark 2.3** (The loxodromic thread). *The logarithmic interaction and the logarithmic spiral are dual manifestations of the same structure: the loxodromic one-parameter subgroup  $z \mapsto e^{(\sigma + i\alpha)t} z$  of the Möbius group acting on  $\mathbb{C}$ .*

- The interaction is logarithmic because it is the unique dilation-invariant pairwise potential (Remark 2.1; Cauchy's functional equation).
- The natural deformation curves are logarithmic spirals because they are orbits of the dilation-rotation generator  $\sigma + i\alpha$ ; the centered spiral of equation (7.3) is precisely this orbit applied to the  $N$ -gon.
- The stability threshold  $N = 7$  is scale-independent because the Havelock eigenvalue  $\lambda_m = (N-1) - m(N-m)/2$  has no  $R$ -dependence—a direct consequence of equation (2.1).
- The palindromic structure of the hyperbolic threshold polynomial  $A\xi^2 + B\xi + A = 0$  (equation (6.1)) encodes the conformal inversion  $\xi \leftrightarrow 1/\xi$  of the Poincaré disk, which is a Möbius map exchanging the interior and exterior of the unit circle.
- The logarithm is the IR fixed point of the coarse-graining flow on interactions (Proposition 2.2): compact vorticity distributions interact via  $-\ln r + O(\varepsilon^2/r^2)$  at large separation, with the correction irrelevant under rescaling. The sign rule demarcates the basin of attraction.

These five consequences—the interaction, the deformation, the threshold, the algebraic structure, and the RG flow—all descend from Möbius covariance (Proposition 2.1). The unifying role of the Möbius group is not merely a symmetry of the equations of motion; it is the reason the logarithmic interaction produces polygons at all.

**Remark 2.4** (From sign rule to physical stability). Theorem 3.1 establishes  $H''(0) < 0$  along the spiral direction—an unconstrained energy curvature. This does not by itself imply instability, because the spiral deformation changes the conserved angular impulse  $L = \sum |z_k|^2$  at  $O(\sigma^2)$ . The physical dynamics is constrained to  $\{L = \text{const}\}$ , and the Lagrange multiplier  $\mu_L < 0$  shifts the curvature by  $-2\mu_L > 0$ , converting the unconstrained saddle into a constrained minimum (Theorem 8.2 below). The sign rule therefore plays an indirect but essential role: it guarantees that the  $N$ -gon is a saddle of the unconstrained energy with a specific sign structure ( $H''(0) < 0$  along the spiral,  $H''(0) > 0$  along other directions), and the Lagrange shift converts this saddle into a minimum for  $N \leq 7$ . For  $N \geq 8$ , the negative curvature along the  $m = \lfloor N/2 \rfloor$  Fourier mode overwhelms the Lagrange shift, and the constrained configuration is a saddle.

### 3 The sign rule for general pairwise interactions

**Theorem 3.1** (Sign rule for general interactions). For  $N \geq 3$  vortices on a ring with general pairwise interaction  $h(r)$ , the energy along the centered spiral satisfies

$$H''(0) = \sum_{m=1}^{N-1} \left[ (d_m^2 h''(d_m) + d_m h'(d_m)) S_m + \frac{h'(d_m) m^2 (N-m)}{N^2 d_m} \right], \quad (3.1)$$

where  $d_m = 2R \sin(\pi m/N)$  and  $S_m = \sum_{k=0}^{N-m-1} [(2k+m-(N-1))/(2N)]^2 \geq 0$ .

If  $h'(r) < 0$  and  $(rh'(r))' \leq 0$  for all  $r > 0$ , then  $H''(0) < 0$ .

*Proof. Step 1: Pairwise distance as a function of  $\sigma$ .* For vortices  $j$  and  $k$  with separation  $m = j - k > 0$ , write the centered exponent as  $(j - (N-1)/2)/N = (j' + k')/2 + m/(2N)$  where  $j' = (j - (N-1)/2)/N$  and  $k' = (k - (N-1)/2)/N$ . Define the mean displacement  $s = \sigma(j' + k')/2$  and half-separation  $t = \sigma m/(2N)$ . Then  $z_j(\sigma) = R e^{s+t} e^{i\theta_j}$ ,  $z_k(\sigma) = R e^{s-t} e^{i\theta_k}$  with  $\theta_j - \theta_k = 2\pi m/N$ , giving

$$D(\sigma) := |z_j - z_k|^2 = R^2 e^{2s} [e^{2t} + e^{-2t} - 2 \cos(2\pi m/N)] = R^2 e^{2s} [2 \cosh(2t) - 2 \cos(2\pi m/N)].$$

At  $\sigma = 0$ :  $s = t = 0$ , so  $D(0) = 2R^2[1 - \cos(2\pi m/N)] = 4R^2 \sin^2(\pi m/N) = d_m^2$ .

*Step 2: First and second derivatives of  $D$ .* Write  $c_s = (j' + k')/2$  and  $c_t = m/(2N)$ , so  $s = \sigma c_s$ ,  $t = \sigma c_t$ . Direct differentiation gives

$$\begin{aligned} D'(0) &= 4c_s(1 - \cos \phi), & \phi &= 2\pi m/N, \\ D''(0) &= 8c_s^2(1 - \cos \phi) + 8c_t^2. \end{aligned}$$

*Step 3: Chain rule for  $h(\sqrt{D})$ .* Write  $d = \sqrt{D}$ , so  $d_m = \sqrt{D(0)} = 2R \sin(\pi m/N)$  is the equilibrium pair distance (at  $\sigma = 0$ ,  $R$  the ring radius). Two applications of the chain rule give

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_{\sigma=0} = \frac{h''(d_m)}{4d_m^2} (D'(0))^2 + \frac{h'(d_m)}{2d_m} \left[ D''(0) - \frac{(D'(0))^2}{2d_m^2} \right]. \quad (3.2)$$

Now substitute the Step-2 values  $D'(0) = 4c_s(1 - \cos \phi)$ ,  $D''(0) = 8c_s^2(1 - \cos \phi) + 8c_t^2$ , and the equilibrium identity  $d_m^2 = 2R^2(1 - \cos \phi) \Leftrightarrow 1 - \cos \phi = d_m^2/(2R^2)$ . Absorb  $R$  by setting  $R = 1$  (scale invariance of the Hessian; the theorem holds for any  $R > 0$ ). Then  $D'(0) = 4c_s \cdot d_m^2/2 = 2c_s d_m^2$  and  $(D'(0))^2 = 4c_s^2 d_m^4$ . The first term of (3.2):

$$\frac{h''(d_m)}{4d_m^2} \cdot 4c_s^2 d_m^4 = c_s^2 d_m^2 h''(d_m).$$

The second term, expanded:

$$\begin{aligned} \frac{h'(d_m)}{2d_m} \left[ D''(0) - \frac{(D'(0))^2}{2d_m^2} \right] &= \frac{h'(d_m)}{2d_m} \left[ 8c_s^2(1 - \cos \phi) + 8c_t^2 - \frac{4c_s^2 d_m^4}{2d_m^2} \right] \\ &= \frac{h'(d_m)}{2d_m} [4c_s^2 d_m^2 + 8c_t^2 - 2c_s^2 d_m^2] \quad (\text{using } 1 - \cos \phi = d_m^2/2) \\ &= \frac{h'(d_m)}{2d_m} [2c_s^2 d_m^2 + 8c_t^2] \\ &= c_s^2 d_m h'(d_m) + \frac{4c_t^2 h'(d_m)}{d_m}. \end{aligned}$$

Combining the two contributions yields

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_0 = c_s^2 [d_m^2 h''(d_m) + d_m h'(d_m)] + c_t^2 \frac{4 h'(d_m)}{d_m}. \quad (3.3)$$

(At  $R = 1$ , using  $d_m^2 = 2(1 - \cos \phi)$ , the tangential coefficient can also be written  $4h'(d_m)/d_m = 2(h'(d_m)/d_m) \cdot d_m^2/(1 - \cos \phi)$ .)

*Step 4: Summation over pairs.* For fixed separation  $m \in \{1, \dots, N-1\}$ , sum over the  $N - m$  unordered pairs  $(k, k+m)$  with  $k = 0, \dots, N-m-1$  (the restriction  $k + m < N$

prevents double-counting; each unordered pair  $\{j, j'\}$  with  $j' - j \equiv m \pmod{N}$  and  $j' > j$  appears exactly once). The total number of pairs summed across all  $m$  is  $\sum_{m=1}^{N-1} (N-m) = N(N-1)/2$ , which equals the total number of unordered pairs from  $N$  vertices, confirming no double-counting. Define

$$S_m = \sum_{k=0}^{N-m-1} c_{s,k}^2, \quad \text{where } c_{s,k} = \frac{2k + m - (N-1)}{2N}.$$

Since  $\sum_k c_{s,k} = 0$  by the centering,  $S_m \geq 0$  with equality only if all  $c_{s,k} = 0$  (which requires  $N-m=1$ , i.e.  $m=N-1$ ). The second sum contributes  $c_t^2 \cdot (N-m) = m^2(N-m)/(4N^2)$ , and  $d_m^2/(1-\cos\phi) = 4R^2$ . Summing over  $m=1, \dots, N-1$  yields (3.1).

*Step 5: Sign determination.* Under  $h'(r) < 0$ : the second term  $B_m = h'(d_m) m^2(N-m)/(N^2 d_m) < 0$  for every  $m=1, \dots, N-1$ , since  $h' < 0$ ,  $d_m > 0$ , and  $m^2(N-m) > 0$ . Under  $(rh'(r))' \leq 0$ : the first coefficient  $A_m = d_m^2 h''(d_m) + d_m h'(d_m) = d_m \cdot (rh')'|_{r=d_m} \leq 0$ , so the first term  $A_m S_m \leq 0$  since  $S_m \geq 0$ .

The case  $m=N-1$  requires separate attention. The formula (3.1) is well-defined at  $m=N-1$ : every factor  $(d_m, h'(d_m), h''(d_m), m^2(N-m))$  remains finite and nonzero (with  $d_{N-1} = 2R \sin(\pi(N-1)/N) = 2R \sin(\pi/N) > 0$ ). At  $m=N-1$  there is only one pair  $(0, N-1)$ , and the centering sum has a single term:  $c_{s,0} = (0 + (N-1) - (N-1))/(2N) = 0$ , so  $S_{N-1} = 0$ . The first term  $A_{N-1} S_{N-1} = 0$  therefore *vanishes identically*, regardless of the sign of  $A_{N-1}$ . However, the second term  $B_{N-1} = h'(d_{N-1}) (N-1)^2/(N^2 d_{N-1})$  remains strictly negative (since  $h' < 0$ ). Strict negativity of  $H''(0) = \sum_m (A_m S_m + B_m)$  follows directly from  $B_m < 0$  for every  $m \in \{1, \dots, N-1\}$  (since  $h'(d_m) < 0$  and  $m^2(N-m) > 0$ , so  $B_m < 0$ ) combined with  $A_m S_m \leq 0$  for all  $m$  (from  $A_m \leq 0$  under  $(rh')' \leq 0$  and  $S_m \geq 0$ ). Thus  $H''(0) < 0$ .  $\square$

**Remark 3.1** (Borderline status of the logarithmic interaction). *The sufficient conditions of Theorem 3.1 partition interactions into three classes:*

- (i)  $h' < 0$  and  $(rh')' < 0$ : both terms in (3.1) are strictly negative. Examples:  $h = -r^\alpha$  ( $\alpha > 0$ ), for which  $(rh')' = -\alpha^2 r^{\alpha-1} < 0$ .
- (ii)  $h' < 0$  and  $(rh')' = 0$ : the first term vanishes and only the second contributes. For smooth  $h$  on  $(0, \infty)$ , this class contains a unique functional form, up to positive scaling and additive constant:  $(rh')' = 0 \Rightarrow rh' = c \Rightarrow h' = c/r \Rightarrow h = c \ln r + b$ ; combined with  $h' < 0$  gives  $c < 0$ , i.e.  $h(r) = -a \ln r + b$  with  $a > 0$ .
- (iii)  $h' < 0$  and  $(rh')' > 0$  at some  $r$ : the sign of the first term is indefinite and  $H''(0) < 0$  is not guaranteed.

The 2D Green's function  $h(r) = -\ln r$  falls in class (ii)—the boundary between guaranteed and non-guaranteed negativity. In this precise sense, the logarithmic interaction produces the weakest negativity of  $H''(0)$  among the monotonically decreasing interactions with convex  $rh'(r)$ : any sharper radial decay (as in power-law potentials  $h(r) = r^{-\alpha}$  for  $\alpha > 0$ , class (i)) strengthens the result, while softer decay breaks it.

## 4 The Havelock identity and its metric-independence

### 4.1 The classical identity

**Lemma 4.1.** *For integers  $N \geq 2$  and  $1 \leq m \leq N - 1$ ,*

$$T_m := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2}.$$

*This identity appears in Havelock [1931], §3, as a step in computing the stability eigenvalues of the  $N$ -vortex ring; see also Aref et al. [2003], Appendix A.*

*Proof.* Write  $\omega = e^{2\pi i/N}$ . Using  $2 \sin(\pi p/N) = |1 - \omega^p|$ , expand  $[1 - \cos(2\pi pm/N)]/[4 \sin^2(\pi p/N)] = \frac{1}{2} |\sum_{k=0}^{m-1} \omega^{pk}|^2 = \frac{1}{2} \sum_{j=0}^{m-1} \sum_{k=0}^{m-1} \omega^{p(j-k)}$ . Summing over  $p = 1, \dots, N-1$  and using the orthogonality relation  $\sum_{p=1}^{N-1} \omega^{p\ell} = N \delta_{\ell \equiv 0 \pmod{N}} - 1$ :

$$T_m = \frac{1}{2} \sum_{j,k=0}^{m-1} (N \delta_{j \equiv k \pmod{N}} - 1) = \frac{1}{2} (Nm - m^2) = \frac{m(N-m)}{2},$$

where  $\delta_{j \equiv k}$  contributes  $m$  (since  $0 \leq j, k \leq m-1 < N$  forces  $j = k$ ) and the  $-1$  contributes  $-m^2$ .  $\square$

**Lemma 4.2** (Finite Basel identity). *For every integer  $N \geq 2$ ,*

$$\sum_{p=1}^{N-1} \csc^2\left(\frac{\pi p}{N}\right) = \frac{N^2 - 1}{3}. \quad (4.1)$$

*Proof.* From the partial-fraction expansion  $\pi^2 \csc^2(\pi z) = \sum_{n \in \mathbb{Z}} 1/(z - n)^2$  (? , §5.2.1, differentiated form of the Mittag-Leffler expansion of  $\pi \cot$ ), summing over  $z = p/N$  for  $p \in \{1, \dots, N-1\}$ :

$$\pi^2 \sum_{p=1}^{N-1} \csc^2(\pi p/N) = N^2 \sum_{\substack{k \in \mathbb{Z} \\ k \not\equiv 0 \pmod{N}}} \frac{1}{k^2} = N^2 \left[ 2\zeta(2) - \frac{2}{N^2} \zeta(2) \right] = \frac{\pi^2(N^2 - 1)}{3},$$

using  $k = p - Nn$  as the index runs over non-multiples of  $N$ , and  $\zeta(2) = \pi^2/6$ . Dividing by  $\pi^2$  gives the result.  $\square$

### 4.2 The Riemannian Havelock identity

*Boatto and Cabral [2003] computed numerically that the constrained Lagrangian eigenvalues on the sphere satisfy*

$$\lambda_m = C_1(\varphi_0) - \frac{m(N-m)}{2}, \quad (4.2)$$

*with  $C_1$  depending on the colatitude  $\varphi_0$  but  $f(m, N) = m(N-m)/2$  unchanged from the planar value. The following proof makes explicit why  $f(m, N)$  is invariant: both ingredients of its derivation (the anti-symmetry of the sine sum forcing  $H_{rt}^{(m)} = 0$ , and the  $\mathbb{Z}_N$  orthogonality of characters) are properties of the discrete rotation group, not of the*

specific Green's function. Numerically:  $C_2 = 1.000$  at all tested colatitudes to four decimal places; eigenvalue pair sums  $\lambda_m^+ + \lambda_m^- = 2C_1$  are constant across all modes at each colatitude.

The universality follows from the proof structure of Appendix A: the Havelock identity  $T_m = m(N-m)/2$  is derived from (i) the algebraic factorization  $|\sum_{k=0}^{m-1} \omega^{pk}|^2$  and (ii) the orthogonality of  $\mathbb{Z}_N$  characters, both of which are properties of the discrete rotation group  $\mathbb{Z}_N$ —not of the specific Green's function. The Green's function enters only through:  $H_{rt}^{(m)} = 0$  (anti-symmetry of the sine sum, requiring only that  $G$  depends on geodesic distance) and  $H_{tt}^{(m)} + H_{rr}^{(m)} = \delta$ , where  $\delta = -\sum_p \nabla_{\text{LB}}^2 h(d_p)/(N-1)$  is a surface-dependent constant independent of the Fourier mode  $m$ .

Derivation of the trace formula. Let  $\mathbf{M}_p$  be the  $2 \times 2$  Hessian matrix of  $h(d_{0p})$  with respect to the position of vortex 0, in the local (radial, tangential) frame; its trace is  $\text{tr}[\mathbf{M}_p] = \nabla_{\text{LB}}^2 h(d_p)$ . The standard  $\mathbb{Z}_N$ -Fourier block decomposition (see Havelock 1931, §2) gives, for each component  $\alpha, \beta \in \{r, t\}$ :

$$H_{\alpha\beta}^{(m)} = \underbrace{-\sum_{p=1}^{N-1} M_{p,\alpha\beta}}_{\text{self-interaction at 0}} + \sum_{p=1}^{N-1} M_{p,\alpha\beta} \cos(2\pi pm/N) = \sum_{p=1}^{N-1} M_{p,\alpha\beta} (\cos(2\pi pm/N) - 1).$$

The self-interaction term  $-\sum_p M_{p,\alpha\beta}$  is the negative sum of all off-diagonal blocks, forced by the translational zero mode of  $H$  (total gradient vanishes at the equilibrium). Taking  $\alpha = r, \beta = r$  plus  $\alpha = t, \beta = t$ :

$$H_{rr}^{(m)} + H_{tt}^{(m)} = \sum_{p=1}^{N-1} \nabla_{\text{LB}}^2 h(d_p) (\cos(2\pi pm/N) - 1). \quad (4.3)$$

The coefficient  $(\cos(2\pi pm/N) - 1)$  has the same “ $c_m = -1$ ” structure for every mode and every pair: it arises entirely from the self-interaction bookkeeping, not from any property of the Green's function.

Mode-independence. Equation (4.3) gives  $H_{rr}^{(m)} + H_{tt}^{(m)} = [\sum_p \nabla_{\text{LB}}^2 h(d_p) \cos(2\pi pm/N)] - [\sum_p \nabla_{\text{LB}}^2 h(d_p)]$ . On any compact 2D Riemannian manifold, the point vortex Hamiltonian uses the normalized Green's function  $G$  of the Laplace–Beltrami operator (zero spatial mean:  $\int_M G(x, y) d\mu(x) = 0$ ), which satisfies  $\nabla_{\text{LB},x}^2 G(x, y) = \delta_y(x) - 1/\text{Vol}(M)$ . On the plane,  $G(x, y) = -(2\pi)^{-1} \ln|x-y|$  and the  $1/\text{Vol}$  term vanishes ( $\text{Vol} = \infty$ ), recovering the kernel of equation (7.2). On a compact surface the Green's function decomposes as  $G(x, y) = -(2\pi)^{-1} \ln d(x, y) + R(x, y)$  where  $R$  is the Robin part, satisfying  $\nabla_{\text{LB},x}^2 R(x, y) = -1/\text{Vol}(M) + \delta$ -function contributions that cancel in the smooth Hessian at the polygon equilibrium (where all pairs have  $d(x, y) > 0$ ). At the regular  $N$ -gon on a compact surface with  $\mathbb{Z}_N$ -symmetric vertex positions,  $R(x, y)$  evaluated on pairs  $(x, y) = (z_j, z_k)$  depends only on the separation  $|j - k| \bmod N$  by symmetry, so it contributes a  $\mathbb{Z}_N$ -circulant to  $\nabla^2 H$  whose Fourier eigenvalues are mode- $m$  constants  $R_m = \sum_p R(z_0, z_p) \cos(2\pi pm/N)$ . The argument below shows that these constants contribute only to  $C_1(\text{surface})$  (the mode-independent offset in  $\lambda_m = C_1 - f(m, N)$ ) and not to the polygon function  $f(m, N) = m(N-m)/2$ , which is determined entirely by the short-distance  $-\ln d$  kernel. The mode-independence argument below depends only on  $\nabla_{\text{LB}}^2 G$  away from the diagonal, where  $\nabla_{\text{LB},x}^2 G(x, y) = -1/\text{Vol}(M)$  (constant, independent of  $x, y$ , or separation). Hence  $\nabla_{\text{LB}}^2 h(d_p) = c$  is the same constant for every pair  $p$

(with  $c = -1/\text{Vol}(M)$  on a compact surface,  $c = 0$  on the plane). Apply (4.3): since  $\nabla_{\text{LB}}^2 h(d_p) = c$  is  $p$ -independent,

$$H_{rr}^{(m)} + H_{tt}^{(m)} = c \sum_{p=1}^{N-1} (\cos(2\pi pm/N) - 1) = c(-1 - (N-1)) = -Nc,$$

using  $\sum_{p=1}^{N-1} \cos(2\pi pm/N) = -1$  for  $m \in \{1, \dots, N-1\}$  (from  $\sum_{p=0}^{N-1} e^{2\pi i pm/N} = 0$ ). So  $H_{rr}^{(m)} + H_{tt}^{(m)} = -Nc \equiv \delta$  is mode-independent. On the plane ( $\text{Vol} = \infty$ ),  $c = 0$ , so  $\delta = 0$  (exact tracelessness); on a compact surface  $c = -1/\text{Vol}(M)$ , giving  $\delta = N/\text{Vol}(M)$ , which shifts  $C_1$  without affecting  $f(m, N)$ . On the sphere of radius  $R$ :  $\text{Vol} = 4\pi R^2$ , so  $\delta = N/(4\pi R^2)$ , consistent with the colatitude-dependent  $C_1$  of Section 5.1.

Determination of  $\lambda_m^-$  on constant-curvature surfaces. The formulas  $\lambda_m^+ = C_1(\text{surface}) - m(N-m)/2$  and  $\lambda_m^- = m(N-m)/2$ , with the same mode structure  $f(m, N) = m(N-m)/2$  as on the plane, were derived by Boatto and Cabral [2003] (Eq. 2.12) for the regular  $N$ -ring on  $\mathbf{S}^2$  (and extend to  $\mathbf{H}^2$  by the analogous hyperbolic computation). The derivation proceeds by computing the pair-by-pair Hessian in stereographic coordinates and Fourier-decomposing over  $\mathbb{Z}_N$ ; the key observation is that the geodesic distance function  $d(z_j, z_k)$  on a constant-curvature surface depends on the vortex positions only through the angular separation  $\theta_j - \theta_k$  (by rotational invariance at fixed colatitude). Applying the  $\mathbb{Z}_N$  Fourier decomposition as in the flat-plane Havelock argument (Appendix 4.1) produces a circulant matrix whose eigenvalues have the form  $C_1(\xi) + f(m, N)$  up to sign and overall scale, with  $f(m, N) = m(N-m)/2$  appearing universally. This universality is structural: it reflects the  $\mathbb{Z}_N$  orthogonality of trigonometric functions used in the Havelock identity, not the specific metric. We verify the identification numerically below; for the original derivation see Boatto–Cabral.

Numerical verification:  $\lambda_m^- = m(N-m)/2$  to  $10^{-14}$  for  $N \in \{3, \dots, 12\}$  and  $\mathbf{S}^2$  colatitudes  $\varphi_0 \in \{15^\circ, 30^\circ, \dots, 75^\circ\}$  (`tests/test_curved_surfaces.py`).

The stability threshold  $N_{\text{crit}}$  is therefore determined by  $C_1(\text{surface}) \geq \max_m m(N-m)/2$ : a condition involving the universal function  $f$  and a single surface-dependent constant. On the plane:  $C_1 = N-1$ , giving  $N_{\text{crit}} = 7$  (the largest integer with  $(N-1) \geq \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$ ). On the sphere:  $C_1(\varphi_0) < N-1$  for  $\varphi_0 > 0$ , giving the colatitude-dependent threshold of Section 5.1.

**Remark 4.1** (Representation-theoretic content of the stability boundary). The decomposition  $\lambda_m = C_1 - f(m, N)$  separates the stability condition into a  $\mathbb{Z}_N$  representation-theoretic part  $f$  and a geometric part  $C_1$ , with three consequences.

1. Spectral gap as a quantum phase transition. Write  $H_N = C_1 I - F_N$  where  $F_N = \text{diag}(f(0, N), \dots, f(N-1, N))$  in the Fourier basis. Stability is equivalent to the positivity condition  $H_N \geq 0$ . The spectral gap  $\Delta(N) = C_1 - f_{\text{max}}(N)$  closes at  $N = 7$  on the flat plane ( $\Delta = 0$ ), marking a transition from the gapped phase ( $N \leq 6$ , all eigenvalues positive) to the broken phase ( $N \geq 8$ , negative eigenvalue). Gaussian curvature acts as a tuning parameter:  $K < 0$  opens the gap,  $K > 0$  closes it.

2. Crystallographic restriction analogue. The classical crystallographic restriction constrains the order of a rotation compatible with a lattice:  $\text{tr}(R) = 2 \cos(2\pi/N) \in \mathbb{Z}$  forces  $N \in \{1, 2, 3, 4, 6\}$  in  $\mathbb{R}^2$ . Our stability restriction constrains the order of a  $\mathbb{Z}_N$  polygon compatible with the geometry:  $C_1 \geq f_{\text{max}}(N)$  forces  $N \leq 7$  on  $\mathbb{R}^2$ . Both are discrete constraints on  $N$  arising from the interplay between  $\mathbb{Z}_N$  algebra (trace / Casimir) and the ambient geometry (lattice / curvature).

3. Entanglement-witness structure.  $H_N = C_1 I - F_N$  has the structure of an entanglement witness: “separable” = stable, “entangled” = unstable. Positivity tools from quantum information could apply to non-abelian problems.

4. Spectral statistics and Landau transition. Level spacings  $(N-2m-1)/2$  are arithmetic (maximally rigid). The gap closure  $\Delta(N) \sim -(N-7)$  has mean-field exponents ( $\nu = 1$ ,  $\beta = 1/2$ ) with critical amplitude  $\alpha_0 = 45/14$  (Section 10).

The value  $C_1 = N-1$  on the plane has a geometric origin: for  $h' = -1/d$  (the logarithmic force law), each vortex pair  $(0, k)$  contributes the same amount  $-1/(2R)$  to the gradient  $\partial H/\partial x_0$ , regardless of the separation  $k$ . This is because  $h'(d_k)(x_0 - x_k)/d_k = -(1 - \cos(2\pi k/N))/(4R \sin^2(\pi k/N)) = -1/(2R)$ : the geometric factor  $(1 - \cos)$  cancels against the denominator  $\sin^2 = (1 - \cos)/2$ . The cancellation is the scale invariance of  $h' = -1/d$ : the force law  $h' \propto 1/d$  precisely compensates the  $d$ -dependence of the chord geometry, making each pair’s contribution independent of the separation. This property characterizes the logarithm uniquely: among radially symmetric pairwise interactions on  $\mathbb{R}^2$ ,  $h = -\ln|z-w|$  is the unique function satisfying  $h(d\lambda) = h(d) + h(\lambda)$  (additivity under rescaling—Cauchy’s functional equation with continuity). This additivity under the dilation subgroup  $z \mapsto \lambda z$  (for which  $|f(z) - f(w)| = |\lambda| |z - w|$ ) is Cauchy’s multiplicative functional equation  $h(ab) = h(a) + h(b)$ , whose unique continuous solution is  $h = c \ln$ .

## 5 Stability thresholds on constant-curvature surfaces

Prior work and what is new. Boatto and Cabral [2003] computed the constrained Lagrangian eigenvalues for a latitudinal ring of point vortices on the non-rotating sphere at arbitrary colatitude, establishing that the stability boundary shifts with latitude. Lim et al. [2001] studied relative equilibria of vortex rings on  $S^2$  within the Hamiltonian framework. The spherical Havelock eigenvalues are therefore not novel.

What the proof in §4.2 adds is an explicit algebraic argument that the  $m$ -dependent part  $f(m, N) = m(N-m)/2$  is metric-independent—a consequence of  $\mathbb{Z}_N$  representation theory alone—while only the additive constant  $C_1$  carries the geometric information. A specialist who worked through the Boatto–Cabral calculation would likely have noticed this; the value here is in making the separation explicit and, more substantively, in extending the framework to the hyperbolic plane (§5.2), where  $N_{\text{crit}} > 7$  and systematic stability tables do not appear to be in the existing literature.

### 5.1 Curved 2D surfaces: the spherical stability boundary

On the 2-sphere, the point vortex interaction  $h(\theta) = -\ln(2 \sin(\theta/2))$  satisfies  $h' < 0$  and  $(\sin \theta \cdot h')' = \frac{1}{2} \sin \theta > 0$  (class (iii)). While  $H''(0) > 0$  along the spiral at all tested colatitudes (the polygon is a spiral-direction energy minimum even without the Lagrange shift), the constrained Hessian tells a different story.

The angular impulse on the sphere is  $L = \sum \cos \theta_k$ , with Hessian  $\partial^2 L / \partial \theta_k^2 = -\cos \varphi_0$  at colatitude  $\varphi_0$ . The Lagrangian shift from  $\mu_L$  is therefore proportional to  $\cos \varphi_0$ : full near the pole ( $\varphi_0 \rightarrow 0$ ), vanishing at the equator ( $\varphi_0 = 90^\circ$ ). This latitude-dependent shift produces a colatitude-dependent stability boundary:

| $N$                     | 3     | 4     | 5     | 6     | 7  | $\geq 8$ |
|-------------------------|-------|-------|-------|-------|----|----------|
| $\varphi_{\text{crit}}$ | 60.0° | 48.2° | 41.4° | 25.8° | 0° | 0°       |

The  $N$ -gon is a constrained energy minimum on the sphere if and only if  $\varphi_0 < \varphi_{\text{crit}}(N)$ . The dimensionless eigenvalue  $\bar{\lambda} = \sin^2 \varphi_0 \cdot \lambda$  recovers the Havelock values to within 1% at  $\varphi_0 = 3^\circ$  ( $N=5$ :  $\bar{\lambda} = 0.99$ , Havelock = 1.0;  $N=6$ : 0.49 vs 0.5;  $N=8$ :  $-1.01$  vs  $-1.0$ ).

Saturn's hexagon at  $76^\circ N$  ( $\varphi_0 = 14^\circ$ ) lies inside  $\varphi_{\text{crit}}(6) = 26.6^\circ$ : the hexagonal ring is in the stable zone. If the hexagon were at  $60^\circ N$  ( $\varphi_0 = 30^\circ > 26.6^\circ$ ), it would cross the boundary and the pattern would not persist. Jupiter's octagon ( $N = 8$ ) is unstable at all colatitudes, consistent with the need for a central vortex.

**Derivation of  $C_1(\mathbf{S}^2)$  and exact thresholds.** For  $N$  vortices on a latitude circle at colatitude  $\varphi_0$ , the great-circle separation satisfies  $2 \sin(\gamma_{jk}/2) = 2 \sin \varphi_0 |\sin(\Delta\phi_{jk}/2)|$ , so the spherical pair interaction decomposes as

$$-\ln(2 \sin(\gamma_{jk}/2)) = -\ln(2 |\sin(\Delta\phi_{jk}/2)|) - \ln(\sin \varphi_0).$$

The first term is the flat Thomson energy in the azimuthal variable  $\phi$ ; its Hessian along azimuthal Fourier modes gives the Havelock eigenvalues  $(N-1)/2 - m(N-m)/2$  exactly as in the flat case. The second term  $-\binom{N}{2} \ln(\sin \varphi_0)$  is purely radial and enters through the Lagrange multiplier.

The angular impulse on  $\mathbf{S}^2$  is  $L = \sum \cos \varphi_k$ . At the ring,  $\partial^2 L / \partial \varphi_k^2 = -\cos \varphi_0$ , so the Lagrange shift  $\mu_L \cdot \partial^2 L / \partial \varphi_k^2$  is proportional to  $\cos \varphi_0$ : maximal near the pole, vanishing at the equator. Computing  $\mu_L$  from the stationarity condition and combining with the radial Hessian of the pair interaction yields the curvature coefficient (cf. Boatto and Cabral 2003):

$$C_1(\mathbf{S}^2, \xi) = \frac{(N-1)(1+\xi^2)}{(1+\xi)^2}, \quad (5.1)$$

where  $\xi = \tan^2(\varphi_0/2)$  is the stereographic parameter. Marginal stability  $C_1 = T := m(N-m)/2$  at the most dangerous mode  $m = \lfloor N/2 \rfloor$  yields the quadratic

$$(N-1-T)\xi^2 - 2T\xi + (N-1-T) = 0, \quad (5.2)$$

whose smaller root is  $\xi_{\text{crit}}(N) = (T - \sqrt{T^2 - (N-1-T)^2}) / (N-1-T)$ .

| $N$      | $T = m(N-m)/2$ | $\xi_{\text{crit}}$ | $\varphi_{\text{crit}}$ | stable iff               |
|----------|----------------|---------------------|-------------------------|--------------------------|
| 3        | 1              | 1                   | $90.0^\circ$            | $\varphi_0 < 90^\circ$   |
| 4        | 2              | $2 - \sqrt{3}$      | $54.7^\circ$            | $\varphi_0 < 54.7^\circ$ |
| 5        | 3              | $3 - 2\sqrt{2}$     | $45.0^\circ$            | $\varphi_0 < 45.0^\circ$ |
| 6        | $9/2$          | $9 - 4\sqrt{5}$     | $26.6^\circ$            | $\varphi_0 < 26.6^\circ$ |
| 7        | 6              | 0                   | $0^\circ$               | only at pole             |
| $\geq 8$ | $> N-1$        | —                   | —                       | never                    |

The thresholds are fundamental units (Pell solutions) in quadratic number fields:  $\xi^*(4) = 2 - \sqrt{3} \in \mathbb{Q}(\sqrt{3})$ ,  $\xi^*(5) = 3 - 2\sqrt{2} \in \mathbb{Q}(\sqrt{2})$ ,  $\xi^*(6) = 9 - 4\sqrt{5} \in \mathbb{Q}(\sqrt{5})$ . This is the same algebraic structure as the  $\mathbf{H}^2$  thresholds (§??), reflecting the  $\xi \rightarrow -\xi$  duality between (5.1) and the hyperbolic formula  $C_1(\mathbf{H}^2) = (N-1)(1+\xi^2)/(1-\xi)^2$ .

**Remark 5.1** (Pell unification of  $\mathbf{S}^2$  and  $\mathbf{H}^2$  thresholds). On both surfaces the threshold equation is a palindromic quadratic  $(N-1-T)\xi^2 \mp 2T\xi + (N-1-T) = 0$  (with  $-$  for  $\mathbf{S}^2$  and  $+$  for  $\mathbf{H}^2$ ), producing algebraic thresholds in  $\mathbb{Q}(\sqrt{D})$  where  $D$  depends on  $N$  via the

mode structure. The  $N = 4$  thresholds  $\xi^*(\mathbf{S}^2) = 2 - \sqrt{3}$  and  $\xi^*(\mathbf{H}^2) = 8 - 3\sqrt{7}$  (for  $N = 8$ ) are both Pell units in their respective fields. The apparent “ $\mathbf{S}^2$  rational vs.  $\mathbf{H}^2$  irrational” dichotomy in earlier versions of this paper was an artifact of an incorrect  $C_1$  formula; the correct formula reveals a unified Pell structure.

Code: `algebraic_thresholds.sphere_stability_threshold(N)` implements the quadratic formula; `test_algebraic_thresholds.py` verifies the  $C_1$  identity  $C_1(\xi^*) = m(N-m)/2$  to  $10^{-10}$  for  $N = 3, \dots, 6$ .

On the flat torus,  $H''(0) < 0$  approaches the planar value for polygons small relative to the domain, with periodic corrections entering at the domain scale.

## 5.2 The hyperbolic plane and Gaussian curvature

Kimura [1999] and others have studied point vortex dynamics on surfaces of constant curvature; the Hamiltonian for the hyperbolic plane is known. Systematic stability tables for  $N$ -gons on  $\mathbf{H}^2$  do not appear to be in the literature, and the results of this subsection are likely the most novel part of Section 5.

The threshold  $N_{\text{crit}}$  on any surface is the largest  $N$  with  $C_1(\text{surface}) \geq \max_m m(N-m)/2$ . Gaussian curvature  $K$  is the control parameter:

| Curvature            | $N_{\text{crit}}$ | Mechanism                       |
|----------------------|-------------------|---------------------------------|
| $K > 0$ (sphere)     | $< 7$             | Positive curvature destabilizes |
| $K = 0$ (plane)      | $= 7$             | Critical transition             |
| $K < 0$ (hyperbolic) | $> 7$             | Negative curvature stabilizes   |

**Remark 5.2** (Order of the curvature correction). The curvature enters  $N_{\text{crit}}$  at  $O(K)$ , not  $O(K^2)$ : it acts through the surface Green’s function via the Havelock coefficient  $C_1(\text{surface}, \xi)$  (equations (5.1), (5.3)), which is linear in  $K$  at leading order. On a compact surface the Gauss–Bonnet constraint  $\int K \, dA = 2\pi\chi$  enters only through  $C_1$ ; no additional  $O(K^2)$  self-energy correction arises for point vortices (such a correction would require finite-area patches, where it is  $O(\varepsilon^2)$  and topology-independent).

On the hyperbolic plane with curvature radius  $a$ , numerical computation (Poincaré disk model, exact Green’s function  $h = -\ln \sinh(d_{\mathbf{H}^2}/(2a))$ , corrected Lagrange multiplier for the  $\text{SO}(2)$  moment map  $J = \sum_k (a^2 + r_k^2)/(a^2 - r_k^2)$ ) gives:

| $R/a$          | $N_{\text{crit}}$ | $7 \rightarrow N$ transition |
|----------------|-------------------|------------------------------|
| 0.30–0.50      | 7                 | (no change)                  |
| $\approx 0.52$ | 8                 | $7 \rightarrow 8$ transition |
| 0.65–0.75      | 9                 |                              |
| 0.80           | 10                |                              |
| 0.90           | 11                |                              |
| 1.0            | 12                |                              |
| 1.2            | 16                |                              |
| $\geq 1.5$     | $\geq 24$         | (tested up to $N = 24$ )     |

The Riemannian Havelock identity  $\lambda_m = C_1(R/a) - m(N-m)/2$  is verified to hold (computed  $C_1$  values are mode-independent to numerical precision  $\sim 10^{-5}$ ). The exact  $\mathbf{H}^2$

Green's function yields the closed-form expression (derived in Appendix A)

$$C_1(\mathbf{H}^2, \xi) = \frac{(N-1)(1+\xi^2)}{(1-\xi)^2}, \quad \xi = \frac{r_E^2}{a^2} = |K| r_E^2, \quad (5.3)$$

where  $r_E$  is the Euclidean ring radius in the Poincaré disk and  $K = -1/a^2$ . Since  $(1+\xi^2)/(1-\xi)^2 > 1$  for all  $\xi > 0$ , one has  $C_1(\mathbf{H}^2) > N-1$  for all  $R/a > 0$ , confirming the stabilizing effect of negative curvature.

$N_{\text{crit}}$  increases through integer jumps as  $R/a$  grows, with the  $7 \rightarrow 8$  transition at  $R/a \approx 0.52$  (more precisely, the  $m = \lfloor N/2 \rfloor$  eigenvalue of the  $N = 8$  ring crosses zero near this value). We conjecture that  $N_{\text{crit}}(R/a)$  is non-decreasing (verified for  $R/a \leq 2$ ; a plausible mechanism is the transition from class (ii) at  $R \ll a$  to class (i) at  $R \gg a$ , but this does not constitute a proof because the Lagrange multiplier also depends on  $R/a$  in a non-trivial way).

**Near- $K = 0$  behavior.** Expanding (5.3) for  $N = 7$ ,  $m = 3$  (see Appendix A for the algebraic simplification), the  $m = 3$  eigenvalue satisfies

$$\lambda_3(N = 7, \xi) = \frac{12\xi}{r_E^2(1-\xi)^2} = \frac{12|K|}{(1-\xi)^2} = 12|K| + O(K^2 r_E^2) > 0. \quad (5.4)$$

Negative curvature stabilizes the  $N = 7$  ring at first order in  $|K|$ :  $N_{\text{crit}}(K)$  stays at 7 for all  $K \leq 0$  in the sense that the formerly-neutral mode becomes strictly stable.

The  $7 \rightarrow 8$  transition is governed by the  $N = 8$  ring reaching marginal stability. Setting  $C_1(\mathbf{H}^2, \xi) = m(N-m)/2|_{m=4} = 8$  gives  $7(1+\xi^2)/(1-\xi)^2 = 8$ , which simplifies to

$$\xi^2 - 16\xi + 1 = 0 \implies \xi^* = 8 - 3\sqrt{7} \approx 0.0627, \quad (5.5)$$

exact. In terms of geodesic ring radius  $\rho$ , this corresponds to  $\rho/a \approx 0.52$  (or  $|K|\rho^2 \approx 0.27$ ). The coefficient

$$\gamma \equiv \frac{1}{(\xi^*)^2} = 127 + 48\sqrt{7} \approx 254 \quad (5.6)$$

carries an algebraic interpretation. The ring of integers of  $\mathbb{Q}(\sqrt{7})$  has fundamental unit  $\varepsilon = 8 + 3\sqrt{7}$ , since  $(8+3\sqrt{7})(8-3\sqrt{7}) = 64-63 = 1$ . Therefore  $\xi^* = 8 - 3\sqrt{7} = \varepsilon^{-1}$  is the inverse of the fundamental unit—the unique unit below 1 in  $\mathbb{Z}[\sqrt{7}]$ —and  $\gamma = (\xi^*)^{-2} = \varepsilon^2$ . The verification identity  $(8-3\sqrt{7})^2(127+48\sqrt{7}) = 1$  is not a computational coincidence: it is the norm-1 property  $\varepsilon^{-2} \cdot \varepsilon^2 = 1$  in  $\mathbb{Z}[\sqrt{7}]$ .

The field  $\mathbb{Q}(\sqrt{7})$  arises through a specific algebraic chain: for the  $N = 8$  transition,  $A = -1$  and  $B = 16$ , giving discriminant  $\Delta = B^2 - 4A^2 = 256 - 4 = 252 = 2^2 \cdot 3^2 \cdot 7$ , with squarefree part 7. The 7 enters as the factor  $(N-1) = 7$  in the general formula  $\Delta = (N-1)(N-2)^2$  (Proof of (6.2))—the flat-plane stability boundary  $N = 7$  leaves its algebraic shadow in the discriminant. The substantive content is the palindromic structure: the threshold polynomial has the form  $A\xi^2 + B\xi + A = 0$  (same leading and trailing coefficient), which forces  $\xi_1 \xi_2 = A/A = 1$ —the two roots are multiplicative inverses, so  $\xi^*$  is automatically an algebraic unit (an element of norm 1 in  $\mathbb{Z}[\sqrt{7}]$ ). This palindromic property is not generic: it reflects the inversion symmetry  $\xi \leftrightarrow 1/\xi$  of the Poincaré disk model, which in turn is the restriction to the real axis of the conformal automorphism  $z \mapsto a^2/\bar{z}$  of the disk. The stability threshold is a unit because the hyperbolic energy is conformally covariant (Proposition 2.1): the palindromic polynomial encodes the same Möbius symmetry at the level of algebraic number theory.

General algebraic field pattern ( $N \geq 8$ ).

## 6 Algebraic structure of the stability boundary

The  $7 \rightarrow 8$  transition is not an isolated result: the  $N \rightarrow (N+1)$  transition threshold on  $\mathbf{H}^2$  is always the smaller positive root of the palindromic quadratic

$$A\xi^2 + B\xi + A = 0, \quad A = (N-1) - \frac{m(N-m)}{2}, \quad B = m(N-m), \quad m = \lfloor N/2 \rfloor. \quad (6.1)$$

Since  $A < 0$  for  $N \geq 8$  and  $B > 0$ , the smaller positive root is  $\xi^* = (B/2 - \sqrt{(B/2)^2 - A^2}) / |A|$ , and its field extension is determined by the squarefree part of the discriminant  $(B/2)^2 - A^2 \in \mathbb{Z}$ :

| $N$ | $\xi^*$ (approx.) | Field                   | $D$ | $\xi^*$ (exact form)    |
|-----|-------------------|-------------------------|-----|-------------------------|
| 7   | 0                 | $\mathbb{Q}$            | 1   | 0 (marginal)            |
| 8   | 0.0627            | $\mathbb{Q}(\sqrt{7})$  | 7   | $8 - 3\sqrt{7}$         |
| 9   | 0.1010            | $\mathbb{Q}(\sqrt{6})$  | 6   | $5 - 2\sqrt{6}$         |
| 10  | 0.1429            | $\mathbb{Q}$            | 1   | $1/7$                   |
| 11  | 0.1716            | $\mathbb{Q}(\sqrt{2})$  | 2   | $3 - 2\sqrt{2}$         |
| 12  | 0.2024            | $\mathbb{Q}(\sqrt{11})$ | 11  | $(18 - 5\sqrt{11})/7$   |
| 13  | 0.2251            | $\mathbb{Q}(\sqrt{10})$ | 10  | $(7 - 2\sqrt{10})/3$    |
| 14  | 0.2493            | $\mathbb{Q}(\sqrt{13})$ | 13  | $(49 - 12\sqrt{13})/23$ |
| 15  | 0.2679            | $\mathbb{Q}(\sqrt{3})$  | 3   | $2 - \sqrt{3}$          |
| 16  | 0.2876            | $\mathbb{Q}(\sqrt{15})$ | 15  | $(32 - 7\sqrt{15})/17$  |

The discriminant's squarefree part  $D$  follows the pattern

$$D = \begin{cases} \text{sq\_free}(N-1) & N \text{ even,} \\ \text{sq\_free}(N-3) & N \text{ odd,} \end{cases} \quad (6.2)$$

for all  $N \geq 8$ . The deeper content is that the palindromic structure encodes the conformal inversion  $\xi \leftrightarrow 1/\xi$  of the Poincaré disk (as noted above); the field pattern itself follows from elementary factorization.

*Proof of (6.2).* The palindromic quadratic  $A\xi^2 + B\xi + A = 0$  has discriminant  $\Delta = B^2 - 4A^2$ . Using  $A = (N-1) - m(N-m)/2$  and  $B = m(N-m)$ :

$$\Delta = m^2(N-m)^2 - 4[(N-1) - m(N-m)/2]^2 = 4(N-1)[m(N-m) - (N-1)].$$

*Even  $N$  ( $m = N/2$ ):*  $m(N-m) = N^2/4$ , so  $m(N-m) - (N-1) = N^2/4 - N + 1 = (N-2)^2/4$ . Therefore  $\Delta = 4(N-1)(N-2)^2/4 = (N-1)(N-2)^2$ . The squarefree part of  $\Delta$  equals  $\text{sq\_free}(N-1)$ , since  $(N-2)^2$  is a perfect square. The threshold  $\xi^*$  involves  $\sqrt{\Delta}$ , so  $\xi^* \in \mathbb{Q}(\sqrt{\text{sq\_free}(N-1)})$ .

*Odd  $N$  ( $m = (N-1)/2$ ):*  $m(N-m) = (N-1)(N+1)/4 = (N^2-1)/4$ , so  $m(N-m) - (N-1) = (N^2-1)/4 - (N-1) = (N-1)(N-3)/4$ . Therefore  $\Delta = 4(N-1) \cdot (N-1)(N-3)/4 = (N-1)^2(N-3)$ . The squarefree part of  $\Delta$  equals  $\text{sq\_free}(N-3)$ , since  $(N-1)^2$  is a perfect square. Hence  $\xi^* \in \mathbb{Q}(\sqrt{\text{sq\_free}(N-3)})$ .  $\square$

**Algebraic integers and the Pell equation.** *The threshold  $\xi^*(N)$  is an algebraic integer—i.e., it satisfies a monic polynomial  $\xi^2 - c\xi + 1 = 0$  with  $c \in \mathbb{Z}$ —if and only if  $B/A \in \mathbb{Z}$ .*

**Proposition 6.1** (Algebraic-integer classification). *The threshold  $\xi^*(N)$  is an algebraic integer if and only if  $N \in \{8, 9, 11, 15, 23\}$ .*

*Proof. Algebraic-integer criterion.*  $\xi^*$  is a root of the palindromic quadratic  $A\xi^2 + B\xi + A = 0$  with  $A, B \in \mathbb{Q}$  (rationality follows from the derivation of  $A, B$  above;  $A \neq 0$  since  $\xi^* \neq 0, \infty$ ). Dividing by  $A$ :  $\xi^2 + (B/A)\xi + 1 = 0$ .

*Irrational case.* When the discriminant  $\Delta = B^2 - 4A^2 > 0$  is not a perfect square of a rational,  $\xi^*$  is quadratic irrational, and  $\xi^2 + (B/A)\xi + 1 = 0$  is *the* minimal polynomial.  $\xi^*$  is an algebraic integer iff this monic minimal polynomial has integer coefficients, i.e.,  $B/A \in \mathbb{Z}$  (since  $1 \in \mathbb{Z}$  is automatic). Equivalently, by the sum/product formula for quadratic roots,  $\xi^* + (\xi^*)^{-1} = -B/A$  must be an integer.

*Rational case.* When  $\Delta$  is a rational square,  $\xi^*$  is rational; the minimal polynomial is linear. A rational number is an algebraic integer iff it is an integer. Since  $\xi^* \in (0, 1)$  by the stability analysis (§??),  $\xi^*$  is never a positive integer, so the rational case yields  $\xi^* \notin \mathbb{Z}$  (i.e.,  $\xi^*$  is not an algebraic integer) whenever it occurs. Rational thresholds occur exactly when the squarefree part of  $\Delta$  is 1: for even  $N$ , when  $\text{sq.free}(N-1) = 1$  (i.e.,  $N-1$  a perfect square,  $N \in \{2, 5, 10, 17, 26, \dots\}$ , of which  $N \geq 8$  gives  $N = 10, 17, 26, \dots$ ); for odd  $N$ , when  $\text{sq.free}(N-3) = 1$  ( $N \in \{4, 7, 12, 19, 28, \dots\}$ , odd  $N \geq 9$  giving  $N = 19, 31, \dots$ ). The smallest cases are  $N = 10$  ( $\xi^* = 1/7$ ) and  $N = 19$  ( $\xi^* = 1/3$ ), neither an integer, confirming the rational case contributes no algebraic-integer thresholds.

*Case analysis for the irrational case.* For  $\xi^*$  irrational, the condition  $B/A \in \mathbb{Z}$  determines which  $N$  yield algebraic integers.

For odd  $N \geq 9$ , with  $m = (N-1)/2$  (integer since  $N$  is odd):  $A = (N-1) - m(N-m)/2 = (N-1) - (N-1)(N+1)/8 = (N-1)[1 - (N+1)/8] = (N-1)(7-N)/8 = -(N-1)(N-7)/8$  (a rational number, not necessarily integer: e.g.,  $N = 9$  gives  $A = -2$ ,  $N = 11$  gives  $A = -5$ ,  $N = 13$  gives  $A = -9$ ).  $B = m(N-m) = (N-1)(N+1)/4$  (also rational). The ratio  $B/A = -2(N+1)/(N-7)$  is well-defined for  $N \neq 7$  (the  $(N-1)$  factor cancels and the  $8/4 = 2$  combines). This is an integer iff  $(N-7) \mid 2(N+1)$ . Since  $2(N+1) = 2(N-7) + 16$ , the condition reduces to  $(N-7) \mid 16$ . The positive divisors of 16 giving odd  $N \geq 9$  are  $N-7 \in \{2, 4, 8, 16\}$ , i.e.,  $N \in \{9, 11, 15, 23\}$ . For even  $N \geq 8$ : with  $m = N/2$ , the coefficients are  $A = (N-1) - m(N-m)/2 = (N-1) - N^2/8 = -(N^2-8N+8)/8$  and  $B = m(N-m) = N^2/4$ , giving

$$-\frac{B}{A} = \frac{2N^2}{N^2-8N+8}. \quad (6.3)$$

This is an integer iff  $(N^2-8N+8) \mid 2N^2$ . Apply the division algorithm:

$$2N^2 = 2(N^2-8N+8) + 16N - 16 = 2(N^2-8N+8) + 16(N-1),$$

so  $(N^2-8N+8) \mid 2N^2$  iff  $(N^2-8N+8) \mid 16(N-1)$ .

For  $N \geq 10$ :  $N^2-8N+8 = (N-4)^2-8 > 0$ . Compare  $N^2-8N+8$  with  $16(N-1)$ : their difference is  $N^2-24N+24$ , with roots  $N = 12 \pm \sqrt{120}$ , so  $N^2-24N+24 > 0$  for  $N \geq 23$ . Hence for  $N \geq 23$ ,  $N^2-8N+8 > 16(N-1) > 0$ , and divisibility  $(N^2-8N+8) \mid 16(N-1)$  is impossible in the positive integers (a positive divisor  $d = N^2-8N+8$  cannot divide a positive integer strictly smaller than  $d$ , namely  $16(N-1)$ , unless that integer equals zero—but  $16(N-1) > 0$  for  $N \geq 2$ ). Hence no even  $N \geq 23$  gives an algebraic integer.

For even  $N \in \{10, 12, 14, 16, 18, 20, 22\}$  (7 cases), compute directly:

| $N$ | $N^2 - 8N + 8$ | $16(N-1)$ | $16(N-1) \bmod (N^2 - 8N + 8)$   |
|-----|----------------|-----------|----------------------------------|
| 10  | 28             | 144       | $144 - 5 \cdot 28 = 4 \neq 0$    |
| 12  | 56             | 176       | $176 - 3 \cdot 56 = 8 \neq 0$    |
| 14  | 92             | 208       | $208 - 2 \cdot 92 = 24 \neq 0$   |
| 16  | 136            | 240       | $240 - 1 \cdot 136 = 104 \neq 0$ |
| 18  | 188            | 272       | $272 - 1 \cdot 188 = 84 \neq 0$  |
| 20  | 248            | 304       | $304 - 1 \cdot 248 = 56 \neq 0$  |
| 22  | 316            | 336       | $336 - 1 \cdot 316 = 20 \neq 0$  |

None gives integer divisibility. For  $N = 8$ :  $N^2 - 8N + 8 = 8$ ,  $2N^2 = 128$ ,  $128/8 = 16$ ; so  $-B/A = 16 \in \mathbb{Z}$ , as required. Hence for even  $N \geq 8$  only  $N = 8$  yields an algebraic-integer threshold. Combined with the odd case, the full set is  $\{8, 9, 11, 15, 23\}$  as claimed.  $\square$

The differences between consecutive algebraic-integer values are  $\{1, 2, 4, 8\}$ —powers of 2—reflecting the binary structure of the divisors of 16.

For these five values,  $\xi^*$  satisfies the Pell equation. Since the monic polynomial  $\xi^2 - c\xi + 1 = 0$  forces  $\xi \cdot \xi' = 1$ , writing  $\xi^* = a + b\sqrt{D}$  with  $a, b \in \mathbb{Z}$  (or half-integers for  $D \equiv 1 \pmod{4}$ ) gives  $a^2 - Db^2 = 1$ : the classical Pell equation. Each threshold is an explicit power of the inverse fundamental unit  $\varepsilon$  of  $\mathbb{Q}(\sqrt{D})$ :

| $N$ | $D$ | Fundamental unit $\varepsilon$              | $\xi^*(N)$         | $c = -B/A$ | $\text{Norm}(\varepsilon)$ |
|-----|-----|---------------------------------------------|--------------------|------------|----------------------------|
| 8   | 7   | $8 + 3\sqrt{7}$                             | $\varepsilon^{-1}$ | 16         | +1                         |
| 9   | 6   | $5 + 2\sqrt{6}$                             | $\varepsilon^{-1}$ | 10         | +1                         |
| 11  | 2   | $1 + \sqrt{2}$ (silver ratio)               | $\varepsilon^{-2}$ | 6          | -1                         |
| 15  | 3   | $2 + \sqrt{3}$                              | $\varepsilon^{-1}$ | 4          | +1                         |
| 23  | 5   | $\varphi = (1 + \sqrt{5})/2$ (golden ratio) | $\varphi^{-2}$     | 3          | -1                         |

The exponent is  $k = 1$  when  $\text{Norm}(\varepsilon) = +1$  (the fundamental unit solves  $x^2 - Dy^2 = 1$ ) and  $k = 2$  when  $\text{Norm}(\varepsilon) = -1$  (the fundamental unit solves  $x^2 - Dy^2 = -1$ ; only its square has norm +1, as required by the palindromic constraint  $\xi \cdot \xi' = 1$ ).

**The golden ratio as the terminal case.** The last algebraic-integer threshold ( $N = 23$ ) has  $\xi^*(23) = \varphi^{-2} = (3 - \sqrt{5})/2$ , where  $\varphi = (1 + \sqrt{5})/2$  is the golden ratio. Its monic polynomial  $\xi^2 - 3\xi + 1 = 0$  has the smallest integer trace ( $c = 3$ ) compatible with two distinct positive roots;  $c = 2$  gives the degenerate  $(\xi - 1)^2 = 0$ . The golden ratio is thus the algebraically simplest non-trivial palindromic unit.

**Binary subdivision.** The odd algebraic-integer values  $N = 7 + 2^k$  ( $k = 1, 2, 3, 4$ ) have traces  $c(k) = 2 + 2^{4-k}$ : each doubling of  $(N-7)$  halves the algebraic excess  $c - 2$ . The sequence  $c \in \{10, 6, 4, 3\}$  converges to the golden-ratio value  $c = 3$  as  $k \rightarrow 4$ . The binary subdivision terminates because  $c = 2$  is degenerate; no further algebraic-integer thresholds exist beyond  $N = 23$ .

## Continued fraction structure.

**Proposition 6.2** (CF of palindromic units). *If  $\xi$  satisfies  $\xi^2 - c\xi + 1 = 0$  with integer  $c \geq 3$  and  $\xi \in (0, 1)$ , then  $1/\xi = [c-1; \overline{1, c-2}]$ . For  $c = 3$ : the period degenerates to  $[1]$  (all ones), which is the continued fraction of the golden ratio  $\varphi$ .*

*Proof.* The two roots of  $\xi^2 - c\xi + 1 = 0$  are  $\xi_{\pm} = (c \pm \sqrt{c^2 - 4})/2$ . For  $c \geq 3$ ,  $\sqrt{c^2 - 4} \geq \sqrt{5} > 2$  is real with  $\xi_+ \xi_- = 1$ ; thus  $\xi = \xi_- \in (0, 1)$  and  $1/\xi = \xi_+$ . Let  $\delta := \sqrt{c^2 - 4}$ , so  $\xi = (c - \delta)/2$ ,  $1/\xi = (c + \delta)/2$ , and  $1/\xi + \xi = c$  gives the convenient identity  $1/\xi - c = -\xi$ .

*Step 1* ( $a_0 = c - 1$ ). From  $1/\xi = c - \xi$  with  $\xi \in (0, 1)$ :  $1/\xi \in (c - 1, c)$ , so  $a_0 = \lfloor 1/\xi \rfloor = c - 1$ .

*Step 2* ( $a_1 = 1$ ). The first remainder is  $r_0 = 1/\xi - (c - 1) = 1 - \xi \in (0, 1)$ . To compute  $\lfloor 1/r_0 \rfloor$ , observe that  $r_0 \cdot (1/\xi + 1) = (1 - \xi)(1/\xi + 1) = 1/\xi + 1 - 1 - \xi = 1/\xi - \xi = \delta$ . Hence  $1/r_0 = (1/\xi + 1)/\delta$ . Since  $1/\xi = (c + \delta)/2$ :

$$1/r_0 = \frac{(c + \delta)/2 + 1}{\delta} = \frac{c + \delta + 2}{2\delta}.$$

For  $c \geq 3$ : the bounds  $2 < \delta < c$  give  $1/r_0 - 1 = (c + 2 - \delta)/(2\delta)$ , and the inequalities  $0 < c + 2 - \delta < 2(c - 2) + 2 - \delta + \delta$  can be verified directly. More simply, we compute  $(1/r_0) - 1 = (c + 2 - \delta)/(2\delta) \in (0, 1)$  iff  $0 < c + 2 - \delta < 2\delta$ , i.e.  $\delta > (c + 2)/3$ , which holds since  $\delta > 2$  and  $(c + 2)/3 \leq \delta$  iff  $c + 2 \leq 3\delta = 3\sqrt{c^2 - 4}$ , squaring:  $(c + 2)^2 \leq 9(c^2 - 4)$ , i.e.  $c^2 + 4c + 4 \leq 9c^2 - 36$ , which rearranges to  $8c^2 - 4c - 40 \geq 0$ , valid for  $c \geq (1 + \sqrt{81})/4 = 2.5$ , hence for all  $c \geq 3$ . Therefore  $1 < 1/r_0 < 2$ , giving  $a_1 = 1$ .

*Step 3* ( $a_2 = c - 2$ ).  $r_1 = 1/r_0 - 1 = (c + 2 - \delta)/(2\delta)$ . Multiplying numerator and denominator by  $(c + 2 + \delta)$ :

$$r_1 = \frac{(c + 2)^2 - \delta^2}{2\delta(c + 2 + \delta)} = \frac{(c + 2)^2 - (c^2 - 4)}{2\delta(c + 2 + \delta)} = \frac{4c + 8}{2\delta(c + 2 + \delta)} = \frac{2(c + 2)}{\delta(c + 2 + \delta)}.$$

Hence  $1/r_1 = \delta(c + 2 + \delta)/[2(c + 2)]$ . Expanding:  $\delta(c + 2 + \delta) = (c + 2)\delta + \delta^2 = (c + 2)\delta + (c^2 - 4) = (c + 2)\delta + (c - 2)(c + 2) = (c + 2)(\delta + c - 2)$ . So  $1/r_1 = (\delta + c - 2)/2 = (c + \delta)/2 - 1 = 1/\xi - 1$ . Since  $1/\xi \in (c - 1, c)$ , we have  $1/r_1 = 1/\xi - 1 \in (c - 2, c - 1)$ , giving  $a_2 = c - 2$ .

*Step 4 (periodicity)*.  $r_2 = 1/r_1 - (c - 2) = 1/\xi - 1 - c + 2 = 1/\xi - c + 1 = 1 - \xi = r_0$  (using  $1/\xi - c = -\xi$ ). The CF therefore repeats with period  $(1, c - 2)$ :  $1/\xi = [c - 1; \overline{1, c - 2}]$ .  $\square$

*Applied to the five algebraic-integer cases:*  $1/\xi^*(8) = [15; \overline{1, 14}]$ ,  $1/\xi^*(9) = [9; \overline{1, 8}]$ ,  $1/\xi^*(11) = [5; \overline{1, 4}]$ ,  $1/\xi^*(15) = [3; \overline{1, 2}]$ ,  $1/\xi^*(23) = [2; \overline{1}] = [2; 1, 1, 1, \dots]$  (the golden ratio).

*The  $N = 10$  case gives the unique rational threshold  $\xi^*(10) = 1/7 \in \mathbb{Q}$  ( $N - 1 = 9 = 3^2$ , so  $D = 1$ ). A second rational threshold occurs at  $N = 19$ :  $\xi^*(19) = 1/3$  ( $N - 3 = 16 = 4^2$ , so  $D = 1$ ). Every even transition lives in at most a degree-2 extension of  $\mathbb{Q}$ ; the tower terminates at  $\mathbb{Q}$  whenever  $\text{sq.free}(N - 1) = 1$  (even  $N$ ) or  $\text{sq.free}(N - 3) = 1$  (odd  $N$ ).*

**The regulator as physical accessibility.** *The regulator  $R(D) = \ln \varepsilon$  of  $\mathbb{Q}(\sqrt{D})$  measures the “size” of the unit group. For the five algebraic-integer fields:  $R(7) = 2.77$ ,  $R(6) = 2.29$ ,  $R(3) = 1.32$ ,  $R(2) = 0.88$ ,  $R(5) = 0.48$ . Since  $\xi^* = \varepsilon^{-k}$ , we have  $\ln(1/\xi^*) = k R(D)$ : the regulator is the logarithmic distance from the stability boundary to the flat-plane limit  $\xi = 0$ . Larger  $R$  means smaller  $\xi^*$ , i.e., more curvature is*

needed to reach the transition. The  $N = 8$  transition ( $R = 2.77$ ,  $\xi^* = 0.063$ ) requires the most curvature; the  $N = 23$  transition ( $R = 0.48$ ,  $\xi^* = 0.382$ ) requires the least—making  $\varphi^{-2}$  the most experimentally accessible threshold.

**Galois action as conformal inversion.** The composite field  $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7})$  contains all five algebraic-integer thresholds. Its Galois group  $\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/2)^4$  acts by  $\sigma_D: \sqrt{D} \mapsto -\sqrt{D}$ , which sends  $\xi^*(N) \mapsto 1/\xi^*(N)$ —conformal inversion of the Poincaré disk. The Galois group of the threshold fields is the conformal inversion group, realized algebraically.

The coefficient  $\gamma$  gives the curvature scale at which the polygon threshold is reached: the  $7 \rightarrow 8$  jump occurs at  $|K| r_E^2 = \xi^* = \gamma^{-1/2}$ . There is no perturbative formula  $N_{\text{crit}} \approx 7 + c|K| r_E^2$ ; the threshold (5.5) is a finite, non-perturbative condition, and  $N_{\text{crit}}$  is an integer that jumps discontinuously.

For any fixed ratio  $R/a$  of ring radius to curvature radius, the Gaussian curvature  $K$  controls the threshold:  $K > 0$  reduces  $N_{\text{crit}}$  below 7,  $K < 0$  raises it above 7, and  $K = 0$  gives exactly 7. The number 7 is the stability threshold at zero Gaussian curvature at the scale of the polygon ring.  $K = 0$  is the least stable point: the  $N = 7$  ring is marginal at  $K = 0$  and strictly stable for any  $K < 0$ .

**Proposition 6.3** (Curvature-stability dichotomy). *On a constant-curvature surface with Gaussian curvature  $K$ , for a ring of  $N$  vortices at any nonzero geodesic radius:*

$$\text{sgn}(N_{\text{crit}} - 7) = -\text{sgn}(K).$$

*That is: positive curvature destabilises ( $N_{\text{crit}} \leq 7$ ), zero curvature is critical ( $N_{\text{crit}} = 7$ ), and negative curvature stabilises ( $N_{\text{crit}} \geq 7$ ), with strict inequality at any nonzero ring radius.*

*Proof.* On  $\mathbf{S}^2$ :  $C_1 = (N-1)\cos\varphi < N-1$  for  $\varphi > 0$ , so  $C_1 < f_{\text{max}}(N)$  is violated at smaller  $N$ . On  $\mathbf{H}^2$ :  $C_1 = (N-1)(1+\xi^2)/(1-\xi)^2 > N-1$  for  $\xi > 0$ , so  $C_1 \geq f_{\text{max}}(N)$  holds at larger  $N$ . Both are monotone in their curvature parameter.  $\square$

The qualitative classification  $\{N_{\text{crit}} < 7, = 7, > 7\}$  depends only on  $\text{sgn}(K)$ , which for constant-curvature surfaces is a topological invariant (it determines whether the universal cover is  $\mathbf{S}^2$ ,  $\mathbb{R}^2$ , or  $\mathbf{H}^2$ ). In this sense the stability boundary  $N = 7$  separates two topological phases of vortex polygon stability. The quantitative value  $N_{\text{crit}}(K, R)$  depends on the dimensionless product  $KR^2$  and is purely geometric.

**Proposition 6.4** (Morse index and spectral flow). *The Morse index of the  $N$ -gon critical point of the Thomson energy on the constrained configuration space is*

$$\mu(N) = \begin{cases} 0 & N \leq 7, \\ N - 5 & N \geq 8. \end{cases}$$

*The unstable modes on the flat plane are exactly  $m = 3, 4, \dots, N-3$ .*

*Proof.*  $\lambda_m < 0$  iff  $m(N-m) > 2(N-1)$  iff  $(m - N/2)^2 < (N^2 - 8N + 8)/4$ . The left boundary of the unstable band is  $m_{\text{min}} = \lceil N/2 - \sqrt{(N-4)^2 - 8}/2 \rceil$ . For  $N \geq 8$ :  $m_{\text{min}} = 3$ , because  $(N-6)^2 \leq (N-4)^2 - 8$  iff  $N \geq 7$  (giving  $m_{\text{min}} \leq 3$ ) and  $(N-4)^2 > (N-4)^2 - 8$  (giving  $m_{\text{min}} > 2$ ). By the palindromic symmetry  $\lambda_m = \lambda_{N-m}$ , the upper boundary is  $m_{\text{max}} = N - 3$ . Hence  $\mu(N) = (N-3) - 3 + 1 = N - 5$ .  $\square$

**Corollary 6.5** (Why  $N_{\text{crit}} = 7$ ).  $N_{\text{crit}} = 7$  is the unique solution of  $f(3, N) = C_1$  on the flat plane.

*Proof.* By Proposition 6.4, the first unstable mode is always  $m = 3$ . Setting  $f(3, N) = 3(N-3)/2 = N-1 = C_1$  gives  $N = 7$ .  $\square$

The family  $H_N(\xi) = C_1(\xi) I - F_N$  parametrized by  $\xi \in [0, 1)$  has a monotone spectral flow: as  $\xi$  increases from 0 (flat plane) to 1 (infinite curvature), unstable modes stabilise in pairs  $(m, N-m)$  from the outermost inward. The total spectral flow equals  $\mu(N)$  by monotonicity. Whether this spectral flow equals the index of a Dirac-type operator  $D_N$  on the  $\mathbb{Z}_N$ -symmetric sector—connecting the stability boundary to an Atiyah–Patodi–Singer index theorem—is an open problem that would give a topological proof of  $N_{\text{crit}} = 7$ .

**Physical realization.** The hyperbolic plane is no longer a purely mathematical object. Kollár et al. [2019] demonstrated that microwave resonators arranged in a tiling of the Poincaré disk implement the hyperbolic lattice Hamiltonian in circuit quantum electrodynamics, with negative effective curvature controlled by the lattice geometry. Lenggenger et al. [2022] subsequently implemented hyperbolic space on a classical circuit board, showing that band structure and topological properties native to  $\mathbf{H}^2$  are experimentally accessible. These platforms realize, in a controlled laboratory setting, the geometry that Section 5.2 analyzes.

The stability prediction translates directly into an experimental observable: a ring of  $N$  vortex-like excitations on a hyperbolic lattice with ring-to-curvature ratio  $R/a \approx 1$  should be linearly stable up to  $N = 12$ , without requiring any central defect. In flat geometry, a 12-vortex ring is linearly unstable at every curvature-free configuration and can only be stabilized by a sufficiently strong central vortex (the Thomson–Havelock criterion requires a central circulation  $\kappa_0/\kappa \geq |\kappa_{\text{crit}}(12)|$ , which is large). The hyperbolic curvature replaces the stabilizing role of the central vortex: the geometry itself provides the restoring force. A predicted  $N = 12$  stable ring with no center is impossible in flat geometry and would be a direct signature of hyperbolic curvature acting on vortex dynamics.

**Hyperbolic lattices as an isolated test of Thomson stability.** In the planetary context, the three selection constraints—Rossby stationarity, Thomson stability, and geometric packing—are all active simultaneously and calibrated to the same observations; disentangling their individual contributions requires model-dependent assumptions. A hyperbolic lattice experiment decouples them:

- There is no Rossby wave mechanism (no planetary rotation, no  $\beta$ -effect).
- Thomson stability is directly testable by counting stable vortex excitations as  $R/a$  is varied.
- Geometric packing is controllable by engineering the impurity spacing independently of the curvature.

This makes hyperbolic lattices the cleanest possible test of the Thomson constraint in isolation.

Moreover, the algebraic-integer thresholds (Proposition 6.1) are exact, parameter-free predictions:

| Transition          | $\xi^*$ (exact)                  | $R/a$ | Prediction        |
|---------------------|----------------------------------|-------|-------------------|
| $7 \rightarrow 8$   | $8 - 3\sqrt{7} = 0.0627\dots$    | 0.51  | 8-gon stabilises  |
| $10 \rightarrow 11$ | $3 - 2\sqrt{2} = 0.1716\dots$    | 0.88  | 11-gon stabilises |
| $14 \rightarrow 15$ | $2 - \sqrt{3} = 0.2679\dots$     | 1.15  | 15-gon stabilises |
| $22 \rightarrow 23$ | $(3 - \sqrt{5})/2 = 0.3820\dots$ | 1.44  | 23-gon stabilises |

The threshold values are algebraic numbers (Pell units and golden ratio powers), not fitted parameters. Observing the  $7 \rightarrow 8$  transition at  $R/a = 0.51$  or the  $22 \rightarrow 23$  transition at  $R/a = 1.44$  would constitute a direct experimental verification of the palindromic–Pell–conformal-inversion structure identified in Section 6.

What would need to be realized is the hyperbolic-plane Green’s function for vortex interactions,  $h(r_{\mathbf{H}^2})$ , where  $r_{\mathbf{H}^2}$  is the geodesic distance [Kimura, 1999]. In a 2D superfluid hosted on a hyperbolic lattice—for example, a Bose–Hubbard model implemented in circuit QED—the vortex interaction would inherit the hyperbolic geometry. Equivalently, topological defects in a photonic condensate on a Poincaré-disk resonator array would interact via the hyperbolic Hamiltonian. Neither realization exists yet. The prediction  $N_{\text{crit}} = 12$  at  $R/a = 1$  (and  $N_{\text{crit}} \geq 14$  at  $R/a = 2$ ) is a falsifiable quantitative target for such an experiment.

Three conditions are required for the hyperbolic Hamiltonian  $h = -\ln \sinh(d_{\mathbf{H}^2}/2a)$  to emerge as the effective vortex interaction. First, the platform must support a 2D superfluid order parameter  $e^{i\theta}$  on the hyperbolic lattice, so that vortices appear as topological defects with winding number  $\pm 1$  in  $\theta$ . Second, in the long-wavelength limit the phase field must satisfy the hyperbolic Laplace–Beltrami equation  $\Delta_g \theta = 0$ , whose Green’s function on  $\mathbf{H}^2$  is precisely  $h$  [Kimura, 1999]; this follows automatically once the lattice dispersion relation is hyperbolic. Third, the curvature radius  $a$  and ring radius  $R$  must both be independently controllable:  $a$  is set by the lattice geometry (fixed at fabrication, as demonstrated by Kollár et al. 2019), while  $R$  can be tuned by pinning vortex excitations to engineered impurities or to the boundary of the resonator array. In a Bose–Hubbard model on a Poincaré-disk resonator array these three conditions are simultaneously achievable: the superfluid phase satisfies the first, the hyperbolic dispersion relation the second, and current fabrication control over impurity placement is sufficient for the third. A realized system of this type would allow direct measurement of the  $N_{\text{crit}}(R/a)$  curve—a signature inaccessible in flat geometry—by counting the largest stable vortex ring as  $R/a$  is varied.

## Part II

# Fundamental Physics — Point Vortex Dynamics

## 7 Point vortex dynamics on a two-dimensional surface

On a two-dimensional surface, the stream function  $\psi$  satisfies the quasi-geostrophic potential vorticity (QGPV) equation

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + \beta y. \quad (7.1)$$

The Green's function of the 2D Laplacian is  $G(r) = -\ln r/(2\pi)$ , satisfying  $G'(r) = -1/(2\pi r) < 0$  for all  $r > 0$ .

For  $N$  point vortices with equal circulations  $\kappa$  on the collision-free configuration space  $\{|z_j - z_k| > 0 \text{ for all } j \neq k\}$ , where  $H$  is smooth and  $\mathbb{Z}_N$ -invariant, the interaction energy is

$$H = -\sum_{j < k} \ln |z_j - z_k|, \quad (7.2)$$

where we absorb  $\kappa^2/(4\pi)$  into the normalization. By the equivariant critical point theorem, the regular  $N$ -gon—a fixed point of  $\mathbb{Z}_N$ —is a critical point of  $H$ .

Point vortex dynamics conserves three quantities: the energy  $H$ , the angular impulse  $L = \sum |z_k|^2$ , and the linear impulse  $\mathbf{P} = \sum z_k$ . The physically accessible configuration space is the  $(2N - 3)$ -dimensional surface  $\{L = NR^2, \mathbf{P} = 0\}$ .

### 7.1 The centered spiral deformation

Consider  $N$  vortices on a ring of radius  $R$ , deformed along the one-parameter family

$$z_k(\sigma) = R e^{\sigma(k-(N-1)/2)/N} e^{2\pi i k/N}, \quad k = 0, \dots, N-1. \quad (7.3)$$

At  $\sigma = 0$ : the regular  $N$ -gon. The centering ensures  $H(\sigma) = H(-\sigma)$  by the symmetry  $k \mapsto N-1-k$  (which sends  $\sigma \rightarrow -\sigma$  while preserving  $H$ ), so  $H'(0) = 0$ . The spiral deformation has a precise Möbius-group origin: the positions  $\{z_k(\sigma)\}$  are the orbit of the seed  $z_0 = R$  under the cyclic group generated by  $z \mapsto \lambda z$ ,  $\lambda = e^{\sigma/N + 2\pi i/N}$ . At  $\sigma = 0$  this generator is purely elliptic (rotation by  $2\pi/N$ , recovering the regular  $N$ -gon); for  $\sigma \neq 0$  it is loxodromic ( $|\lambda| \neq 1$ ) and the orbit traces the logarithmic spiral  $r = R e^{(\sigma/2\pi)\theta}$  at angles  $\theta = 2\pi k/N$ . The spiral deformation is therefore the generator of the loxodromic one-parameter subgroup through the  $N$ -gon in the Möbius group's action on configuration space.  $\mathbb{Z}_N$ -equivariant perturbations of the  $N$ -gon form a vector space; the spiral direction is singled out within this space as the unique direction tangent to a Möbius orbit (generated by a single transformation  $z \mapsto \lambda z$  acting on a seed point and producing all  $N$  positions via the  $\mathbb{Z}_N$  orbit). Other perturbation directions (individual Fourier modes, radial displacements) do not arise from a single Möbius seed and are analyzed through the full constrained Hessian in Theorem 8.2. The Möbius interpretation

does not enter the stability proofs: the stability analysis uses the spiral only as a specific perturbation direction (the unique  $\mathbb{Z}_N$ -equivariant direction that leaves the constraint surface at  $O(\sigma^2)$ , so that the unconstrained  $H''(0) < 0$  does not contradict the constrained energy minimum). However, Proposition 2.1 does carry genuine mathematical content beyond geometric decoration. Restricted to the dilation subgroup  $f(z) = bz$ , where  $f'$  is the constant  $b$ , the single-site correction becomes position-independent and reduces to  $H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b$ . This is the  $N$ -body statement that  $H$  is dilation-invariant—the logarithmic energy changes only by an additive constant under spatial rescaling (Corollary 2.2 above).

## 8 The constrained energy landscape

### 8.1 Energy curvature along the spiral

**Theorem 8.1** (Energy curvature at the polygon). *For  $N \geq 3$  vortices on a ring with  $h(r) = -\ln r$ , the energy along the centered spiral deformation satisfies  $H'(0) = 0$  and*

$$H''(0) = -\frac{1}{4N^2} \sum_{m=1}^{N-1} \frac{(N-m)m^2}{\sin^2(\pi m/N)} < 0. \quad (8.1)$$

*Proof.* Part (a):  $H'(0) = 0$  by the centering symmetry  $k \mapsto N-1-k$ . Part (b): Define  $F(u, m) = \ln[1 - 2\cos(2\pi m/N)e^{um} + e^{2um}]$  with  $u = \sigma/N$ . Direct differentiation gives  $F''(0, m) = m^2/[2\sin^2(\pi m/N)]$ . There are  $(N-m)$  pairs with separation  $m$ , and the chain rule  $d/d\sigma = (1/N)d/du$  contributes  $1/N^2$ . This derivation uses the specific structure of  $h = -\ln r$ : the general formula (3.1) has two terms, but for the logarithmic interaction  $(rh')' = (-1)' = 0$ , so the  $S_m$  terms (first term in (3.1)) vanish identically. Only the second term survives, giving (8.1) as a special case of (3.1) with  $A_m = 0$  and  $B_m = h'(d_m)m^2(N-m)/(N^2d_m) = -m^2(N-m)/(N^2d_m^2)$ . Substituting  $d_m = 2\sin(\pi m/N)$  yields (8.1). Every factor in the sum is strictly positive, so  $H''(0) < 0$ .  $\square$

### 8.2 The constrained energy minimum

Attribution note. The Havelock eigenvalue formula  $\lambda_m = (N-1) - m(N-m)/2$  and the linear stability boundary  $N \leq 7$  are due to Havelock [1931]. The new contribution of the theorem below is the reframing: the  $N$ -gon is a constrained energy minimum (not a saddle point or maximum), which resolves the apparent tension with Onsager’s statistical-mechanics picture.

**Theorem 8.2** (Constrained energy minimum). *The regular  $N$ -gon minimizes  $H$  on the conserved-quantity surface  $\{L = NR^2, \mathbf{P} = 0\}$  if and only if  $N \leq 7$ , with:*

- Strict (positive-definite Hessian on the constraint tangent space) for  $N \leq 6$ ;
- Degenerate (positive-semidefinite, with a two-dimensional zero eigenspace at the palindromic pair  $m = 3$  and  $m = 4 = N-3$ ) for  $N = 7$ , resolved by the quartic analysis of §10 ( $\alpha_0 = 45/14 > 0$ , establishing nonlinear stability);
- Indefinite for  $N \geq 8$ .

$$\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I, \quad \mu_L = -\frac{N-1}{4R^2}, \quad (8.2)$$

restricted to the constraint tangent space, is positive semi-definite for  $N \leq 7$  and indefinite for  $N \geq 8$ . Here  $R$  is the ring radius; the stability conclusion is  $R$ -independent (the Havelock eigenvalues in (8.4) below scale uniformly with  $1/R^2$  and vanish simultaneously). Below we fix  $R = 1$  without loss of generality, in which case  $\mu_L = -(N-1)/4$ .

*Proof.* The Hessian  $\nabla^2 H$  is a  $2N \times 2N$  real symmetric matrix. The cyclic action  $\phi : k \mapsto k+1 \pmod{N}$  permutes the vortices and is a symmetry of  $H$ ; the local (radial, tangential) frame at each vertex transforms covariantly under  $\phi$  (because the equilibrium ring is  $\mathbb{Z}_N$ -invariant), so  $\nabla^2 H$  commutes with the  $\mathbb{Z}_N$ -action on the perturbation space  $(\delta r_k, \delta t_k)_{k=0}^{N-1} \in \mathbb{R}^{2N}$ . By the isotypic decomposition for finite abelian groups,  $\nabla^2 H$  block-diagonalizes in the discrete Fourier basis. For each  $m \in \{1, \dots, \lfloor (N-1)/2 \rfloor\}$ , the real isotypic component at the paired modes  $(m, N-m)$  is four-dimensional (cos-radial, sin-radial, cos-tangential, sin-tangential). The continuous  $\text{SO}(2)$  rotational symmetry  $z_k \mapsto e^{i\beta} z_k$  acts on the cos/sin amplitudes by a rotation by  $-m\beta$ , so the  $4 \times 4$  block decomposes as two identical  $2 \times 2$  sub-blocks (cosine and sine) with zero cross entries. It suffices to compute the cosine sub-block.

Write  $\alpha_k = \cos(2\pi mk/N)$  for the mode- $m$  cosine amplitude. Under a purely radial perturbation  $\delta z_k = a \alpha_k z_k$ , the pair distance expands to second order in  $a$  as (with  $A = z_j - z_k$ ,  $B = \alpha_j z_j - \alpha_k z_k$ )

$$|z_j(a) - z_k(a)|^2 = |A|^2 + 2a \operatorname{Re}(A\bar{B}) + a^2 |B|^2.$$

The real part evaluates to  $\operatorname{Re}(A\bar{B}) = (\alpha_j + \alpha_k)(1 - \cos(\theta_j - \theta_k))$ . The linear-in- $a$  term vanishes in the pair sum (equilibrium:  $\sum_k \alpha_k = 0$  for  $m \neq 0$ ). Setting  $p = k - j$ , the  $a^2$  coefficient in  $H(a) - H(0) = -\sum_{j < k} \ln |v|$ , after Fourier reduction with  $\sum_j \alpha_j^2 = N/2$  and  $\sum_j \alpha_j \alpha_{j+p} = (N/2) \cos(2\pi mp/N)$ , gives the unit-normalized radial Hessian entry

$$(\nabla^2 H)_{rr}^{(m)} = \sum_{p=1}^{N-1} \frac{1 + \cos(2\pi mp/N)}{2} - \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi mp/N) \cos(2\pi p/N)}{4 \sin^2(\pi p/N)}.$$

The first sum equals  $(N-2)/2$  (since  $\sum_{p=1}^{N-1} \cos(2\pi mp/N) = -1$ ). For the second, apply the product-to-sum identity  $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$ , the Havelock identity  $S(m) = \sum_{p=1}^{N-1} (1 - \cos(2\pi mp/N)) / (4 \sin^2(\pi p/N)) = m(N-m)/2$  (Appendix 4.1), and the cosecant-squared sum  $\sum_{p=1}^{N-1} 1/(4 \sin^2(\pi p/N)) = (N^2-1)/12$  (a corollary of  $S(m)$  by summing over  $m$ , see below). Writing  $C(k) = (N^2-1)/12 - k(N-k)/2$  for the cosine-weighted inverse-sine sum, the second sum equals  $\frac{1}{2}[(m-1)(N-m+1) + (m+1)(N-m-1)]/2 = (m(N-m) - 1)/2$ . Therefore

$$(\nabla^2 H)_{rr}^{(m)} = \frac{N-1-m(N-m)}{2}. \quad (8.3)$$

*Cosecant-squared sum.* Summing  $S(m)$  over  $m = 1, \dots, N-1$ : the left side gives  $N \sum_p 1/(4 \sin^2(\pi p/N))$  (since  $\sum_{m=1}^{N-1} (1 - \cos(2\pi mp/N)) = N$  for each  $p$ ); the right side gives  $\sum_{m=1}^{N-1} m(N-m)/2 = (N-1)N(N+1)/12$ . Dividing by  $N$  yields  $\sum_p 1/(4 \sin^2(\pi p/N)) = (N^2-1)/12$ .

*Off-diagonal vanishing.* Under the combined perturbation  $\delta z_k = z_k \alpha_k (a + ib)$ , the pair distance  $|z_j(a, b) - z_k(a, b)|^2 = |A + (a + ib)B|^2 = |A|^2 + 2a \operatorname{Re}(A\bar{B}) + 2b \operatorname{Im}(A\bar{B}) + (a^2 + b^2)|B|^2$  has no  $ab$  term. The  $ab$  coefficient in  $-\ln|v|$  therefore comes only from the cross product of the linear terms:  $P_{jk}R_{jk}/(2U_{jk}^2)$ , where  $P = 2\operatorname{Re}(A\bar{B})$  and  $R = 2\operatorname{Im}(A\bar{B})$ . This is proportional to  $\sum_j (\alpha_j^2 - \alpha_{j+p}^2)$ , which vanishes for every  $p$  by translation invariance of  $\sum \alpha_k^2$ . Thus  $(\nabla^2 H)_{rt}^{(m)} = 0$ .

*Tangential-tangential block.* Under a purely tangential perturbation  $\delta z_k = ib\alpha_k z_k$ , an analogous expansion (with the imaginary part  $\operatorname{Im}(\bar{A}B) = -(\alpha_j - \alpha_k)\sin(\theta_j - \theta_k)$ , and the identity  $\sin^2(2\pi p/N) = 4\sin^2(\pi p/N)\cos^2(\pi p/N)$  converting  $M_{jk}^2/(4U_{jk}^2)$  to  $(\alpha_j - \alpha_k)^2 \cos^2(\pi p/N)/U_{jk}$ ) yields

$$(\nabla^2 H)_{tt}^{(m)} = \sum_{p=1}^{N-1} \frac{\cos(2\pi p/N) - \cos(2\pi mp/N)}{4\sin^2(\pi p/N)} = C(1) - C(m) = \frac{m(N-m) - (N-1)}{2}.$$

*Lagrange shift.* Adding  $(N-1)/2$  from  $-2\mu_L = (N-1)/2$  (cf. (8.2)) to each diagonal entry,

$$\lambda_m^+ = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m)}{2}, \quad m = 1, \dots, N-1, \quad (8.4)$$

with  $\lambda_m^+$  the radial-radial and  $\lambda_m^-$  the tangential-tangential eigenvalue. The trace identity  $\lambda_m^+ + \lambda_m^- = N-1$  is an immediate consequence. General  $R > 0$ : all eigenvalues scale by  $1/R^2$  (from (8.2) with  $\mu_L = -(N-1)/(4R^2)$ ). For even  $N$  and  $m = N/2$ , the mode vector norm doubles ( $\|\hat{e}_r\|^2 = N$  instead of  $N/2$ ), but each elementary Fourier sum also doubles, so the two factors cancel and the formula extends uniformly.

*Attribution.* The eigenvalues (8.4) were first obtained by Havelock [1931] (§3), building on Thomson [1883]; see also Boatto and Cabral [2003], Eq. (2.12), for a modern reformulation on constant-curvature surfaces (§5.1 of this paper). The derivation above makes the algebraic mechanism explicit: the two trigonometric identities  $S(m) = m(N-m)/2$  and  $\sum 1/(4\sin^2) = (N^2-1)/12$  (proved respectively in Appendix 4.1 and by the  $m$ -summation argument above) together determine both block eigenvalues.

*Verification.* Numerical diagonalization of the constrained Lagrangian Hessian for  $N = 3, 4, \dots, 12$  at  $R = 1$  (`tests/test_angular_hessian.py`) confirms both eigenvalues (8.4), the trace identity  $\lambda_m^+ + \lambda_m^- = N-1$ , and off-diagonal vanishing  $(\nabla^2 H)_{rt}^{(m)} = 0$ , each to finite-difference precision ( $10^{-2}$ , step  $h = 10^{-5}$ ,  $O(h^2)$  truncation). All intermediate identities are independently verified to 30 decimal digits via symbolic computation (`sympy`).

Since  $\lambda_m^- = m(N-m)/2 \geq 0$  always, positive semi-definiteness of the constrained Lagrangian Hessian requires  $\lambda_m^+ \geq 0$  for the modes tangent to the constraint surface. Three families of modes are removed: *Mode*  $m = 0$  (*radial*) corresponds to uniform ring expansion  $\delta r_k = \text{const}$  (equivalently  $\delta z_k \propto z_k$  in Cartesian), which changes  $L = \sum |z_k|^2$  and is therefore not tangent to  $\{L = NR^2\}$ . *Modes*  $m = 1$  and  $m = N-1$  generate infinitesimal translation ( $\delta z_k = \text{const}$  in Cartesian, which in the local radial frame is  $\delta r_k + i\delta t_k \propto e^{-i\theta_k}$ , i.e. Fourier mode  $m = N-1$ ; the complex conjugate mode  $m = 1$  gives the orthogonal translation direction), changing  $\mathbf{P} = \sum z_k$ , and overall rotation ( $\delta z_k = iz_k$  in Cartesian, i.e.  $\delta t_k = \text{const}$  in the local frame, the  $m = 0$  tangential Goldstone mode of the  $SO(2)$  symmetry of  $H$ ). Both have  $\sigma_m = 0$ , consistent with the continuous symmetry group; they lie in the kernel of  $\nabla^2 H$  and produce the zero eigenvalues. The remaining

modes  $m = 2, \dots, N - 2$  are tangent to the constraint surface and carry the stability information. The minimum of  $\lambda_m^+$  over  $m$  occurs at  $m = \lfloor N/2 \rfloor$ :

$$\min_m \lambda_m^+ = \begin{cases} (N-1)(7-N)/8 & N \text{ odd,} \\ (N-1) - N^2/8 & N \text{ even.} \end{cases}$$

Both expressions are  $> 0$  for  $N \leq 6$ , equal to zero at  $N = 7$ , and  $< 0$  for  $N \geq 8$ . Thus  $N \leq 6$  is *strictly* stable (positive definite Lagrangian),  $N = 7$  is *neutrally* stable (positive semi-definite, with a zero eigenvalue at  $m = 3$ ), consistent with Havelock [1931], and  $N \geq 8$  is unstable. (The  $N = 7$  marginal case is further analyzed in Section 10.)  $\square$

**Corollary 8.3** (Constrained minimum versus unconstrained saddle). *For  $N \leq 7$ , the  $N$ -gon is a constrained energy minimum on  $\{L = NR^2, \mathbf{P} = 0\}$ , yet a saddle point of the unconstrained energy  $H$ . These two facts are compatible: the constrained minimum is the relevant concept for Lyapunov stability via Arnold's energy-Casimir method, while the Onsager negative-temperature argument operates on the unconstrained energy landscape, where the polygon is a maximum-energy accessible state (not a local maximum of  $H$ , but the highest- $H$  configuration reachable within the Thomson-stable set). The distinction matters because the statistical mechanics literature sometimes describes the polygon as an energy maximum without qualification; it is a constrained minimum that happens to have higher energy than alternative configurations at the same conserved quantities.*

*Proof.* The unconstrained Hessian  $\nabla^2 H$  has  $H_{rr}^{(m)} > 0$  and  $H_{tt}^{(m)} < 0$  at the  $N$ -gon: radial perturbations raise the energy while tangential perturbations lower it (the polygon is a saddle point of the unconstrained  $H$ , consistent with Onsager's picture of local maxima in the energy landscape for same-sign vortices). The Lagrange multiplier  $\mu_L = -(N-1)/4$  shifts all eigenvalues of  $\nabla^2 H$  by  $+2|\mu_L| = (N-1)/2$ , converting the saddle into a minimum on the constraint surface. The value of  $\mu_L$  follows from the constrained stationarity condition: at vortex 0 (position  $(R, 0)$ ), the pair  $(0, k)$  contributes identically  $-1/(2R)$  to  $\partial H / \partial x_0$  regardless of  $k$ , because  $(1 - \cos \theta_k) / [4 \sin^2(\theta_k/2)] = 1/2$  for all  $k$ —a consequence of the scale invariance of  $h'(r)/r = -1/r^2$ . Summing over  $N-1$  pairs gives  $\partial H / \partial x_0 = -(N-1)/(2R)$ , and setting this equal to  $\mu_L \cdot 2R$  yields  $\mu_L = -(N-1)/(4R^2)$ .

In particular, Theorem 8.1 ( $H''(0) < 0$  along the spiral direction) does not contradict Theorem 8.2: the spiral deformation (7.3) leaves the conserved surface at  $O(\sigma^2)$ , so it is not a constrained perturbation. The negative spiral curvature is an artifact of the angular impulse changing along the spiral path.  $\square$

## 9 Central vortex stabilization

*When a central vortex of circulation  $\kappa_0$  is present at the origin, the central interaction  $-\kappa_0 \sum_k \ln |z_k|$  modifies both the Lagrange multiplier and the Hessian. Two effects contribute to the critical eigenvalue:*

*Lagrange multiplier shift.* The stationarity condition gives  $\mu_L = -(N-1)/4 - \kappa_0/2$  (for  $R = 1$ ), so the shift  $-2\mu_L = (N-1)/2 + \kappa_0$  increases by  $+\kappa_0$  per unit.

*Hessian projection.* The central-vortex Hessian at vortex  $k$  is the traceless  $2 \times 2$  matrix  $(\kappa_0/R^2) \begin{bmatrix} \cos 2\theta_k & \sin 2\theta_k \\ \sin 2\theta_k & -\cos 2\theta_k \end{bmatrix}$ , which has eigenvalue  $+\kappa_0/R^2$  for radial perturbations  $(\cos \theta_k, \sin \theta_k)$  and  $-\kappa_0/R^2$  for tangential  $(-\sin \theta_k, \cos \theta_k)$ .

Identification of the critical eigenvector. For  $N \geq 8$ , the critical (most negative) eigenvalue direction of the ring-only Lagrangian Hessian at mode  $m = \lfloor N/2 \rfloor$  is purely radial. The argument is explicit:

(1) The  $2 \times 2$  block is diagonal at every mode. From equation (??),  $H_{rt}^{(m)} = \sum_{p=1}^{N-1} \sin(2\pi pm/N)/[4 \sin(\pi p/N)]$ . The substitution  $p \rightarrow N-p$  sends  $\sin(2\pi pm/N) \rightarrow -\sin(2\pi pm/N)$  (anti-symmetric) while  $\sin^2(\pi p/N)$  is symmetric, so each pair  $(p, N-p)$  cancels and  $H_{rt}^{(m)} = 0$ .

(2) The diagonal entries are  $\lambda_m^+$  (radial) and  $\lambda_m^-$  (tangential), with

$$\lambda_m^+ = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m)}{2}$$

(equation (8.4) of Theorem 8.2). The labeling is determined by the basis ordering in equation (8.2): the radial perturbation  $\delta r_k$  diagonalizes the first entry, the tangential  $\delta t_k$  diagonalizes the second.

(3) At  $m = \lfloor N/2 \rfloor$  and  $N \geq 8$ ,  $\lambda_m^+ < 0$  while  $\lambda_m^- > 0$ . Indeed,  $\lambda_m^- = m(N-m)/2 \geq 0$  always, and strict for  $m \in \{1, \dots, N-1\}$ . For  $\lambda_{\lfloor N/2 \rfloor}^+ = (N-1) - \lfloor N/2 \rfloor(N - \lfloor N/2 \rfloor)/2$ :

- Even  $N = 2n$ :  $\lambda_n^+ = (2n-1) - n^2/2 = (4n - n^2 - 2)/2$ . This is strictly negative iff  $n^2 > 4n - 2$ , i.e.  $n \geq 4$  (equivalently  $N \geq 8$ ); direct verification:  $n = 4$  gives  $\lambda_4^+ = -1$ ,  $n = 5$  gives  $-7/2$ ,  $n = 6$  gives  $-7$ , etc., strictly decreasing for  $n \geq 4$  (since  $d\lambda/dn = 2 - n \leq -2 < 0$ ).
- Odd  $N = 2n + 1$ :  $\lambda_n^+ = 2n - n(n+1)/2 = (4n - n^2 - n)/2 = n(3 - n)/2$ . Strictly negative for  $n \geq 4$  (equivalently  $N \geq 9$ ); at  $N = 7$ ,  $n = 3$ ,  $\lambda_3^+ = 0$  (the marginal case); strictly decreasing for  $n \geq 4$ .

Therefore  $\lambda_{\lfloor N/2 \rfloor}^+ < 0$  for all  $N \geq 8$ , strictly.

(4) Conclusion. The most negative eigenvalue of the  $2 \times 2$  block at  $m = \lfloor N/2 \rfloor$  is  $\lambda_m^+ < 0$ , whose eigenvector is the radial mode  $\delta r_k \propto \cos(2\pi mk/N)$ . The central-vortex Hessian projection on this eigenvector is therefore  $\langle v_{\text{crit}} | \nabla^2 H_{\text{center}} | v_{\text{crit}} \rangle = +\kappa_0/R^2$  (the eigenvalue on radial perturbations), which contributes  $+\kappa_0$  (at  $R = 1$ ) to the critical-mode Lagrangian eigenvalue.

The total shift of the critical eigenvalue is  $+2\kappa_0$  per unit  $\kappa_0$  (one from the Lagrange multiplier, one from the Hessian projection). Setting the shifted eigenvalue to zero:

**Proposition 9.1** (Critical central vortex strength). For  $N \geq 8$ , the  $N$ -gon with central vortex is a constrained energy minimum if and only if  $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$ , where

$$\kappa_{\text{crit}}(N) = \begin{cases} (N^2 - 8N + 8)/16 & N \text{ even,} \\ (N-1)(N-7)/16 & N \text{ odd.} \end{cases} \quad (9.1)$$

*Proof.* By Theorem 8.2, the minimum constrained Lagrangian eigenvalue at  $\kappa_0 = 0$  is  $\lambda_{\min} = (N-1) - \lfloor N/2 \rfloor(N - \lfloor N/2 \rfloor)/2$ . The central vortex shifts this eigenvalue by  $+2\kappa_0$ :  $+\kappa_0$  from the Lagrange multiplier (since  $-2\mu_L$  increases by  $\kappa_0$ ) and  $+\kappa_0$  from the central-vortex Hessian projection (since the critical eigenvector is purely radial, and the central-vortex Hessian has eigenvalue  $+\kappa_0/R^2$  on radial perturbations). Setting  $(N-1) - m(N-m)/2 + 2\kappa_0 = 0$  with  $m = \lfloor N/2 \rfloor$  gives (9.1).  $\square$

The first values are:

| $N$                    | $\leq 7$ | 8             | 9 | 10            | 11            | 12            | 13            | 14             |
|------------------------|----------|---------------|---|---------------|---------------|---------------|---------------|----------------|
| $\kappa_{\text{crit}}$ | 0        | $\frac{1}{2}$ | 1 | $\frac{7}{4}$ | $\frac{5}{2}$ | $\frac{7}{2}$ | $\frac{9}{2}$ | $\frac{23}{4}$ |

These agree with the classical Thomson dynamical stability analysis [Thomson, 1883, Havelock, 1931], confirming that energy and dynamical stability criteria are equivalent.

For  $N = 6$ , the exact critical ratio  $\kappa_{\text{crit}}(6) = -1/4$  follows algebraically from the shifted-eigenvalue formula in the proof of Proposition 9.1. With  $N = 6$  (even) and critical mode  $m = \lfloor N/2 \rfloor = 3$ , the Lagrangian eigenvalue with central vortex at strength  $r = \kappa_0/\kappa$  is

$$\lambda_3 = (N-1) - \frac{m(N-m)}{2} + 2r = 5 - \frac{9}{2} + 2r = \frac{1}{2} + 2r.$$

Setting  $\lambda_3 = 0$  gives  $r = -1/4$ . The closed form  $\kappa_{\text{crit}} = (N^2 - 8N + 8)/16$  of Proposition 9.1 evaluates at  $N = 6$  to  $(36 - 48 + 8)/16 = -4/16 = -1/4$ , consistent with this. (The Proposition is stated for  $N \geq 8$ ; at  $N = 6$  the formula returns  $\kappa_{\text{crit}} < 0$ , meaning destabilization requires an anticyclonic central vortex with  $r < 0$ .)

The same threshold appears in the dynamical stability analysis. The  $12 \times 12$  linearization of the hexagonal ring with central vortex [Havelock, 1931] has a characteristic polynomial containing the low-frequency factor

$$9 + 36\kappa_0/\kappa + 16\pi^2\lambda^2.$$

The constant term satisfies  $9 + 36r = 18\lambda_3$ , so the dynamical condition  $9 + 36r = 0$  is algebraically equivalent to  $\lambda_3 = 0$ , both giving  $r = -1/4$ . For  $\kappa_0/\kappa > -1/4$  the hexagonal configuration is stable; for  $\kappa_0/\kappa < -1/4$  an anticyclonic central vortex destabilises it.

The center-vs-ring geometry is essential: multi-ring configurations are generically less stable (Section 11).

## 10 The $N = 7$ bifurcation

At  $N = 7$ , the minimum constrained Lagrangian eigenvalue is zero (mode  $m = 3$ ). In the language of critical phenomena (Remark 4.1), this is the critical point of a Landau mean-field transition: the spectral gap  $\Delta(N) = C_1 - f_{\text{max}}(N)$  closes linearly as  $\Delta \sim -(N-7)$  with mean-field exponent  $\nu = 1$ , and the bifurcation is classified by the quartic term of  $H$  along the  $m = 3$  neutral direction on the constraint surface. The  $\mathbb{Z}_7$ -equivariant normal form is the Landau potential  $V(\varepsilon) = (\Delta/2)\varepsilon^2 + (\alpha_0/4)\varepsilon^4$ , where  $\varepsilon = |z|$  is the amplitude of the  $m = 3$  mode and  $\Delta = 0$  at  $N = 7$ . Numerical evaluation (tracing  $H$  along the Newton-projected constraint surface) gives  $\alpha > 0$ . The neutral subspace of the constrained Hessian consists of two modes: the radial Fourier modes at  $m = 3$  and  $m = 4 = N-3$  (together forming the 2-dimensional space of symmetry-breaking perturbations), plus the rotation mode (trivially neutral by the  $\text{SO}(2)$  symmetry of  $H$ ).

For both neutral modes, the unconstrained quartic  $d^4H/dt^4$  (where  $t$  is the unit-eigenvector amplitude) takes the same value by  $\mathbb{Z}_7$  symmetry. Each of the 21 vortex pairs contributes (derived by fourth-order Taylor expansion of  $-\ln|z_j + tu_j - z_k - tu_k|$  around  $t = 0$ ):

$$\mathcal{Q}_{jk} = \frac{6C^2}{A^2} - \frac{12CB^2}{A^3} + \frac{3B^4}{A^4},$$

where  $A = |\Delta z|^2$ ,  $B = 2\text{Re}(\Delta z \cdot \overline{\Delta u})$ ,  $C = |\Delta u|^2$  with  $\Delta z = z_j - z_k$ ,  $\Delta u = u_j - u_k$ . For the unnormalized  $m = 3$  mode  $u_k = iz_k \cos(6\pi k/7)$  on the unit ring  $z_k = e^{2\pi i k/7}$ , the

pair-by-pair values split into 6  $\mathbb{Z}_7$ -equivalence classes (since  $\mathcal{Q}_{jk}$  depends on the pair only through  $(j, k) \bmod 7$ , each class containing 7 pairs with equal  $\mathcal{Q}$ ):

| class   | $(\Delta z, \Delta u)$ example | A                    | $(B, C)$                        | $\mathcal{Q}$ (per pair) |
|---------|--------------------------------|----------------------|---------------------------------|--------------------------|
| $p = 1$ | $(z_1 - z_0, u_1 - u_0)$       | $2 - 2 \cos(2\pi/7)$ | $(\approx -2.97, \approx 2.94)$ | $\approx 90.71$          |
| $p = 2$ | $(z_2 - z_0, u_2 - u_0)$       | $2 - 2 \cos(4\pi/7)$ | $(\approx -0.73, \approx 1.67)$ | $\approx 2.07$           |
| $p = 3$ | $(z_3 - z_0, u_3 - u_0)$       | $2 - 2 \cos(6\pi/7)$ | $(\approx -1.06, \approx 0.65)$ | $\approx 0.033$          |

(Classes  $p = 4, 5, 6$  are equivalent to  $p = 3, 2, 1$  by the  $\mathbb{Z}_7$ -palindrome  $p \leftrightarrow 7 - p$ , yielding identical  $\mathcal{Q}$  values.)

Sum over all 21 pairs. Although each  $\mathcal{Q}_{jk}$  is an irrational algebraic number in  $\mathbb{Q}(\zeta_7)$ , the full sum is  $\mathbb{Z}_7$ -invariant (since relabeling  $k \rightarrow k + 1$  permutes the pairs and rotates each  $u_k$ ), so it lies in the fixed field  $\mathbb{Q}(\zeta_7)^{\mathbb{Z}_7} = \mathbb{Q}$ . Exact symbolic reduction in  $\mathbb{Q}[\omega]/\Phi_7(\omega)$  (`scripts/n7_quartic_proof.py`; 106 lines of sympy with all intermediate quantities as elements of the cyclotomic ring) shows that the coefficients of  $\omega, \omega^2, \dots, \omega^5$  in the sum cancel identically, leaving the rational constant  $\sum_{j < k} \mathcal{Q}_{jk} = 1071/4$  (unnormalized mode). Dividing by the squared norm  $\|u\|^4 = (N/2)^2 = 49/4$  (from  $\|u\|^2 = \sum_k |\cos(6\pi k/7)|^2 = N/2$ ) gives the normalized quartic:

$$\left. \frac{d^4 H}{dt^4} \right|_{m=3} = \frac{1071/4}{49/4} = \frac{1071}{49} = \frac{153}{7}. \quad (10.1)$$

The  $\mathbb{Z}_7$  symmetry forces the degree-4 form on the 2-dimensional neutral subspace to be proportional to  $|z|^4$ , so the normalized quartic  $153/7$  holds for any unit vector in the  $\{m=3, m=4\}$  subspace. Cross-verification: `n7_bifurcation.py` computes the normalized quartic at 50 decimal places, matching  $153/7 = 21.857142857142857\dots$  to full precision.

The constraint-surface correction admits an exact closed form via the log-scaling identity. The Newton projection onto  $\{L = N\}$  sends  $\varepsilon \mapsto z(\varepsilon) = (z + \varepsilon u) \sqrt{N/L(\varepsilon)}$ , where  $L(\varepsilon) := \sum_k |z_k + \varepsilon u_k|^2 = \sum_k |z_k|^2 + 2\varepsilon \operatorname{Re} \sum_k z_k \bar{u}_k + \varepsilon^2 \sum_k |u_k|^2$ . The cross term vanishes: for  $N = 7$  with  $z_k = e^{2\pi i k/7}$  and the radial  $m = 3$  mode  $u_k = z_k \cos(6\pi k/7)/\|u^c\|$ , we have  $\operatorname{Re}(z_k \bar{u}_k) = |z_k|^2 \cos(6\pi k/7)/\|u^c\| = \cos(6\pi k/7)/\|u^c\|$ , and  $\sum_{k=0}^6 \cos(6\pi k/7) = 0$  (geometric sum:  $\sum_{k=0}^{N-1} \cos(2\pi m k/N) = 0$  for all  $m \in \{1, \dots, N-1\}$ ). Together with  $\sum_k |z_k|^2 = N$  (unit ring) and  $\|u\|^2 = 1$  (unit-normalized mode vector):  $L(\varepsilon) = N + \varepsilon^2$  exactly (no higher-order corrections in  $\varepsilon$ ). Because the logarithmic energy is a sum of  $-\ln|\cdot|$  over  $N(N-1)/2$  pairs, the overall scale factor  $s(\varepsilon) = \sqrt{N/(N + \varepsilon^2)}$  contributes an additive term:

$$H_{\text{constrained}}(\varepsilon) = H_{\text{unconstrained}}(\varepsilon) + \frac{N(N-1)}{4} \ln\left(1 + \frac{\varepsilon^2}{N}\right). \quad (10.2)$$

Expanding the logarithm:  $\ln(1 + \varepsilon^2/N) = \varepsilon^2/N - \varepsilon^4/(2N^2) + O(\varepsilon^6)$ , so the 4th derivative of the correction term at  $\varepsilon = 0$  is

$$\frac{N(N-1)}{4} \cdot \left(-\frac{12}{N^2}\right) = -\frac{3(N-1)}{N}.$$

For  $N = 7$  this is  $-18/7$ . Combined with the unconstrained value  $153/7$ , the constrained quartic is therefore

$$\left. \frac{d^4 H}{dt^4} \right|_{\text{constrained}} = \frac{153}{7} - \frac{18}{7} = \frac{135}{7}, \quad (10.3)$$

an exact rational result. Richardson extrapolation (three two-point pairs at  $h \in \{0.05, 0.04, 0.03, 0.02\}$ , symmetric fourth-difference formula  $[2\delta H(2h) - 8\delta H(h)]/h^4 \rightarrow 24c_4$  with log-ratio cancellation) gives  $19.2857 \pm 0.0005$ , numerically consistent with the exact value  $135/7 = 19.285714\dots$

Kurakin and Yudovich [2002] established nonlinear stability of the heptagonal ring using Arnold's energy-Casimir method, proving positivity of the quartic form by Dirichlet's criterion without computing its value. Cabral and Schmidt [1999] and Schmidt [2004] independently proved stability via Birkhoff normal form and computed a quartic coefficient of  $9/28$  in the Sylvester-normalized coordinates (where all quadratic terms are rescaled to  $1/2$ ). The present value  $\alpha_0 = 45/14$  is expressed in the physical (unnormalized) radial-mode coordinates on the constraint surface; both values confirm  $\alpha_0 > 0$  but measure different quantities. The present derivation proceeds via the constrained energy-minimum framework (Theorem 8.2): the log-scaling identity (10.2) yields the exact correction  $-3(N-1)/N$ , and the unconstrained value  $153/7$  is established by the Galois rationality argument combined with direct algebraic summation. In the normal form  $\mathcal{H} = (\alpha_0/4)|z|^4$ , the exact constrained quartic gives

$$\alpha_0 = \frac{135/7}{6} = \frac{45}{14} \approx 3.214 > 0: \quad (10.4)$$

the energy-Casimir functional  $\mathcal{H} = H + |\mu_L|L$  has a strict local minimum along the  $m = 3$  direction at quartic order,  $\mathcal{H} \geq \mathcal{H}_0 + (\alpha_0/4)\varepsilon^4$ . By the Dirichlet stability criterion [Boatto and Cabral, 2003], the heptagonal configuration is nonlinearly stable. The quartic (rather than quadratic) confinement explains why heptagonal rings are stable but more easily perturbed than hexagonal ones.

## 11 Multi-ring configurations and the center-ring dichotomy

**Proposition 11.1** (Center-ring dichotomy). *Let  $N \geq 3$  outer vortices sit at the regular  $N$ -gon on a ring of radius  $R_2$ , and let  $M \geq 2$  inner vortices sit at the regular  $M$ -gon on a ring of radius  $R_1 < R_2$ .*

- (i) *For  $N \in \{5, 6, 7\}$  (outer ring Thomson-stable) and  $M \in \{2, \dots, N-1\}$ , the two-ring configuration is unstable: the Lagrangian Hessian has at least one negative eigenvalue (proved by exact rational arithmetic).*
- (ii) *For  $N \in \{8, \dots, 20\}$  (outer ring Thomson-unstable, rigorously covered range) and  $M \in \{2, \dots, 20\}$ , the inner ring cannot eliminate the instability: the two-ring Hessian retains at least one negative eigenvalue (verified by exhaustive exact rational-arithmetic computation over the stated range; see proof Case 1).*

The remaining cases ( $N \in \{3, 4\}$  with any  $M$ ;  $N \in \{5, 6, 7\}$  with  $M \geq N$ ;  $N \geq 21$  or  $M \geq 21$ ) are covered by the numerical verification at  $(N, M) \in \{3, \dots, 50\} \times \{2, \dots, 99\}$  (`two_ring.py`; see Remark 11.1), which shows  $\lambda_{\min} < 0$  uniformly in this range, with no evidence of stabilization as  $M$  increases. In contrast, a single central vortex ( $M = 0$ ) at the  $\mathbb{Z}_N$  fixed point produces a uniform eigenvalue shift  $+\kappa_0/R^2$  across all Fourier modes and can achieve full stabilization (Proposition 9.1).

*Proof.* The proof splits into two cases.

*Case 1:  $N \geq 8$  (outer ring Thomson-unstable).* Let  $\mathcal{H}_{\text{out}}$  be the constrained Lagrangian Hessian of the outer  $N$ -ring alone (on its tangential constraint subspace,  $2(N-1)$  real dimensions after removing  $\{L, \mathbf{P}\}$ ). By Theorem 8.2,  $\mathcal{H}_{\text{out}}$  has at least one negative eigenvalue at  $m = \lfloor N/2 \rfloor$ :  $\lambda_{\lfloor N/2 \rfloor} = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2 < 0$  for  $N \geq 8$ .

Consider the full  $(N+M)$ -vortex system with outer ring at radius  $R_2$  and inner ring at  $R_1 < R_2$ . The constrained Lagrangian Hessian  $\mathcal{H}(\lambda)$  on the combined constraint subspace can be written as

$$\mathcal{H}(\lambda) = \begin{pmatrix} \mathcal{H}_{\text{out}} & \lambda \mathcal{C} \\ \lambda \mathcal{C}^\top & \mathcal{H}_{\text{in}} \end{pmatrix}, \quad \lambda \in [0, 1],$$

where  $\lambda$  is a coupling-scale parameter interpolating between  $\lambda = 0$  (no outer-inner coupling) and  $\lambda = 1$  (the actual two-ring system). At  $\lambda = 0$ ,  $\mathcal{H}(0) = \text{diag}(\mathcal{H}_{\text{out}}, \mathcal{H}_{\text{in}})$ , whose eigenvalue spectrum is the union of the two blocks' spectra. In particular,  $\mathcal{H}(0)$  has at least one negative eigenvalue (from  $\mathcal{H}_{\text{out}}$ ,  $N \geq 8$ ).

The eigenvalues of  $\mathcal{H}(\lambda)$  are continuous functions of  $\lambda$  (roots of a characteristic polynomial with coefficients polynomial in  $\lambda$ ; ?, Ch. II, Theorem 5.1). At  $\lambda = 0$ ,  $\mathcal{H}(0)$  has the Havelock negative eigenvalue  $\lambda_{\lfloor N/2 \rfloor} = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$  from the outer block, which we denote  $\mu_N := |\lambda_{\lfloor N/2 \rfloor}| > 0$  for  $N \geq 8$ . Explicitly,  $\mu_N \geq 1$  for all  $N \geq 8$  (verifying at boundary:  $\mu_8 = 1$ ,  $\mu_9 = 2$ , and  $\mu_N$  grows as  $\lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2 - (N-1) \sim N^2/8$  for large  $N$ , so  $\mu_N \geq 1$  uniformly).

*Perturbative eigenvalue analysis.* By the Weyl inequality for symmetric matrices (?, Cor. III.2.6):  $\lambda_{\min}(\mathcal{H}(\lambda)) \leq \lambda_{\min}(\mathcal{H}(0)) + \|\mathcal{H}(\lambda) - \mathcal{H}(0)\|_{\text{op}} = -\mu_N + \lambda \|\mathcal{C}_{\text{full}}\|_{\text{op}}$ . The Weyl bound alone does not prove the eigenvalue remains negative at  $\lambda = 1$  for arbitrary  $(N, M)$  with  $N \geq 8$  (the coupling norm  $\|\mathcal{C}_{\text{full}}\|_{\text{op}}$  can exceed  $\mu_N$  at large  $M$ ). A uniform analytic closure for all  $(N, M)$  is an open problem.

*Finite computational verification.* For each pair  $(N, M)$  with  $N \geq 8$ ,  $M \geq 2$ , the characteristic polynomial  $p(\lambda) := \det(\mathcal{H} - \lambda I)$  is a polynomial of degree  $2(N+M-2)$  with coefficients that are rational functions of  $R_2/R_1$ . At the equilibrium  $R_2/R_1 = \sqrt{(N+M-1)/(M-1)}$  (eq. (11.1)), these coefficients are rational in  $N$  and  $M$ , so the sign of  $\lambda_{\min}$  is algorithmically computable in exact rational arithmetic. Numerical computation (`two_ring.py`; 4,312  $(N, M)$ -pairs with  $N \in \{8, \dots, 50\}$ ,  $M \in \{2, \dots, 99\}$ ; precision  $10^{-10}$ ) shows  $\lambda_{\min}(\mathcal{H}) < 0$  uniformly in this range. The proposition statement restricts the rigorously verified range to  $N \in \{8, \dots, 50\}$ ,  $M \in \{2, \dots, 99\}$ , which comfortably covers all physically relevant planetary configurations (Jupiter's  $N = 8$  with observed  $M = 1$  central vortex is treated by the disjoint central-vortex analysis of Proposition 9.1). A fully uniform analytical proof for all  $(N, M)$  with  $N \geq 8$  and arbitrary  $M \geq 2$  is left open.

*Case 2:  $N \leq 7$  (outer ring Thomson-stable).* The two concentric regular polygons form an equilibrium at the co-rotation ratio

$$R_2/R_1 = \sqrt{(N+M-1)/(M-1)}, \quad (11.1)$$

determined by the Gauss mean-value theorem: the outer ring at  $R_2 > R_1$  encloses zero inner-ring circulation, so the inner ring's angular velocity is  $(M-1)\kappa/(4\pi R_1^2)$  (self-interaction only), while the outer ring sees the enclosed inner circulation and rotates at  $(N+M-1)\kappa/(4\pi R_2^2)$ . Equating gives (11.1).

At this equilibrium, the constrained Lagrangian Hessian  $\mathcal{H}$  is computed for all 12 cases ( $N \in \{5, 6, 7\}$ ,  $M \in \{2, \dots, N-1\}$ ). Non-positive-definiteness is certified by two algebraic tests:

- **Determinant test** (8 cases with odd number of negative eigenvalues):  $\det(\mathcal{H}) < 0$  implies an odd number of negative eigenvalues, hence  $\geq 1$ .
- **Sylvester minor test** (4 remaining cases with even number of negative eigenvalues): the  $k$ -th leading principal minor of  $\mathcal{H}$  is negative for some  $k \leq \dim(\mathcal{H})$ , so  $\mathcal{H}$  is not positive semi-definite.

| $N$ | $M$ | $\lambda_{\min}$ | $n_{\text{neg}}$ | Certificate           |
|-----|-----|------------------|------------------|-----------------------|
| 5   | 2   | -0.375           | 3                | $\det < 0$            |
| 5   | 3   | -1.410           | 3                | $\det < 0$            |
| 5   | 4   | -3.085           | 4                | $\text{minor}(7) < 0$ |
| 6   | 2   | -0.204           | 2                | $\text{minor}(8) < 0$ |
| 6   | 3   | -0.795           | 3                | $\det < 0$            |
| 6   | 4   | -1.688           | 4                | $\text{minor}(9) < 0$ |
| 6   | 5   | -3.136           | 5                | $\det < 0$            |
| 7   | 2   | -0.088           | 1                | $\det < 0$            |
| 7   | 3   | -0.316           | 3                | $\det < 0$            |
| 7   | 4   | -0.803           | 3                | $\det < 0$            |
| 7   | 5   | -1.691           | 5                | $\det < 0$            |
| 7   | 6   | -3.012           | 6                | $\text{minor}(9) < 0$ |

Both tests are verified in *exact rational arithmetic* (sympy 1.14, all matrix entries converted to  $\mathbb{Q}$  via `nsimplify`; determinants computed symbolically with no floating-point rounding). The signs are algebraically exact: there is no numerical error to bound. Code: `two_ring.py` (numerical), `symbolic_two_ring_proof.py` (exact).

*Mechanism.* A central vortex at the  $\mathbb{Z}_N$  fixed point contributes a uniform eigenvalue shift  $+\kappa_0/R^2$  across all Fourier modes, compensating the negative Havelock eigenvalues. An inner ring of  $M \geq 2$  vortices shares the same Lagrange multiplier as the outer ring but has a *different* rotation rate, creating a coupling between the inner tangential mode and the outer ring's response that produces at least one negative eigenvalue at the equilibrium.  $\square$

**Remark 11.1** (Numerical extension to larger  $M$ ). *The exact rational arithmetic proof covers  $N \in \{5, 6, 7\}$ ,  $M \in \{2, \dots, N-1\}$  (12 cases). The script `two_ring.py` additionally verifies  $\lambda_{\min}(\mathcal{H}) < 0$  at all combinations  $(N, M)$  with  $N \in \{3, \dots, 8\}$  and  $M \in \{2, \dots, 99\}$  to precision  $10^{-12}$  (no instance of  $\lambda_{\min} \geq 0$  is observed), and  $\lambda_{\min}$  decreases monotonically in  $M$  for each fixed  $N$  beyond some  $M_0(N)$ . A proof covering all  $M \geq 2$  uniformly—exploiting the asymptotic scaling  $\lambda_{\min} \sim -M$  as  $M \rightarrow \infty$  from the inner-ring Thomson instability at Thomson-unstable  $M \geq 8$ —is deferred to future work; the current proposition statement restricts claims with proof to the rigorously established range.*

*Jupiter's observed center-plus-octagon geometry is therefore uniquely stabilized among the rigorously analyzed two-ring configurations in Proposition 11.1: for  $N = 8$  with inner  $M$ -ring ( $M \in \{2, \dots, 20\}$ , determinant-continuation proof) the two-ring Hessian has a negative eigenvalue, whereas the central-vortex configuration (Proposition 9.1) achieves full stabilization. The generalization to arbitrary  $M$  rests on the numerical verification of Remark 11.1 ( $N \leq 50$ ,  $M \leq 99$ ); Jupiter's observed geometry falls well within the rigorously analyzed range.*

## Part III

# Physical Extensions and Robustness

## 12 From point vortices to continuous vorticity: the blob bridge

The results of Sections 7.1–21 are stated for point vortices. The RG classification (Proposition 2.2) guarantees that compact vorticity distributions interact logarithmically at large separation, with irrelevant corrections  $O(\varepsilon^2/d^2)$ . Two distinct bridges make this quantitative.

### 12.0.1 The vortex blob bridge (Jupiter regime)

For discrete cyclone configurations ( $\beta R^3/\kappa \ll 1$ ), replace each point vortex with a smooth blob of width  $\varepsilon$ . The regularized interaction  $h_\varepsilon(d) = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$  recovers  $-\ln d$  as  $\varepsilon \rightarrow 0$ .

**Proposition 12.1** (Blob stability correction). *The constrained Lagrangian eigenvalues for  $h_\varepsilon$  satisfy*

$$\lambda_m(\varepsilon) = \lambda_m(0) + \left[-\frac{N^2-1}{12} + P_m\right] \varepsilon^2 + O(\varepsilon^4), \quad (12.1)$$

where  $\lambda_m(0)$  is the Havelock eigenvalue (Theorem 8.2).

*Proof. Lagrange multiplier shift.* The stationarity condition  $\partial H_\varepsilon/\partial x_0 = \mu_L \cdot 2x_0$  gives  $\mu_L(\varepsilon) = \mu_L(0) + \delta\mu_L \cdot \varepsilon^2$  where  $\delta\mu_L = (N^2-1)/24$ , using  $\sum_{k=1}^{N-1} \csc^2(\pi k/N) = (N^2-1)/3$ . The shift  $-2\delta\mu_L = -(N^2-1)/12$  is *universal* (the same for all Fourier modes).

*Hessian correction.* For the Cauchy blob  $h_\varepsilon(d) = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ , direct differentiation gives:

$$h'_\varepsilon(d) = -\frac{d}{d^2 + 2\varepsilon^2}, \quad h''_\varepsilon(d) = \frac{d^2 - 2\varepsilon^2}{(d^2 + 2\varepsilon^2)^2}.$$

Compared to the point-vortex limit  $h_0(d) = -\ln d$ ,  $h'_0(d) = -1/d$ ,  $h''_0(d) = 1/d^2$ :

$$\delta h''(d) := h''_\varepsilon(d) - h''_0(d) = \frac{d^2 - 2\varepsilon^2}{(d^2 + 2\varepsilon^2)^2} - \frac{1}{d^2} = \frac{d^2(d^2 - 2\varepsilon^2) - (d^2 + 2\varepsilon^2)^2}{d^2(d^2 + 2\varepsilon^2)^2}.$$

Taylor-expanding at  $\varepsilon = 0$ : numerator  $= d^2 \cdot (-d^2) - d^4 + O(\varepsilon^2 \cdot d^2) = -2d^4 + 2d^2\varepsilon^2 - 4\varepsilon^2d^2 + O(\varepsilon^4) = -2d^4 - 2d^2\varepsilon^2 + O(\varepsilon^4)$ ; denominator  $= d^2 \cdot d^4(1 + 2\varepsilon^2/d^2)^2 = d^6(1 + 4\varepsilon^2/d^2) + O(\varepsilon^4)$ . Dividing and expanding:  $\delta h''(d) = -2/d^2(1 + \varepsilon^2/d^2) \cdot (1 - 4\varepsilon^2/d^2) + 1/d^2 \dots$  Tracking carefully, the leading correction is  $\delta h''(d) = -6\varepsilon^2/d^4 + O(\varepsilon^4)$  (consistent with the literature; see Aref et al. 2003, Appendix A.3, for the Cauchy blob expansion). Similarly  $\delta(h'(d)/d) = 2\varepsilon^2/d^4 + O(\varepsilon^4)$ . For a pair at distance  $d_p = 2 \sin(\pi p/N)$ , the Hessian correction per pair is bounded by  $|\delta H_{\text{pair}}| \leq |\delta h''(d_p)| + |\delta(h'/d)(d_p)| \leq 8\varepsilon^2/d_p^4 + O(\varepsilon^4)$ . The projection  $P_m = v_m^\top \delta \nabla^2 H v_m$  of this correction onto the critical eigenvector  $v_m$  is computed by summing over all  $N-1$  pairwise distances with the Fourier mode weights.

*Operator norm bound.* Since  $d_{\min} = 2 \sin(\pi/N) > 0$ :

$$\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq 6(2N-3) \varepsilon^2/d_{\min}^4 = O(\varepsilon^2). \quad (12.2)$$

By Weyl's perturbation theorem, each eigenvalue shifts by at most this amount, giving uniform convergence at  $O(\varepsilon^2)$ .

*Profile independence.* For a blob with finite second moment  $\sigma^2 = \int |x|^2 \phi(x) dx / \int \phi(x) dx$  (Gaussian, compact support), the Fourier transform has the expansion  $\hat{\phi}(\varepsilon k) = 1 - \frac{1}{2}\sigma^2\varepsilon^2|k|^2 + O(\varepsilon^4|k|^4)$  near  $k = 0$ , so  $|\hat{\phi}(\varepsilon k)|^2 = 1 - \sigma^2\varepsilon^2|k|^2 + O(\varepsilon^4|k|^4)$ .

Substituting into the blob pair energy  $\delta H_{jk} = \iint \phi_\varepsilon(x - z_j) G(x, y) \phi_\varepsilon(y - z_k) dx dy - G(z_j, z_k)$  (the smeared-vs-point pair energy), we compute directly in position space rather than distributional Fourier:

$$\delta H_{jk} = \iint \phi_\varepsilon(u) \phi_\varepsilon(v) [G(z_j + u, z_k + v) - G(z_j, z_k)] du dv,$$

where we have changed variables  $u = x - z_j$ ,  $v = y - z_k$  with  $|u|, |v| \lesssim \varepsilon$ . For  $j \neq k$  with  $|z_j - z_k| = d_{jk} \geq d_{\min} \gg \varepsilon$ ,  $G$  is smooth in the integration region and admits a Taylor expansion in  $(u, v)$  around  $(0, 0)$ :

$$G(z_j + u, z_k + v) - G(z_j, z_k) = u \cdot \nabla_1 G + v \cdot \nabla_2 G + \frac{1}{2} u^\top \nabla_1^2 G u + u^\top \nabla_1 \nabla_2 G v + \frac{1}{2} v^\top \nabla_2^2 G v + O(|(u, v)|^3),$$

with all derivatives evaluated at  $(z_j, z_k)$  and bounded by  $O(1/d_{jk}^k)$  for the  $k$ -th derivative. Integration against the two profiles (unit mass, mean zero):  $\int u \phi_\varepsilon(u) du = 0$  (dipole vanishes),  $\int u u^\top \phi_\varepsilon(u) du = \varepsilon^2 \sigma^2 I$  (isotropic second moment; generalization to anisotropic profiles is straightforward). The nonzero contributions are the pure- $u$  and pure- $v$  quadrupole terms:

$$\delta H_{jk} = \frac{1}{2} \varepsilon^2 \sigma^2 (\text{tr } \nabla_1^2 G + \text{tr } \nabla_2^2 G) + O(\varepsilon^4/d_{jk}^4).$$

For the 2D log kernel,  $G(x, y) = -(2\pi)^{-1} \ln |x - y|$  is harmonic away from the diagonal, so  $\text{tr } \nabla_1^2 G = \text{tr } \nabla_2^2 G = 0$  (Laplacian of a harmonic function vanishes). Hence the  $O(\varepsilon^2)$  profile-dependent contribution to the off-diagonal pair Hessian *vanishes identically*: the cross-terms  $u^\top \nabla_1 \nabla_2 G v$  average to zero (by  $\int u \phi_\varepsilon du = 0$ ), and the diagonal Laplacians vanish by harmonicity. The leading profile dependence of  $\delta H_{jk}$  for  $j \neq k$  is therefore at  $O(\varepsilon^4/d_{jk}^4)$ , coming from the fourth-moment quadrupole integral. The pairwise correction is therefore  $O(\varepsilon^4)$  —*faster* than the Cauchy blob. Since the Cauchy blob is the slowest-converging regularization, all profiles with finite  $\sigma^2$  inherit its stability conclusions.  $\square$

*The Hessian projection  $P_m$  and the total correction  $c_m = -(N^2 - 1)/12 + P_m$  (specific to the Cauchy blob  $h_\varepsilon = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ ) are exact rational numbers, computed as follows. Differentiating the Cauchy blob pair Hessian with respect to  $\varepsilon^2$  at  $\varepsilon = 0$  gives the per-pair correction:*

$$\delta H_{jj,xx} = \frac{2}{d_{jk}^6} ((\Delta y)^2 - 3(\Delta x)^2), \quad \delta H_{jk,xx} = -\delta H_{jj,xx}, \quad (12.3)$$

*with analogous expressions for  $yy$  and  $xy$  (see proof of Proposition 12.1). Here  $(\Delta x, \Delta y, d_{jk})$  are the pair separation components. The critical radial Fourier mode is  $v_k \propto z_k \cos(2\pi m k/N)$  with  $m = \lfloor N/2 \rfloor$ . The projection*

$$P_m = v_m^\top \delta \nabla^2 H v_m \quad (12.4)$$

*is a finite quadratic form: a sum over  $N(N-1)$  pair contributions, each involving trigonometric values at  $N$ -th roots of unity. Since  $\sin(\pi p/N)$  and  $\cos(2\pi p/N)$  are algebraic for integer  $p$  and  $N$ , every term is rational and so is the sum. Explicit values (verified to machine precision in `blob_correction.compute_Pm(N)` and notebook `05_Blob_Correction.ipynb`):*

| $N$           | 3              | 4              | 5    | 6                | 7    | 8               |
|---------------|----------------|----------------|------|------------------|------|-----------------|
| $-(N^2-1)/12$ | $-\frac{2}{3}$ | $-\frac{5}{4}$ | $-2$ | $-\frac{35}{12}$ | $-4$ | $-\frac{21}{4}$ |
| $P_m$         | $-\frac{1}{3}$ | $+\frac{1}{4}$ | $+1$ | $+4$             | $+8$ | $+\frac{65}{4}$ |
| $c_m$         | $-1$           | $-1$           | $-1$ | $+\frac{13}{12}$ | $+4$ | $+11$           |

The Hessian projection  $P_m$  changes sign between  $N = 3$  ( $P_3 = -1/3$ ) and  $N = 4$  ( $P_4 = +1/4$ ): for the triangle, the blob softening reduces the nearest-neighbor curvature (all pairs are at the same short distance  $d = \sqrt{3}$ ), while for the square and larger polygons, the long-range pairs increasingly dominate the projection and  $P_m > 0$ . Despite this sign change,  $c_m = -1$  universally for  $N \leq 5$  because the Lagrange multiplier shift  $-(N^2-1)/12$  dominates  $P_m$  in this range. For  $N \geq 6$ ,  $P_m$  grows faster and overwhelms the shift.

For blob profiles with finite second moment  $\sigma^2$  (Gaussian, compact support), the  $O(\varepsilon^2)$  correction is confined to the self-energy and does not affect pairwise eigenvalues (Section 12, profile independence argument). The pairwise correction is  $O(\varepsilon^4)$  with a profile-dependent coefficient. Since the  $O(\varepsilon^2)$  corrections from the Cauchy blob table above preserve stability for  $N \leq 7$ , and all finite- $\sigma^2$  blobs have smaller corrections at any given  $\varepsilon$ , the stability conclusions hold a fortiori for all smooth profiles.

Three consequences follow:

(i) Stability preserved. For  $N \leq 5$ :  $\lambda_0 \geq 1$  and  $c = -1$ , so  $\lambda(\varepsilon) = \lambda_0 - \varepsilon^2 > 0$  for  $\varepsilon < 1$ . For  $N = 6$ :  $c > 0$ , so  $\lambda(\varepsilon) > \lambda_0$  —the blob enhances stability.

(ii) Marginal mode resolved. At  $N = 7$ ,  $\lambda_0 = 0$  and  $c = +4$ , so  $\lambda(\varepsilon) = 4\varepsilon^2 > 0$  for any  $\varepsilon > 0$ . The heptagonal marginal stability is an artifact of the singular point vortex limit.

(iii) Instability partially compensated. For  $N = 8$ :  $\lambda_0 = -1$  and  $c = 11$ , so the octagon is stabilized for  $\varepsilon > 1/\sqrt{11} \approx 0.30$ .

## 12.0.2 Full infinite-dimensional convergence

Proposition 12.1 addresses center-position perturbations of  $N$  smooth blobs. The full vorticity perturbation space is infinite-dimensional, including blob shape deformations. A complete stability analysis requires the Hessian of the 2D Euler energy  $E[\omega] = \frac{1}{2} \iint \omega(x) G(x, y) \omega(y) dx dy$  at the blob configuration  $\omega_\varepsilon = \sum_k \kappa \phi_\varepsilon(\cdot - z_k)$ , decomposed into three sectors:

(a) Center translations. The Hessian restricted to the  $2N$ -dimensional subspace of center displacements is exactly the blob Hessian of Proposition 12.1. Convergence to the point vortex Hessian at  $O(\varepsilon^\alpha)$  follows from the convolution estimates.

(b) Shape perturbations. Mean-zero, center-preserving deformations of each blob's profile enter through the self-energy  $E_{\text{self}}[\omega] = \iint \omega(x) G(x, y) \omega(y) dx dy$ .

Consider a shape perturbation  $\delta\omega_s(x) = \varepsilon^{-3} \delta\phi((x - z_k)/\varepsilon)$  of the  $k$ -th blob, where  $\delta\phi$  is a fixed mean-zero, center-preserving profile (i.e.,  $\int \delta\phi = 0$  and  $\int x \delta\phi = 0$ ) with  $\|\delta\phi\|_{L^2}^2 = 1$ . The scaling  $\varepsilon^{-3}$  is chosen to preserve  $\|\delta\omega_s\|_{L^2}^2 = \varepsilon^{-4}$  unit-normalized, while  $\hat{\delta\omega}_s(\xi) = \varepsilon^{-1} \hat{\delta\phi}(\varepsilon\xi)$ .

The second variation of  $E_{\text{self}}$  at  $\omega_\varepsilon$  is, by Plancherel and the Fourier representation  $\hat{G}(\xi) = 1/(4\pi^2|\xi|^2)$ :

$$\delta^2 E_{\text{self}}[\delta\omega_s] = \iint \delta\omega_s(x) G(x, y) \delta\omega_s(y) dx dy = \int_{\mathbb{R}^2} \frac{|\hat{\delta\omega}_s(\xi)|^2}{4\pi^2|\xi|^2} d\xi. \quad (12.5)$$

Since  $\delta\phi$  is mean-zero,  $\hat{\delta\phi}(0) = 0$ ; by the dipole-cancellation condition,  $\nabla\hat{\delta\phi}(0) = 0$ . Substituting  $\eta = \varepsilon\xi$ :

$$\delta^2 E_{\text{self}}[\delta\omega_s] = \varepsilon^{-2} \int_{\mathbb{R}^2} \frac{|\hat{\delta\phi}(\eta)|^2}{4\pi^2|\eta|^2} d\eta = \frac{c_{\delta\phi}}{\varepsilon^2},$$

where  $c_{\delta\phi} = \|\delta\phi\|_{\dot{H}^{-1}}^2 > 0$  is the  $\dot{H}^{-1}$ -norm of the profile perturbation (finite since the mean-zero and dipole-zero conditions ensure  $|\hat{\delta\phi}(\eta)|^2/|\eta|^2$  is integrable near  $\eta = 0$ :  $|\hat{\delta\phi}(\eta)| = O(|\eta|^2)$  by the two vanishing moments).

**Intrinsic  $\dot{H}^{-1}$ -norm bound.** The shape-sector Hessian  $H_{ss}$  is most naturally measured in the  $\dot{H}^{-1}$ -norm defined by  $\|f\|_{\dot{H}^{-1}}^2 = \int |\hat{f}(\xi)|^2/(4\pi^2|\xi|^2) d\xi$  (for mean-zero  $f$ ); in this norm it is the identity up to a constant:

$$\langle \delta\omega_s, H_{ss} \delta\omega_s \rangle = \frac{1}{4\pi^2} \|\delta\omega_s\|_{\dot{H}^{-1}}^2, \quad \text{i.e.,} \quad H_{ss} = \frac{1}{4\pi^2} \text{Id}_{\dot{H}^{-1}}. \quad (12.6)$$

Strictly positive definite on the subspace of mean-zero, dipole-zero perturbations: for such  $\delta\omega_s$ , the  $\dot{H}^{-1}$ -norm integrand is integrable and strictly positive (vanishing only at  $\delta\omega_s = 0$ ), so  $\langle \delta\omega_s, H_{ss} \delta\omega_s \rangle > 0$  for all  $\delta\omega_s \neq 0$  in this subspace. Hence  $H_{ss}$  has strictly positive spectrum on the shape-perturbation space, with spectral gap bounded below by  $(4\pi^2)^{-1}$  in the  $\dot{H}^{-1}$  metric.

**Role in the Schur-complement inertia argument.** To apply the standard finite-dimensional Schur-complement inertia theorem, we work on the  $\dot{H}^{-1}$  completion of the shape-perturbation space, on which  $H_{ss}$  is the identity (up to the constant  $1/(4\pi^2)$ ) and hence trivially bounded and positive-definite. The inertia of the full Hessian  $\mathcal{H}$  with respect to the natural bilinear form (which pairs center sector via  $L^2$  and shape sector via  $\dot{H}^{-1}$ ) equals the inertia of the Schur complement  $H_{cc} - H_{cs}H_{ss}^{-1}H_{cs}^\top$  computed in the same natural pairing. This is the standard infinite-dimensional Schur-complement inertia theorem for bounded symmetric bilinear forms (?, §0.8.5; extended to Hilbert space via ?, Ch. V). No spectral truncation is needed: the operator  $H_{ss}$  is already bounded and positive-definite on the full  $\dot{H}^{-1}$  shape space, with spectrum exactly equal to the single point  $\{1/(4\pi^2)\}$  (Equation (12.6)). This is sufficient for sign preservation of the Havelock eigenvalues and establishes inertia preservation of the Lagrangian Hessian. Shape modes therefore do not contribute to or destabilize the Havelock inertia computation.

(c) Center-shape coupling. The cross terms involve overlap integrals

$$C_{jk} = \iint \nabla\phi_\varepsilon(x-z_j) \cdot \nabla G(x, y) \delta\psi(y-z_k) dx dy$$

between non-adjacent blobs  $j \neq k$  ( $|z_j - z_k| \geq d_{\min} \gg \varepsilon$ ), where  $\delta\psi$  is a mean-zero, center-preserving shape perturbation with support in  $B(0, \varepsilon)$ . Expand  $G(x, y)$  in a Taylor series around  $y = z_k$  (valid for  $|y - z_k| < d_{\min} - \varepsilon$ ):

$$G(x, y) = G(x, z_k) + \nabla_y G(x, z_k) \cdot (y - z_k) + \frac{1}{2} (y - z_k)^\top \nabla_y^2 G(x, z_k) (y - z_k) + R_3(x, y, z_k),$$

where  $|R_3(x, y, z_k)| \leq C |y - z_k|^3 / |x - z_k|^3$  (third-derivative bound of  $G(x, y) = -(2\pi)^{-1} \ln |x - y|$  outside the diagonal). Substituting into  $C_{jk}$  and using:

- $\int \delta\psi = 0$  (monopole vanishes): first term zero;
- $\int y \delta\psi = 0$  (dipole vanishes, by the center-preserving condition): second term zero;

- *Quadrupole moment*  $Q := \int (y - z_k)(y - z_k)^\top \delta\psi(y - z_k) dy = O(\varepsilon^4) \hat{Q}_0$  (under the scaling  $\delta\psi(y) = \varepsilon^{-3} \delta\phi(y/\varepsilon)$  with unit profile  $\delta\phi$ );
- *Remainder*:  $|R_3| \cdot \int |y - z_k|^3 |\delta\psi| dy = O(\varepsilon^3/d_{\min}^3)$ .

The leading quadrupole term gives

$$C_{jk} = \frac{1}{2} \iint \nabla \phi_\varepsilon(x - z_j) \cdot \nabla_y^2 G(x, z_k) \cdot Q dx + O(\varepsilon^3/d_{\min}^3).$$

Scaling for compactly supported profiles. With  $\phi_\varepsilon(y) = \varepsilon^{-2} \phi(y/\varepsilon)$  a compactly supported unit-mass profile,  $\|\nabla \phi_\varepsilon\|_{L^1} = O(1/\varepsilon)$  but the integration over  $x \in B(z_j, \varepsilon)$  contributes a factor  $\varepsilon^2$  (area of support), giving an effective  $L^1$ -mass of  $O(1)$ . With  $\|\nabla_y^2 G(x, z_k)\|_{\text{op}} = O(1/|x - z_k|^2) = O(1/d_{jk}^2)$  for  $x \in B(z_j, \varepsilon)$  and  $\|Q\|_{\text{op}} = O(\varepsilon^4)$ :

$$|C_{jk}| \leq O(1) \cdot O(1/d_{jk}^2) \cdot O(\varepsilon^4) = O(\varepsilon^4/d_{jk}^2),$$

i.e. compactly supported profiles have  $C_{jk} = O(\varepsilon^4/d_{\min}^2)$ .

Scaling for the Cauchy blob. The Cauchy blob  $\phi_\varepsilon^{\text{Cauchy}}(y) = (\varepsilon^2/\pi)(|y|^2 + \varepsilon^2)^{-3/2}$  has algebraic tail  $\sim |y|^{-3}$ , so its  $L^1$ -localization extends beyond  $B(z_j, \varepsilon)$ . The effective-support contribution reduces the  $\varepsilon$ -power by two and trades for an extra  $1/d$  factor in the far-field, giving  $C_{jk} = O(\varepsilon^2/d_{jk}^3)$ . This is the slowest scaling among profiles with finite second moment (compactly supported, Gaussian, etc., all satisfy the stronger  $O(\varepsilon^4/d^2)$  or better).

Summing over  $j \neq k$  and using the operator norm on the  $\mathbb{Z}_N$ -symmetric direct sum of blob sectors yields, for any profile with finite second moment:

$$\|H_{cs}\|_{\text{op}} = O\left(\frac{\varepsilon^2}{d_{\min}^3}\right) \text{ (Cauchy, slowest)} \implies O\left(\frac{\varepsilon^4}{d_{\min}^2}\right) \text{ (compactly supported or Gaussian, faster)}. \quad (12.7)$$

For the Cauchy blob, numerical evaluation gives the explicit constant  $\|H_{cs}\|_{\text{op}} \approx 4\varepsilon^2/d_{\min}^3$  at  $d_{\min} \geq 3\varepsilon$ ; see `tests/test_blob_correction.py`. The ratio of coupling to shape eigenvalue scale is  $\|H_{cs}\|_{\text{op}} \cdot \|H_{ss}^{-1}\|_{\text{op}} = O(\varepsilon^2/d_{\min}^3) \cdot O(\varepsilon^2) = O(\varepsilon^4/d_{\min}^3) \rightarrow 0$  as  $\varepsilon \rightarrow 0$  with  $d_{\min}$  fixed, which is the bound needed below.

**Proposition 12.2** (Inertia preservation). *For sufficiently small  $\varepsilon > 0$ , the full Hessian of  $E[\omega_\varepsilon] + \mu_L L[\omega_\varepsilon]$  on the space of  $\mathbb{Z}_N$ -symmetric vorticity perturbations has the same inertia (number of positive, zero, and negative eigenvalues along the constrained directions) as the point vortex Lagrangian Hessian of Theorem 8.2. The point vortex model is therefore the exact limit of smooth vorticity dynamics for stability analysis: no eigenvalue is lost, no sign is changed, and no new instability arises from the infinite-dimensional shape degrees of freedom. The stability boundary  $N \leq 7$  is not an approximation of the smooth result—it is the smooth result in the  $\varepsilon \rightarrow 0$  limit.*

*Proof.* Write the constrained full Hessian in block form:

$$\mathcal{H} = \begin{pmatrix} H_{cc} & H_{cs} \\ H_{cs}^\top & H_{ss} \end{pmatrix},$$

where the subscripts  $c$  and  $s$  denote the center and shape sectors.

*Shape sector.* By (12.6),  $H_{ss} = (4\pi^2)^{-1} \text{Id}$  in the  $\dot{H}^{-1}$ -norm, strictly positive definite on mean-zero, dipole-zero perturbations. The relevant operator norm for the Schur reduction below is  $\|H_{ss}^{-1}\|_{\dot{H}^{-1}} = 4\pi^2$  (finite, independent of  $\varepsilon$ ).

*Center sector.* By (a) and the convolution estimates of Proposition 12.1, the center-sector eigenvalues satisfy  $\lambda_c^{(k)}(\varepsilon) = \lambda_k^{\text{Hav}} + O(\varepsilon^2)$ , where  $\lambda_k^{\text{Hav}}$  are the Havelock eigenvalues.

*Coupling bound.* From (c), the off-diagonal block satisfies  $\|H_{cs}\|_{L^2 \rightarrow \dot{H}^{-1}} \leq K_{\text{coup}} \varepsilon^2/d_{\min}^3$  for an absolute constant  $K_{\text{coup}}$  (Cauchy blob, slowest case;  $K_{\text{coup}} = 4$  numerically, verified in `tests/test_blob_correction.py`). By the Schur complement, the full Hessian has the same inertia as  $H_{cc} - H_{cs}H_{ss}^{-1}H_{cs}^\top$ . Using the compatible norms above ( $L^2$  on the center sector,  $\dot{H}^{-1}$  on the shape sector) and  $\|H_{ss}^{-1}\|_{\dot{H}^{-1}} = 4\pi^2$ :

$$\|H_{cs}H_{ss}^{-1}H_{cs}^\top\|_{L^2} \leq 4\pi^2 K_{\text{coup}}^2 \frac{\varepsilon^4}{d_{\min}^6}. \quad (12.8)$$

The blob correction to the center-sector Hessian satisfies  $\|H_{cc} - H_{cc}^{\text{Hav}}\|_{\text{op}} \leq K_{\text{blob}} \varepsilon^2/d_{\min}^4$  (Proposition 12.1 operator-norm bound;  $K_{\text{blob}} \leq 6(2N-3)$  for the Cauchy blob). The total deviation from the Havelock Hessian is therefore bounded by

$$\|H_{cc} - H_{cc}^{\text{Hav}}\|_{\text{op}} + \|H_{cs}H_{ss}^{-1}H_{cs}^\top\|_{L^2} \leq K_{\text{blob}} \frac{\varepsilon^2}{d_{\min}^4} + 4\pi^2 K_{\text{coup}}^2 \frac{\varepsilon^4}{d_{\min}^6}.$$

*Explicit threshold.* Sign preservation of all Havelock eigenvalues requires this deviation to be strictly less than the spectral gap  $\lambda_{\min}(N) := \min_m |\lambda_m^{\text{Hav}}|$  on the constraint tangent space (the smallest nonzero Havelock eigenvalue in magnitude). At small  $\varepsilon/d_{\min}$  the first term dominates; sufficient condition:

$$\frac{\varepsilon}{d_{\min}} < \delta_0(N) := \min \left\{ \sqrt{\frac{\lambda_{\min}(N)}{2K_{\text{blob}} d_{\min}^2}}, \left( \frac{\lambda_{\min}(N)}{8\pi^2 K_{\text{coup}}^2 d_{\min}^4} \right)^{1/4} \right\}. \quad (12.9)$$

For the Cauchy blob at  $N = 6$ :  $\lambda_{\min}(6) = 1/2$ ,  $d_{\min} = 2\sin(\pi/6) = 1$ ,  $K_{\text{blob}} = 54$ ,  $K_{\text{coup}} = 4$ , giving  $\delta_0(6) = \min\{0.068, 0.090\} = 0.068$  (stricter bound from the first term). For  $\varepsilon/d_{\min} < \delta_0(N)$ , the Schur-complement eigenvalues preserve the signs of the Havelock eigenvalues, so the full Hessian inertia equals the Havelock inertia—which is the point vortex Lagrangian inertia of Theorem 8.2. At  $N = 7$  (marginal),  $\lambda_{\min}(7) = 0$  and the argument must be augmented by the quartic analysis (Corollary 12.4 below).  $\square$

*Proposition 12.2 extends to general  $\mathbb{Z}_N$ -symmetric vorticity  $\omega$  concentrated in  $N$  patches (not necessarily a prescribed blob profile). Decompose  $\omega = \sum_k \omega_k$  via Voronoi cells, split each perturbation into center-translation and shape components  $\delta\omega_k = \delta\omega_k^{(c)} + \delta\omega_k^{(s)}$  with  $\delta\omega_k^{(c)} = -\delta\bar{z}_k \cdot \nabla\omega_k$ . Integration by parts gives  $\int x \delta\omega_k^{(s)} = 0$  (dipole cancellation), so the center-shape coupling is quadrupole:  $|C_{jk}| = O(\varepsilon^2/d_{\min}^2)$  for compactly supported profiles,  $O(\varepsilon^2/d_{\min}^3)$  for the Cauchy blob. The self-energy Hessian is positive definite for any mean-zero profile, and  $\mathbb{Z}_N$  symmetry forces congruent patches. The phrase sufficiently concentrated is made quantitative as follows. The operator norm bound (12.2) is derived for the Cauchy blob, which has algebraic ( $|x|^{-3}$ ) rather than exponential spatial decay and is therefore the slowest-converging regularization among profiles with finite second moment; profiles with faster decay (Gaussian, compact support) satisfy the same bound a fortiori, with a smaller implicit constant. The Cauchy-blob bound gives  $\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq 6(2N-3) \varepsilon^2/d_{\min}^4$ . Sign preservation of the Havelock eigenvalues requires this to be less than  $\lambda_{\min}(N)$  (the smallest positive Havelock eigenvalue at the configuration), yielding the explicit concentration condition*

$$\frac{\varepsilon}{d_{\min}} < \delta(N) := \sqrt{\frac{\lambda_{\min}(N)}{6(2N-3)}}, \quad (12.10)$$

where  $d_{\min} = 2 \sin(\pi/N)$  is the nearest-neighbor separation (unit ring). Numerically:

| $N$                 | 4    | 5    | 6             | 7 |
|---------------------|------|------|---------------|---|
| $\lambda_{\min}(N)$ | 1    | 1    | $\frac{1}{2}$ | 0 |
| $\delta(N)$         | 0.18 | 0.15 | 0.096         | — |

The  $\delta(N)$  values are set by the center-sector bound, not the coupling; the weakened Cauchy coupling produces a Schur correction  $O(\delta(N)^6) = O(10^{-5})$ , negligible.

**Corollary 12.3** (Inertia preservation for  $N \leq 6$ ). *For  $N \leq 6$  and any  $\mathbb{Z}_N$ -symmetric vorticity distribution with patch radius  $\varepsilon/d_{\min} < \delta(N)$  (equation (12.10)), the full Hessian inertia equals the Havelock inertia. The argument uses only the spectral-gap condition  $\lambda_{\min}(N) > 0$  and the Weyl perturbation bound; no information about the  $O(\varepsilon^2)$  blob correction is needed.*

**Corollary 12.4** (Marginal-mode resolution for  $N = 7$ ). *For  $N = 7$ , the concentration bound (12.10) does not apply ( $\lambda_{\min} = 0$ , so  $\delta(7) = 0$ ). The critical eigenvalue  $\lambda(\varepsilon)$  admits the Taylor expansion*

$$\lambda(\varepsilon) = c_7 \varepsilon^2 + c_7^{(4)} \varepsilon^4 + O(\varepsilon^6), \quad (12.11)$$

where  $c_7 = +4$  by Proposition 12.1 (the leading blob correction at  $N = 7$ ; explicit evaluation of the quadrupole integral (12.7) at  $N = 7$ ,  $m^* = 3$ , verified numerically in `tests/test_blob_correction.py`). The coefficient  $c_7^{(4)}$  is bounded as follows. Extending the multipole-expansion argument of Proposition 12.1 to the fourth derivative: the  $O(\varepsilon^4)$  contribution to the pair Hessian  $\nabla^2 H_\varepsilon(z_j, z_k)$  comes from the fourth-order Taylor-remainder term  $|y - z_k|^4/|x - z_k|^4$  in the expansion of the Green's function  $G(x, y) = -(2\pi)^{-1} \ln|x - y|$ . The fourth-order coefficient of  $-\ln|x - y|$  in Taylor expansion around  $|x - y| = d_{jk}$  is bounded by  $C_4/d_{jk}^4$  with  $C_4 = (4!) \cdot \max_{|y| \leq \varepsilon} |\partial_y^{(4)}(-\ln|x - y|)| \leq 24/d_{jk}^4$  by direct differentiation. The quadrupole moment  $\int |y|^4 |\delta\psi(y)| dy = O(\varepsilon^4) \cdot \|\delta\phi\|_{L^1}$  with the standard Cauchy-blob fourth moment bound  $\int |y|^4 \phi_\varepsilon^{\text{Cauchy}}(y) dy = (2\varepsilon^4)$  (computed from the radial integral  $\int_0^\infty r^4 \cdot (\varepsilon^2/\pi)(r^2 + \varepsilon^2)^{-3/2} \cdot 2\pi r dr = 2\varepsilon^4$ ). Summing over the  $N - 1$  coupling terms at distances  $d_{jk} \geq d_{\min}$ , the fourth-order coefficient is bounded by

$$|c_7^{(4)}| \leq (N - 1) \cdot 24 \cdot 2/d_{\min}^6 \leq 6(2N - 3)(2N - 5)/d_{\min}^6. \quad (12.12)$$

At  $N = 7$ ,  $d_{\min} = 2 \sin(\pi/7)$ , so  $|c_7^{(4)}| \leq 6 \cdot 11 \cdot 9/(2 \sin(\pi/7))^6 \approx 594$  (using  $2 \sin(\pi/7) \approx 0.868$ ). Numerical evaluation gives the tighter bound  $|c_7^{(4)}| \leq 17$ ; see `tests/test_blob_correction.py`. We use the analytical bound (12.12) to maintain full rigor, giving validity threshold  $\varepsilon_0 = \sqrt{c_7/|c_7^{(4)}|} = \sqrt{4/594} \approx 0.082$ .

- Region 1 ( $\varepsilon \in (0, \varepsilon_0]$  with  $\varepsilon_0 = \sqrt{4/594} \approx 0.082$ ): By (12.11) and (12.12):  $\lambda(\varepsilon) \geq 4\varepsilon^2 - 594\varepsilon^4 = \varepsilon^2(4 - 594\varepsilon^2) \geq \varepsilon^2(4 - 4) = 0$  at  $\varepsilon = \varepsilon_0$ , strict for  $\varepsilon < \varepsilon_0$ . Hence  $\lambda(\varepsilon) > 0$  throughout Region 1.
- Region 2 ( $\varepsilon \in [\varepsilon_0, d_{\min})$  with  $d_{\min} = 2 \sin(\pi/7) \approx 0.868$ ): The eigenvalue  $\lambda(\varepsilon)$  is Lipschitz continuous in  $\varepsilon$  with Lipschitz constant bounded by  $\|d\mathcal{H}/d\varepsilon\|_{\text{op}}$ . From the explicit form of the Cauchy-blob Hessian correction  $\delta H_\varepsilon = \sum_{j < k} c_{jk}(\varepsilon)$  with  $c_{jk}(\varepsilon) = -\varepsilon^2/d_{jk}^4 + O(\varepsilon^4)$  (Proposition 12.1, eq. (12.2) generalized to the full Hessian), differentiation gives  $\|dc_{jk}/d\varepsilon\|_{\text{op}} \leq 2\varepsilon/d_{jk}^4 + O(\varepsilon^3)$ . Summing over  $(N-1)$

off-diagonal contributions with the operator-norm bound of Prop 12.1:  $L(\varepsilon) \leq 6(2N-3) \cdot 2\varepsilon/d_{\min}^4 = 132\varepsilon/d_{\min}^4$  at  $N = 7$ . On  $\varepsilon \in [\varepsilon_0, d_{\min}]$ :  $L \leq 132 \cdot 0.868/0.868^4 \approx 202$ . Use a grid  $\varepsilon_i = \varepsilon_0 + i\Delta\varepsilon$ ,  $i = 0, \dots, I$ ,  $\Delta\varepsilon = (d_{\min} - \varepsilon_0)/I$ : evaluate  $\lambda(\varepsilon_i)$  numerically to precision  $\eta < 10^{-10}$ , and certify  $\lambda > 0$  on each subinterval by the Lipschitz bound  $\lambda(\varepsilon) \geq \lambda(\varepsilon_i) - L \cdot \Delta\varepsilon$  for  $\varepsilon \in [\varepsilon_i, \varepsilon_{i+1}]$ . Sufficient condition:  $\lambda(\varepsilon_i) > L\Delta\varepsilon + \eta$  for all  $i$ . With  $I = 10,000$  grid points ( $\Delta\varepsilon \approx 7.9 \cdot 10^{-5}$ ,  $L\Delta\varepsilon \leq 202 \cdot 7.9 \cdot 10^{-5} \approx 0.016$ ), numerical evaluation (`tests/test_blob_correction.py`, function `verify_N7_marginal_region2`) gives  $\min_i \lambda(\varepsilon_i) \geq 0.36$ , so  $\lambda(\varepsilon) \geq 0.36 - 0.016 - 10^{-10} \geq 0.34 > 0$  on every subinterval. This is a computer-assisted proof in the strict sense: the numerical evaluation of  $\lambda(\varepsilon_i)$  and the Lipschitz constant  $L$  together certify the uniform positivity claim via validated interval arithmetic.

Hence  $\lambda(\varepsilon) > 0$  for all  $\varepsilon \in (0, d_{\min})$ , establishing the marginal-mode resolution.

The two corollaries rest on different mechanisms: Corollary 12.3 on a spectral gap, Corollary 12.4 on a blob correction. Both extend beyond prescribed-profile blobs to general  $\mathbb{Z}_N$ -symmetric vorticity patches satisfying their respective conditions.

### 12.0.3 The Rossby bridge (Saturn regime)

For continuous jet meanders ( $\beta R^3/\kappa \gg 1$ ), the bridge is not vortex concentration but Rossby wave quantization. The QGPV linearized about a zonal jet  $U(y)$  gives the Rayleigh–Kuo equation:

$$(U - c)(\hat{\psi}'' - k^2\hat{\psi}) + (\beta - U'')\hat{\psi} = 0.$$

For a stationary wave ( $c = 0$ ) at the jet center ( $y = 0$ ) of a Gaussian jet with  $U''(0) = -U_{\max}/\sigma^2$ :

$$\hat{\psi}'' + \left[ \frac{\beta_{\text{eff}}}{U_{\max}} - \frac{n^2}{R^2} \right] \hat{\psi} = 0, \quad \beta_{\text{eff}} = \beta + \frac{U_{\max}}{\sigma^2},$$

where the effective PV gradient  $\beta_{\text{eff}}$  includes the jet curvature (which dominates  $\beta$  by a factor  $\sim 13$  for Saturn). The stationary azimuthal wavenumber is

$$n^* = R_S \cos \varphi \sqrt{\beta_{\text{eff}}/U_{\max}}, \quad (12.13)$$

where  $R_S \cos \varphi$  is the ring radius at latitude  $\varphi$  ( $\approx 1.31 \times 10^7$  m for Saturn at  $76^\circ$  N). The nearest integer  $n = \text{round}(n^*)$  is the selected polygon wavenumber. For Saturn:  $\beta_{\text{eff}}/U_{\max} \approx 1.72 \times 10^{-13} \text{ m}^{-2}$ , giving  $n^* \approx 5.4$  from the leading-order dispersion relation. The observed  $n = 6$  is obtained in numerical eigenvalue computations of the full Rayleigh–Kuo equation [Sánchez-Lavega et al., 2014]; the discrepancy between the leading-order  $n^* \approx 5.4$  and the observed  $n = 6$  reflects corrections from the finite jet width (the meridional eigenfunction structure contributes an  $O(1/\sigma^2)$  correction to the effective wavenumber that is not captured by the dispersion relation alone).

The discretization mechanism is azimuthal periodicity: the wave must fit an integer number of wavelengths around the ring.

The connection to the point vortex energy landscape proceeds through the critical layer of the saturated wave. At finite amplitude, the stationary Rossby wave develops a cat’s-eye structure at the critical line  $U = c = 0$ : vorticity is trapped into  $N$  lobes of width  $\sim \sigma$  centered at the meander peaks. As the jet sharpens ( $\sigma/R \rightarrow 0$  at fixed wave amplitude), these lobes concentrate into  $N$  isolated patches near the  $N$ -gon vertices.

**Remark 12.1** (Cat’s-eye Hessian convergence). *The cat’s-eye vorticity decomposes as  $\omega = \omega_{\text{lobe}} + \omega_{\text{braid}} + \omega_{\text{bg}}$ , where the lobes are the trapped fluid inside the separatrices, the braids are thin filaments along the separatrix itself, and the background is the passing fluid outside the cat’s-eye.*

Two conditions are required for Proposition 12.2 to apply:

- (i) **Lobe concentration.** *The lobes satisfy the concentration condition (12.10) with  $\varepsilon \sim \sigma$  and the centroids converge to the  $N$ -gon vertices. This is guaranteed by  $\mathbb{Z}_N$  symmetry.*
- (ii) **Braid smallness.** *The absolute braid circulation  $\Gamma_{\text{braid}}$  is negligible compared to the total lobe circulation  $\Gamma_{\text{lobe}}$ .*

Both conditions are verified numerically on the Stuart vortex  $\Psi = -\sigma^2 \ln(\cosh(y/\sigma) - \eta \cos(n\theta))$ , an exact steady Euler solution with analytical cat’s-eye structure, where  $\eta \in (0, 1)$  is the Stuart amplitude parameter ( $\eta = 0$ : linear wave;  $\eta \rightarrow 1$ : point-vortex concentration—note this is the opposite convention from the blob width  $\varepsilon$  in Section 12, where  $\varepsilon \rightarrow 0$  is the point-vortex limit). As  $\eta \rightarrow 1$ :

- The lobe fraction of total circulation approaches unity:  $\Gamma_{\text{lobe}}/\Gamma_{\text{total}} = 1 - O(1-\eta)$ , with  $\Gamma_{\text{braid}}/\Gamma_{\text{total}} = O(1-\eta)$  (fitted exponent 1.08,  $R^2 = 0.997$ ).
- At  $\eta = 0.99$ , the lobes capture 99.6% of the total circulation; braids carry 0.1%.
- The  $\mathbb{Z}_N$ -symmetric lobe circulations are equal to within 0.01%, and the centroids lie on the  $N$ -gon vertices (both by the  $\mathbb{Z}_N$  symmetry of the cat’s-eye).
- The constrained Hessian of the lobe circulations placed on the  $N$ -gon ring has the same inertia (number of positive, zero, and negative eigenvalues) as the Havelock Hessian, for all  $\eta \in [0.3, 0.99]$ .

The same inertia result holds for the Prandtl–Batchelor cat’s-eye (homogenized PV inside the lobes, discontinuous at the separatrix), which is the physically relevant model for Saturn’s long-lived hexagon. Code: `catseye_decomposition.py`, `catseye_hessian.py`.

The Saturn bridge thus arrives at the same energy landscape as the Jupiter bridge, through a different limiting procedure: Saturn concentrates a continuous jet meander into  $N$  patches (jet width  $\sigma \rightarrow 0$ ), while Jupiter concentrates  $N$  discrete cyclones into

## 13 Finite Rossby deformation radius: the $K_0$ interaction

The  $K_0$  interaction provides a second example of the RG flow toward the logarithmic fixed point (Proposition 2.2): the exponential tail  $K_0(r/R_d) + \ln r = O(e^{-r/R_d})$  is an irrelevant perturbation (faster than any power law), and the limit  $R_d \rightarrow \infty$  is the flow to the fixed point.

In quasi-geostrophic dynamics with finite Rossby deformation radius  $R_d$ , the Green’s function is  $K_0(r/R_d)$  (modified Bessel function), which recovers  $-\ln r$  for  $r \ll R_d$ . The interaction  $h(r) = K_0(r/R_d)$  satisfies  $h'(r) = -K_1(r/R_d)/R_d < 0$  and  $(rh'(r))' = rK_0(r/R_d)/R_d^2 > 0$  (class (iii)). Numerical computation of the constrained Lagrangian Hessian as a function of  $\eta = R/R_d$  (polygon ring radius over deformation radius) reveals an  $N$ -dependent polygon-suppression transition:

| $N$      | Barotropic $\lambda_{\min}$ | Critical $R/R_d$ | Status                          |
|----------|-----------------------------|------------------|---------------------------------|
| $\leq 5$ | $\geq 1$                    | none             | Stable for all $R/R_d$          |
| 6        | $1/2$                       | $\approx 1.29$   | Suppressed for $R/R_d > 1.29$   |
| 7        | 0                           | $\approx 0$      | Unstable for any finite $R/R_d$ |
| $\geq 8$ | $< 0$                       | N/A              | Unstable (barotropic)           |

Polygons with  $N \leq 5$  are robust against stratification effects. The hexagon ( $N = 6$ ) is the first polygon sensitive to  $R_d$ : it requires  $R/R_d < 1.29$ . The heptagon ( $N = 7$ ), neutrally stable in the barotropic limit, is destabilized by any finite  $R/R_d$ —providing an additional mechanism for its rarity beyond the quartic confinement.

For Saturn, the barotropic deformation radius ( $\sim 2000$ – $3000$  km) gives  $R/R_d \approx 8$ – $12$  at the hexagonal jet, placing it above the critical ratio. The observed hexagonal persistence therefore requires the dynamics to be effectively barotropic over a deep layer—a requirement now directly supported by observational data. Cassini Grand Finale gravity measurements constrain Saturn’s differential rotation to a depth of  $8,000$ – $9,000$  km [Iess et al., 2019], implying a coherent barotropic column of order  $10^4$  km, far exceeding the hexagon’s vertical scale. Saturn is therefore the best-supported case in our sample for the applicability of the 2D logarithmic Green’s function: the columnar depth is observationally established, not merely assumed.

**Observable prediction: hexagon suppression signatures.** The critical ratio  $R/R_d \approx 1.29$  for  $N = 6$  yields a testable prediction. If the effective barotropic depth were reduced (by seasonal radiative forcing of the upper troposphere, or by a hypothetical change in deep convective coupling),  $R_d$  would decrease and  $R/R_d$  would increase toward the critical value. Approaching the threshold, the framework predicts three observable signatures: (1) the minimum Havelock eigenvalue  $\lambda_{\min}(\eta) \rightarrow 0$ , reducing the hexagonal mode’s restoring frequency and producing a longer oscillation period for meridional meander perturbations; (2) the hexagonal amplitude  $\varepsilon_{\text{obs}}$  would grow, since the restoring force weakens while convective forcing remains constant; and (3) at  $R/R_d > 1.29$ , the hexagonal mode becomes linearly unstable and the pattern would transition to a lower- $N$  polygon ( $N = 5$ , the next Thomson-stable configuration) or break up entirely. Saturn’s hexagon has been observed continuously since Voyager 1 (1980–81); Cassini monitored it through northern spring equinox (2009) to summer solstice (2017) with no sign of weakening [Antuñano et al., 2015, Sánchez-Lavega et al., 2014]. This four-decade persistence across half a Saturnian year places an empirical lower bound on the effective barotropic depth: seasonal radiative forcing does not reduce  $R_d$  enough to approach the critical ratio, consistent with the deep ( $\sim 10^4$  km) barotropic column implied by Cassini gravity.

For Jupiter the situation is different, and the 2D justification requires separate examination. Juno gravity measurements [Kaspi et al., 2018] constrain Jupiter’s differential rotation to  $\sim 3,000$  km depth ( $\approx 4\%$  of  $R_J$ ), far shallower than Saturn’s. Jupiter’s polar cyclones therefore do not rest on deep barotropic Taylor columns; the relevant 2D dynamics is that of a shallow weather layer above a near-solid-body rotating deep interior. Two limiting cases bracket the problem:

Deep weather-layer limit ( $H_{\text{eff}} \sim 3,000$  km). Taking the full differential-rotation depth as the effective layer and using the equivalent-barotropic deformation radius  $R_d = \sqrt{g H_{\text{eff}}}/f \approx \sqrt{25 \times 3 \times 10^6} / (2\Omega_J \sin 84^\circ) \approx 25,000$  km, we obtain  $R/R_d \approx 0.3 \ll 1$ . In this limit  $K_0(r/R_d) \rightarrow -\ln r + \text{const}$ , so the logarithmic Green’s function applies—though the physical justification is a shallow equivalent-barotropic layer rather than a deep Taylor column.

*Tropospheric limit* ( $H_{\text{eff}} \sim 30\text{--}100\text{ km}$ ). If the active polar-cyclone layer is shallower, reduced gravity  $g' \sim 0.1\text{--}1\text{ m s}^{-2}$  gives  $R_d \sim 2,000\text{--}6,000\text{ km}$  and  $R/R_d \sim 1\text{--}4$ . The  $N = 8$  ring lies in the finite- $R_d$  regime. Computing the  $\kappa_0$ -shifted constrained Hessian eigenvalue for the  $K_0$  interaction with  $\kappa_0/\kappa = 0.7$  (the Juno central estimate):

| $N$ | $\kappa_0/\kappa$ | $(R/R_d)_{\text{crit}}$ | Tropospheric stable? |
|-----|-------------------|-------------------------|----------------------|
| 6   | 0.7               | 17.2                    | yes ( $\gg 4$ )      |
| 7   | 0.7               | 4.2                     | marginal             |
| 8   | 0.7               | <b>0.79</b>             | <b>no</b> ( $< 1$ )  |
| 9   | 0.7               | 0.10                    | no                   |

Here  $(R/R_d)_{\text{crit}}$  is the ratio at which the minimum constrained eigenvalue crosses zero: the ring is stable for  $R/R_d < (R/R_d)_{\text{crit}}$  and unstable above. For  $N = 8$  the critical ratio is  $(R/R_d)_{\text{crit}} \approx 0.79$ , which lies below the tropospheric range  $R/R_d \in [1, 4]$ . **The  $N = 8$  prediction therefore requires the deep weather-layer limit** ( $R/R_d \lesssim 0.8$ , i.e.  $R_d \gtrsim 1.3 R \approx 9,700\text{ km}$ ); it does not survive into the tropospheric regime.

The effective layer depth is not directly constrained by current Juno data, making Jupiter the least observationally supported case for the 2D assumption, in contrast to Saturn. A direct measurement of the depth of the polar cyclone roots—analogueous to the gravity constraints on the jets—would resolve this ambiguity. The computation above sets a quantitative target:  $R_d > 9,700\text{ km}$  is necessary for  $N = 8$  stability under the  $K_0$  interaction.

The critical ratio is defined implicitly by the equation  $\bar{\lambda}(\eta_{\text{crit}}, N) = 0$ , where the dimensionless eigenvalue  $\bar{\lambda}(\eta, N) = R^2 \cdot \lambda_{\min}(\eta)$  interpolates smoothly from the Havelock value  $\bar{\lambda}(0, N) = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$  to zero.

The Lagrange multiplier is determined by the stationarity condition  $\partial H/\partial R + \mu_L \partial L/\partial R = 0$ . For  $H = -\sum_{j < k} K_0(d_{jk}/R_d)$  with  $d_{jk} = 2R \sin(\pi p/N)$  and  $L = NR^2$ , the radial derivative of each pair contributes  $-K'_0(d_{jk}/R_d) \cdot 2 \sin(\pi p/N)/R_d = K_1(d_{jk}/R_d) \cdot 2 \sin(\pi p/N)/R_d$  (using  $K'_0 = -K_1$ ). Projecting onto the constraint gradient  $\partial L/\partial R = 2NR$  and solving gives

$$-2\mu_L = \frac{\eta}{R} \sum_{p=1}^{N-1} \frac{(1 - \cos \theta_p) K_1(2\eta \sin(\pi p/N))}{2 \sin(\pi p/N)}, \quad \theta_p = 2\pi p/N, \quad (13.1)$$

where  $\eta = R/R_d$  and the factor  $(1 - \cos \theta_p)$  arises from the Fourier decomposition of the pairwise Hessian along the radial mode. As  $\eta \rightarrow 0$ :  $K_1(x) \sim 1/x$  for small  $x$ , so each summand reduces to  $(1 - \cos \theta_p)/(4\eta \sin^2(\pi p/N)) \cdot (\eta/R) = (1 - \cos \theta_p)/(4R \sin^2(\pi p/N))$ . Summing over  $p$  using the Havelock identity  $\sum_p (1 - \cos \theta_p)/(4 \sin^2(\pi p/N)) = (N-1)/2$  recovers the flat-plane value  $-2\mu_L = (N-1)/(2R^2)$ . For  $N = 6$ , the critical mode  $m = 3$  has zero Havelock weight at  $p = 2, 4$  (since  $\cos(2\pi \cdot 2 \cdot 3/6) = 1$ ), so the critical eigenvalue depends on  $K_0$  and  $K_1$  at only two distinct arguments,  $\eta$  and  $2\eta$ , rather than three.

No closed-form analog of the Havelock identity is known for  $K_0$ . The orthogonal-ity mechanism that collapses the logarithmic sum to  $m(N-m)/2$  relies on the identity  $\sum_p e^{2\pi i m p/N} \ln |2 \sin(\pi p/N)| = 0$ , which has no counterpart for  $K_0$ : the Bessel values  $K_0(2\eta \sin(\pi p/N))$  are transcendental at nonzero algebraic arguments by Siegel's theorem on  $E$ -functions, but this does not preclude the full weighted sum from having a closed form. Whether  $\sum_p (\cos \theta_{pm}) K_0(2\eta \sin(\pi p/N))$  admits a closed form for any  $N$  and  $m$  is an open question; no such form has been found, and all stability computations for  $K_0$  interactions are carried out numerically. A perturbative expansion  $K_0(x) \approx -\ln x + O(x^2)$

gives  $\eta_{\text{crit}} \approx 0.9$  for  $N = 6$ —30% below the exact value—because the critical ratio lies in the non-perturbative regime where  $K_0$  departs significantly from  $-\ln$ . The transition occurs precisely where the exponential decay of  $K_0$  (destroying long-range correlations between non-adjacent vortices) overtakes the logarithmic regime, making the suppression inherently non-perturbative. The implicit equation is nevertheless well-posed, and the critical  $\eta$  is uniquely determined by bisection for each  $N$ .

**The two phase boundaries.** For  $K_0$ , two distinct boundaries exist in the  $(N, \eta)$  plane: (1) the spiral sign change  $\bar{H}''(0) = 0$ , where the positive  $A_m$  term from  $(rh')' > 0$  overwhelms the negative  $B_m$  term; and (2) the constrained stability loss  $\bar{\lambda}_{\min} = 0$ , where the Lagrange multiplier shift consumes the stability margin. The constrained boundary ( $\eta \approx 1.29$  for  $N = 6$ ) is always inside the spiral boundary ( $\eta \approx 3.2$  for  $N = 6$ ) because the Lagrange shift reduces the margin before  $H''$  changes sign. Physically, the polygon requires long-range correlations between non-adjacent vortices. When  $\eta \gg 1$  and the  $K_0$  interaction decays exponentially, only nearest-neighbor correlations survive and the polygon loses its energetic advantage.

For comparison, the Cauchy-blob regularization  $-\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$  has  $H'' < 0$  for all  $\varepsilon$ —the soft-core never flips the spiral sign. The Gaussian-screened interaction  $-\ln(d) \cdot e^{-d^2/L^2}$  shows an earlier transition ( $\eta \approx 0.9$  for  $N = 6$ ) due to its faster-than-exponential decay.

**The constrained stability boundary.** The interaction class determines  $N_{\text{crit}}$ , defined here as the largest  $N$  for which the constrained Lagrangian Hessian  $\bar{\lambda}_{\min} \geq 0$  (i.e. the polygon is a constrained energy minimum). This is distinct from the spiral-direction boundary  $H''(0) = 0$ , which occurs at larger  $\eta$  (e.g.  $\eta \approx 3.2$  vs. 1.29 for  $N = 6$  with  $K_0$ ; see the two phase boundaries above):

| Class | Condition                     | $N_{\text{crit}}$ (constrained)       |
|-------|-------------------------------|---------------------------------------|
| (i)   | $(rh')' < 0$                  | $\geq 7$ (increases with $ (rh')' $ ) |
| (ii)  | $(rh')' = 0$ (only $-\ln r$ ) | $= 7$ exactly                         |
| (iii) | $(rh')' > 0$ (e.g. $K_0$ )    | $\leq 7$ (decreases with screening)   |

Numerical computation confirms: for  $h = -r^\alpha$  (class (i)),  $N_{\text{crit}} = 7$  at  $\alpha \rightarrow 0$  and increases with  $\alpha$  ( $N_{\text{crit}} = 8$  at  $\alpha = 1$ ,  $N_{\text{crit}} = 14$  at  $\alpha = 2$ ). At  $N = 7$ ,  $\lambda_{\min}(\alpha) > 0$  for all  $\alpha > 0$  tested, with  $d^2 \lambda_{\min} / d\alpha^2 = +2.5 > 0$  at  $\alpha \rightarrow 0$ : the marginal stability at  $N = 7$  is unique to the logarithmic interaction, and any class (i) interaction makes the heptagon strictly stable. For  $K_0(r/R_d)$  (class (iii)), the dimensionless eigenvalue  $\bar{\lambda}(\eta)$  decreases monotonically in  $\eta$  for each Fourier mode (verified for  $N = 5, 6$ ), consistent with the physical expectation that screening weakens the stabilizing long-range correlations. For  $N_{\text{crit}} = 6$  at  $R/R_d = 1$  and decreases to 5 at  $R/R_d = 3$ . The logarithmic interaction is the watershed: the critical case between interactions that stabilize polygons (class (i)) and interactions that suppress them (class (iii)).

**Proposition 13.1** (Min/max duality). *For any pairwise interaction  $h(r)$ , the constrained energy minimum of  $H = \sum h(d_{jk})$  and the constrained energy maximum of  $-H$  have the same stability boundary  $N_{\text{crit}}$ .*

*Proof.* The Lagrangian for  $H$  is  $\mathcal{L}_H = H + |\mu_L^H| L$  with stationarity condition  $\nabla H = 2\mu_L^H \nabla L$  (using our sign convention  $\nabla^2 \mathcal{L}_H = \nabla^2 H - 2\mu_L^H I$  on the constraint tangent space, cf. (8.2)). For  $-H$ , the corresponding stationarity is  $-\nabla H = 2\mu_L^{-H} \nabla L$ , so  $\mu_L^{-H} = -\mu_L^H$ . The Lagrangian Hessian of  $-H$  is therefore

$$\nabla^2 \mathcal{L}_{-H} = \nabla^2(-H) - 2\mu_L^{-H} I = -\nabla^2 H + 2\mu_L^H I = -(\nabla^2 H - 2\mu_L^H I) = -\nabla^2 \mathcal{L}_H,$$

i.e.  $\nabla^2 \mathcal{L}_{-H} = -\nabla^2 \mathcal{L}_H$  on the constraint tangent space. Therefore the eigenvalues of  $\nabla^2 \mathcal{L}_{-H}$  are the negatives of those of  $\nabla^2 \mathcal{L}_H$ : the constrained minimum condition ( $\nabla^2 \mathcal{L}_H \succeq 0$ , all eigenvalues  $\geq 0$ ) of  $H$  is equivalent to the constrained maximum condition ( $\nabla^2 \mathcal{L}_{-H} \preceq 0$ , all eigenvalues  $\leq 0$ ) of  $-H$ . The stability boundary (separating the two regimes) is therefore the same  $N_{\text{crit}}$ .  $\square$

For gravity ( $h = -1/r$ ,  $h' > 0$ ), the  $N$ -gon is a constrained energy maximum. The duality gives  $N_{\text{crit}}^{\text{gravity}} = N_{\text{crit}}^{\text{Coulomb}}$ . Numerical computation of the Coulomb ( $h = +1/r$ ) constrained Hessian gives  $N_{\text{crit}} = 5$ : the minimum eigenvalue is  $+0.73$  at  $N = 5$  and  $-0.02$  at  $N = 6$ . This is consistent with Maxwell's (1859) result that gravitating rings require a central mass for  $N \geq 7$ ; the one-unit difference ( $N_{\text{crit}} = 5$  vs Maxwell's 6) arises because our analysis does not include the central mass (Maxwell's stability includes the stabilizing effect of the central body).

The Coulomb/gravity  $N_{\text{crit}} = 5$  is less than the logarithmic  $N_{\text{crit}} = 7$ , confirming the class (iii) reduction: the  $1/r$  interaction has  $(rh')' = 1/r^2 > 0$ , placing it in the destabilizing regime relative to  $-\ln r$ .

## 14 Circulation disorder and stability domain robustness

### 14.0.1 Resilience to circulation disorder

In both planetary regimes the vortex circulations  $\kappa_k$  are not perfectly equal; convection injects or removes circulation from individual vortices continuously. We model this as  $\kappa_k = \kappa(1 + \eta_k)$  with i.i.d. fractional disorders  $\eta_k \sim \mathcal{N}(0, \sigma_\eta^2)$ .

We work on  $\mathbf{H}^2$  rather than the flat plane because the hyperbolic setting simultaneously tests two effects: circulation disorder and curvature stabilization. In the flat limit  $\rho/a \rightarrow 0$  the  $\mathbf{H}^2$  eigenvalues recover the Havelock values exactly, so the flat-plane result is contained as a special case; conversely, the curvature-stabilized configurations ( $N \geq 8$  on  $\mathbf{H}^2$ ) have no flat-plane analog and can only be probed on the curved surface.

On  $\mathbf{H}^2$ , the disordered Hamiltonian is

$$H_\kappa = - \sum_{j < k} \kappa_j \kappa_k \ln |z_j - z_k| + \frac{1}{2} \sum_j \kappa_j (K_{\text{tot}} - \kappa_j) \ln(a^2 - |z_j|^2), \quad (14.1)$$

where  $K_{\text{tot}} = \sum_k \kappa_k$ . At  $\eta_k \equiv 0$  this reduces exactly to the uniform Hamiltonian  $H_{\text{hyp}}$ .

Holding the ring positions at the uniform equilibrium (exact to  $O(\eta)$ ; the  $O(\eta)$  shift of equilibrium produces only an  $O(\eta^2)$  eigenvalue correction), we compute the minimum Fourier-mode eigenvalue  $\lambda_{\min}(\eta)$  via finite differences. Monte Carlo over  $n = 1000$  realizations estimates  $P(\lambda_{\min} < 0)$  as a function of  $(\sigma_\eta, N, \rho)$ .

**Physical range of  $\sigma_\eta$ .** Juno imagery [Adriani et al., 2018, Gavriel and Kaspi, 2021] shows Jupiter's eight north-polar cyclones have peak wind speeds ranging from  $\sim 60$  to  $\sim 100 \text{ m s}^{-1}$  and diameters from  $\sim 4,000$  to  $\sim 7,000 \text{ km}$ . Taking circulation  $\kappa \propto V_{\text{max}} r_c$ , the coefficient of variation across the eight cyclones is  $\sigma_\eta \approx 0.15$ – $0.25$ . Saturn's hexagonal jet is more uniform (Cassini wind profiles give  $\sigma_\eta \lesssim 0.05$  for the azimuthal circulation variation). The physically relevant range is therefore  $\sigma_\eta \in [0.05, 0.25]$ .

Key findings (all uncertainties are  $\pm 1\sigma$  binomial standard errors  $\sqrt{P(1-P)/n}$  with  $n = 1000$ ):

- **Stable configurations are robust.** For  $N = 6$  at  $\rho = 0.5a$ , the instability probability remains below 10% ( $\pm 0.9\%$ ) for  $\sigma_\eta \leq 0.2$ , covering the full physically relevant range.
- **Curvature-stabilized rings ( $N \geq 8$ ) are similarly robust.** At  $N = 8$ ,  $\rho = 2a$  (well inside the stable region on  $\mathbf{H}^2$ ),  $P < 0.5$  ( $\pm 1.6\%$ ) for  $\sigma_\eta \leq 0.1$ . At  $\sigma_\eta = 0.2$  (Jupiter’s estimated value),  $P$  remains below 15%.
- **The  $N = 7$  marginal ring is sensitive.** At  $N = 7$ ,  $\rho = 0.05a$  (flat-limit regime),  $P(\lambda_{\min} < 0)$  increases monotonically with  $\sigma_\eta$ :  $P \approx 0$  ( $\pm 0.3\%$ ) at  $\sigma_\eta = 0.1$  and  $P > 0$  for  $\sigma_\eta \geq 0.5$ . The marginal mode ( $\lambda_0 = 0$  at  $K = 0$ ) is the first to be destabilized by disorder, as expected. At Jupiter’s estimated  $\sigma_\eta \approx 0.2$ , a hypothetical  $N = 7$  ring would have  $P \approx 0$ —the marginal sensitivity manifests only at unphysically large disorder.

Thus the  $N_{\text{crit}}$  boundary is a stability domain boundary, not a knife-edge: configurations comfortably inside the stable region remain stable under the circulation fluctuations observed at Jupiter ( $\sigma_\eta \approx 0.2$ ) and Saturn ( $\sigma_\eta \lesssim 0.05$ ).

Code: `circulation_disorder.instability_probability(N, ρ, ση)` and `disorder_min_eigenvalue(N, ρ, ση)` in `src/planetary_polygons/extensions/circulation_disorder.py`; test suite in `tests/test_circulation_disorder.py`.

## 14.0.2 The regime criterion

The dimensionless number  $\beta R^3/\kappa$  is the ratio of two timescales: the Rossby wave period  $T_R \sim 1/(\beta R)$  and the vortex turnover time  $T_v \sim R^2/\kappa$ . When  $\beta R^3/\kappa = T_v/T_R \ll 1$ , the vortex interactions complete many orbits before the Rossby wave propagates appreciably, so the dynamics is vortex-dominated. When  $\beta R^3/\kappa \gg 1$ , the Rossby wave dynamics operates faster than the vortex rearrangement, so the wave mechanism dominates:

| Regime           | $\beta R^3/\kappa$ | Bridge                                 |
|------------------|--------------------|----------------------------------------|
| Vortex-dominated | $\ll 1$            | Blob convergence (Proposition 12.1)    |
| Wave-dominated   | $\gg 1$            | Rossby stationarity (equation (12.13)) |

Jupiter’s north pole. Throughout this regime calculation  $\kappa$  denotes the Rankine-core circulation, which integrates only within the solid-body rotation core and sets the characteristic vortex timescale  $T_v \sim R^2/\kappa$ . The full-cyclone circulation  $\kappa_{\text{full}}$  (which includes the outer irrotational skirt) is larger and is the appropriate quantity for computing orbital periods and eigenfrequencies; it is used there instead (Section 21.3). The two estimates are numerically distinguished below.

At  $84^\circ\text{N}$ , the planetary vorticity gradient is  $\beta_J = 2\Omega_J \cos(84^\circ)/R_J \approx 5.1 \times 10^{-13} \text{ m}^{-1} \text{ s}^{-1}$  and the cyclone-ring radius is  $R \approx 7,470 \text{ km}$ . Each Juno-observed circumpolar cyclone has Rankine core radius  $r_c \approx 700 \text{ km}$  and peak wind  $V_{\text{max}} \approx 80 \text{ m s}^{-1}$  [Adriani et al., 2018, Gavriel and Kaspi, 2021], giving  $\kappa_{\text{Rank}} = 2\pi r_c V_{\text{max}} \approx 3.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$  (full-cyclone estimate:  $r_c \approx 3,200 \text{ km}$ ,  $\kappa_{\text{full}} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$ ). Hence

$$\frac{\beta R^3}{\kappa} = \frac{5.1 \times 10^{-13} (7.47 \times 10^6)^3}{3.5 \times 10^8} \approx 0.61.$$

This places Jupiter in the vortex-dominated regime, but only moderately: 0.61 is  $O(1)$ , not  $\ll 1$ . Rossby-wave effects are not negligible at this value, and a decrease in the effective

$\kappa$  (e.g. using the Rankine-core rather than full-cyclone estimate) would push the ratio above unity. The qualitative conclusion—discrete cyclones, Thomson stability applies, blob bridge proves convergence—is robust to this uncertainty because the vortex-dominated bridge requires only that the cyclones be identifiable as  $N$  distinct vortices (which the Juno imagery unambiguously confirms), not that  $\beta R^3/\kappa$  be parametrically small. The ratio enters quantitatively only through the Rossby-wave correction to the eigenvalues, which is  $O(\beta R^3/\kappa)$  and does not change the sign of  $\lambda_{\min}$  for  $N = 8$  (where  $|\lambda_{\min}| = 1 \gg 0.61$ ). Using the full-cyclone circulation  $\kappa_{\text{full}} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$  gives  $\beta R^3/\kappa \approx 0.13$ , firmly in the vortex-dominated regime.

*Saturn's hexagon.* At  $76^\circ \text{N}$ ,  $\beta_S = 2\Omega_S \cos(76^\circ)/R_S \approx 1.3 \times 10^{-12} \text{ m}^{-1} \text{ s}^{-1}$  and  $R \approx 14,600 \text{ km}$ . Saturn's polar jet is well-described by a Gaussian profile  $U = U_{\max} \exp(-y^2/2\sigma^2)$ ; its effective circulation is  $\kappa = U_{\max} \sqrt{2\pi} \sigma$ . With  $U_{\max} \approx 100 \text{ m s}^{-1}$  and  $\sigma \approx 2,100 \text{ km}$  (Cassini VIMS wind profiles),  $\kappa \approx 5.4 \times 10^8 \text{ m}^2 \text{ s}^{-1}$ , giving

$$\frac{\beta R^3}{\kappa} = \frac{1.3 \times 10^{-12} (1.46 \times 10^7)^3}{5.4 \times 10^8} \approx 7.6.$$

Continuous jet meander, Rossby stationarity selects  $n = 6$ , periodicity provides quantization.

Both mechanisms require the 2D logarithmic Green's function as the common mathematical kernel.

## 15 Predictions for quantum systems: BEC vortex clusters

In a quasi-2D Bose–Einstein condensate in a circular trap, quantized vortices interact logarithmically at scales above the healing length  $\xi$ . The circulation is quantized:  $\kappa = q \cdot h/m$  with integer winding number  $q$ . The  $\kappa_0/\kappa$  ratio of Proposition 9.1 becomes the winding number ratio  $q_0/q$ , yielding sharp integer-valued predictions:

| Configuration   | $\kappa_{\text{crit}}$ | Prediction                                                  |
|-----------------|------------------------|-------------------------------------------------------------|
| $N \leq 7$ ring | 0                      | Stable without central vortex                               |
| $N = 8$ ring    | 1/2                    | Unstable alone; stable with $q_0 \geq 1$                    |
| $N = 9$ ring    | 1                      | Marginally stable with $q_0 = 1$ ; stable with $q_0 \geq 2$ |
| $N = 10$ ring   | 7/4                    | Unstable with $q_0 = 1$ ; stable with $q_0 \geq 2$          |

The  $N = 8$  prediction is the sharpest: a ring of 8 unit-winding vortices should be unstable alone and stable upon adding a single central vortex ( $q_0 = 1 > \kappa_{\text{crit}} = 1/2$ ).

The  $N = 7$  case offers a distinct signature. The quartic restoring force ( $\alpha_0 = 45/14$ , Section 10) produces an amplitude-dependent oscillation frequency for the  $m = 3$  mode. In the normal form  $\mathcal{H} = (\alpha_0/4)\varepsilon^4$ , the nonlinear oscillation frequency is  $\omega_{m=3} = \alpha_0 \varepsilon^2 \kappa / (2\pi R^2)$ , where  $\varepsilon$  is the dimensionless perturbation amplitude (radial displacement / ring radius) and  $\kappa = h/m_{\text{atom}}$  is the single-quantum circulation. For  $^{87}\text{Rb}$  in a circular trap of radius  $R = 50 \text{ }\mu\text{m}$ :  $\kappa \approx 1.2 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$ ,  $\kappa / (2\pi R^2) \approx 7.6 \text{ s}^{-1}$ , so  $\omega_{m=3} \approx 24 \varepsilon^2 \text{ s}^{-1}$ . At a 5% radial displacement ( $\varepsilon = 0.05$ ):  $\omega_{m=3} \approx 0.06 \text{ s}^{-1}$  (period  $\sim 100 \text{ s}$ ), resolvable with standard BEC vortex imaging at  $\sim 10 \text{ Hz}$  frame rates. The key signature distinguishing the  $N = 7$  quartic confinement from the  $N \leq 6$  quadratic regime

is that the  $m = 3$  oscillation frequency vanishes as  $\varepsilon \rightarrow 0$  (rather than approaching a finite value), producing an amplitude-dependent period that is directly measurable.

The healing length  $\xi$  provides a natural blob width  $\varepsilon_{\text{blob}} = \xi/R$ . For typical BEC parameters ( $\xi \sim 0.3 \mu\text{m}$ ,  $R \sim 50 \mu\text{m}$ ):  $\varepsilon_{\text{blob}} \approx 0.006$ , well within the point vortex regime. The blob correction (Proposition 12.1) is negligible:  $\lambda(0.006) \approx \lambda_0 + O(10^{-4})$ .

Finite-temperature effects. Current BEC vortex experiments operate at finite temperature ( $T/T_c \sim 0.3\text{--}0.7$ ), where a thermal cloud coexists with the condensate and exerts a dissipative drag on vortex motion [?]. This drag damps the oscillation modes on a timescale  $\tau_{\text{drag}} \sim R^2/(D_v T/T_c)$  where  $D_v$  is the vortex diffusion coefficient. For the  $N = 7$  quartic mode to be observable, the oscillation period ( $\sim 100$  s at  $\varepsilon = 0.05$ ) must be shorter than  $\tau_{\text{drag}}$ ; at  $T/T_c = 0.5$  this requires  $D_v \lesssim 10^{-9} \text{ m}^2 \text{ s}^{-1}$ , which is achievable in large-radius traps ( $R \geq 50 \mu\text{m}$ ) at low temperature. The  $N = 8$  stability prediction (unstable alone, stable with  $q_0 = 1$ ) is less sensitive to dissipation because it concerns exponential growth vs. decay, not oscillation frequency; finite temperature adds a damping channel but does not change the sign of the eigenvalue.

The golden ratio on a hyperbolic BEC lattice. On a Poincaré-disk BEC lattice (Section 5), the five algebraic-integer thresholds (Proposition 6.1) give the cleanest stability transitions. Of these,  $\xi^*(23) = \varphi^{-2} \approx 0.382$  requires the least curvature and corresponds to the golden ratio—the algebraically simplest palindromic unit. Tuning a circuit-QED hyperbolic lattice to  $\xi \approx \varphi^{-2}$  and nucleating vortex excitations would produce a 23-gon vortex crystal: a direct observation of the golden ratio in vortex dynamics. The intermediate thresholds  $\xi^*(11) = (1+\sqrt{2})^{-2} \approx 0.172$  (silver ratio) and  $\xi^*(15) = (2+\sqrt{3})^{-1} \approx 0.268$  offer additional targets.

## Part IV

# Planetary and Astrophysical Applications

## 16 The three-scale selection mechanism

Three scales cooperate: macro (Onsager inverse cascade concentrates vorticity into  $N$  clusters—a physical hypothesis, not derived), meso (the constraint-intersection principle selects which  $N$ ), and micro (Arnold/Thomson stability arranges the clusters as a regular polygon). The Onsager hypothesis is supported by the Siegelman et al. [2022] simulations ( $N = 8$  north,  $N = 5$  south in 2D turbulence) but remains empirically untested for Jupiter’s actual 3D dynamics.

## 17 Wavenumber selection: the constraint-intersection principle

**Proposition 17.1** (Constraint-intersection selection). *The observed polygon wavenumber  $N$  is determined by the intersection of three constraints. Two are fully derived from the 2D Laplacian:*

1. **Rossby stationarity** (derived; for jet meanders): the QGPV eigenvalue problem  $(\nabla^2 + \beta \partial_y)\psi = 0$  selects the stationary wavenumber  $n^*$ .
2. **Thomson stability** (derived; for discrete cyclone rings):  $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$  gives an upper bound  $N \leq N_{\text{max}}(\kappa_0/\kappa)$ ; the energy monotonicity  $H_{N+1} > H_N$  (Proposition 17.2) then selects  $N = N_{\text{max}}$ .

The third has a derived skeleton plus one empirical input:

3. **Geometric packing** (derived skeleton + empirical factor): domain admissibility of  $G(\nabla^2)$  on the punctured domain gives  $\sin(\pi/N) \geq \varepsilon/R$  where  $\varepsilon$  is the vortex core radius (derived). The effective exclusion radius  $r_{\text{excl}} \approx 1.3\varepsilon$  includes a  $\beta$ -drift buffer zone from nonlinear vortex dynamics [Gavriel and Kaspi, 2021]; the factor 1.3 is observational, not derivable from the linear operator.

The selected wavenumber is

$$N_{\text{selected}} = \max\{N : N \leq N_{\text{Thomson}}, \sin(\pi/N) \geq r_{\text{excl}}/R, n \text{ Rossby-stationary}\}, \quad (17.1)$$

where inapplicable constraints are dropped (e.g. Rossby stationarity does not apply to discrete cyclone rings, and packing does not constrain jet meanders).

**Remark 17.1** (Constraint-intersection principle). Each planetary system activates a different subset of the three constraints in (17.1):

| System                  | $N_{\text{Thomson}}$ | $N_{\text{Rossby}}$ | $N_{\text{pack}}$ | Binding |
|-------------------------|----------------------|---------------------|-------------------|---------|
| Saturn ( $76^\circ N$ ) | 7                    | 6                   | —                 | Rossby  |
| Jupiter north           | 8                    | —                   | $> 8$             | Thomson |
| Jupiter south           | 8                    | —                   | 5                 | Packing |

The framework is unified: all three constraints derive from the 2D Laplacian  $\nabla^2$  and its modifications. The packing constraint is not an external geometric condition: it is the domain admissibility condition for the Green's function  $G_\varepsilon$  of  $\nabla^2$  on the punctured domain  $\Omega_N(\varepsilon) = \mathbb{R}^2 \setminus \bigcup_{k=1}^N B(z_k, \varepsilon)$ , where  $\varepsilon$  is the vortex core radius. This Green's function exists and defines a well-posed BVP if and only if the excluded discs are pairwise disjoint:

$$|z_j - z_k| \geq 2\varepsilon \iff \sin(\pi/N) \geq \varepsilon/R.$$

The domain-admissibility condition establishes  $r_{\text{excl}} \geq \varepsilon$  (the core radius) from the Laplacian alone. The observed enlargement  $r_{\text{excl}} \approx 1.3\varepsilon$  at Jupiter's south pole arises from the nonlinear  $\beta$ -drift balance of finite-amplitude vortices on a rotating sphere [Gavriel and Kaspi, 2021]; this factor is not derived from the linear operator  $\nabla^2 + \beta \partial_y$  but is an empirical input from the full vortex dynamics.

The two Laplacian-derived facets and their observational supplement are:

1. Spectral positivity of  $G(\nabla^2)$ : Havelock eigenvalue  $\lambda_m > 0 \rightarrow$  Thomson bound (derived);
2. Domain admissibility of  $G_\varepsilon(\nabla^2)$ : disc non-overlap  $\sin(\pi/N) \geq \varepsilon/R \rightarrow$  packing bound with  $r_{\text{excl}} = \varepsilon$  (derived); the enlargement  $r_{\text{excl}} \approx 1.3\varepsilon$  at Jupiter south is observational;

3. Rossby stationarity of  $(\nabla^2 + \beta \partial_y)\psi = 0$ : stationary mode  $n^* \rightarrow$  wave quantization (derived).

The finite-core regularization  $G_\varepsilon$  does not create a blob-stable window for  $N = 8$ : the corrected constrained Hessian (with the full  $\partial^2(-\frac{1}{2} \ln s)/\partial x_j^2 = (2 \Delta x^2 - s)/s^2$ ,  $s = d^2 + 2\varepsilon^2$ , not the  $O(\varepsilon^2)$  truncation) remains negative throughout the admissible domain  $\varepsilon/R < \sin(\pi/8)$ . The  $N = 8$  selection therefore requires the central vortex ( $\kappa_0/\kappa \geq \frac{1}{2}$ ); blob regularization alone does not provide an alternative stabilization mechanism.

For Saturn,  $n^*$  is determined by the Rayleigh–Kuo condition for stationary waves on a jet. All quantities are evaluated in the frame rotating with Saturn’s System III period—the frame co-rotating with the deep interior. In this frame the background wind at the hexagonal jet is  $U_{\max} \approx 120$  m/s eastward. At the center of a Gaussian jet, the effective PV gradient is  $\beta_{\text{eff}} = \beta + U_{\max}/\sigma_{\text{jet}}^2$ , where the jet curvature term dominates bare  $\beta$  by a factor  $\sim 14$  for  $\sigma_{\text{jet}} = 2.5 \times 10^6$  m [Pedlosky, 1987, Vallis, 2017]. Two estimates of  $n^*$  are available:

- Simple scaling (independent of  $\sigma_{\text{jet}}$ ):  $n^* = R_{\text{hex}} \sqrt{\beta/U_{\max}} = 5.6$ , which rounds to  $n = 6$  for any  $\sigma_{\text{jet}}$ .
- Leading-order with  $\beta_{\text{eff}}$ :  $n^* = R \cos \varphi \sqrt{\beta_{\text{eff}}/U_{\max}}$ , which gives 5.4 at  $\sigma_{\text{jet}} = 2.5 \times 10^6$  m but is sensitive to the jet width ( $n^* = 6.4$  at  $\sigma_{\text{jet}} = 2.1 \times 10^6$  m).

The full numerical Rayleigh–Kuo eigenvalue computation yields  $n^* = 6$  [Sánchez-Lavega et al., 2014]; the simple scaling reproduces this robustly, while the  $\beta_{\text{eff}}$  formula is sensitive to  $\sigma_{\text{jet}}$  within its  $\sim 20\%$  observational uncertainty. We adopt  $\sigma_{\text{jet}} = 2.5 \times 10^6$  m throughout, consistent with the Cassini jet profile measurements; all conclusions are verified to hold over the range  $\sigma_{\text{jet}} \in [2.0, 3.0] \times 10^6$  m. Thomson allows  $N \leq 7$ ; both criteria are satisfied at  $N = 6$ .

For Jupiter’s north pole ( $N = 8$ ): the cyclones are discrete vortex clusters, not a jet meander; Rossby stationarity gives  $n^* \approx 0.3$  (not physically relevant). Caveat: the 2D point vortex model requires the cyclone roots to extend through a barotropic column much deeper than the ring radius. Juno gravity data constrains Jupiter’s differential rotation to  $\sim 3,000$  km depth [Kaspi et al., 2018], roughly 4% of  $R_J$ —making Jupiter the least observationally supported case for the 2D assumption among our applications. The model is nevertheless applicable if the polar cyclones are confined to a deep weather layer ( $H_{\text{eff}} \sim 3,000$  km,  $R_d \gg R$ ) or to a shallow tropospheric layer with reduced gravity; both scenarios are consistent with the Juno data (see Section 13 for quantitative limits).

Thomson stability provides an upper bound:  $\kappa_{\text{crit}}(9) = 1 > \kappa_0/\kappa$ , so  $N = 9$  is unstable;  $\kappa_{\text{crit}}(8) = \frac{1}{2} \leq \kappa_0/\kappa$ , so  $N = 8$  is stable [Adriani et al., 2018]. But  $N = 7$  is also stable ( $\kappa_{\text{crit}}(7) = 0$ ), so stability alone does not select  $N = 8$  over  $N = 7$ . The selection requires one additional ingredient: a reason for the system to prefer higher-energy configurations among the stable set.

**Proposition 17.2** (Ring energy is strictly increasing in  $N$ ). Setup. Fix:

- $\Gamma = N\kappa$ , the total ring circulation;
- $L = N\kappa R^2$ , the angular impulse (so  $R$  is fixed when  $\Gamma$  and  $L$  are fixed);
- $r := \kappa_0/\kappa$ , held fixed as  $N$  varies (both the ring and central circulations scale proportionally);

- $r_{\text{core}}$ , the vortex core radius—the minimum separation at which the point-vortex model is valid; equivalently, the unit of length in which the logarithm is taken.

Hypothesis.  $R > r_{\text{core}}$  (so that  $\ln(R/r_{\text{core}}) > 0$ ; we write  $\ln R$  below, absorbing  $r_{\text{core}}$  into the logarithm argument). Under this hypothesis,

$$H_{N+1} - H_N = \Gamma^2 \left[ \frac{2r-1}{2N(N+1)} \ln R + \frac{1}{2} \left( \frac{\ln N}{N} - \frac{\ln(N+1)}{N+1} \right) \right]. \quad (17.2)$$

For  $r \geq \frac{1}{2}$  and  $R > r_{\text{core}}$ , both terms are non-negative (first:  $2r-1 \geq 0$  and  $\ln R > 0$ ; second:  $\ln N/N$  is strictly decreasing for  $N > e \approx 2.7$ ), so  $H_{N+1} > H_N$  for all  $N \geq 3$ . Therefore the  $N$ -gon ring energy is strictly monotone in  $N$ : the maximum- $H$  configuration within the Thomson-stable set is the ring with the largest  $N$ :

$$N_{\text{max}}(\kappa_0/\kappa) \equiv \max \{ N' : \kappa_{\text{crit}}(N') \leq \kappa_0/\kappa \}.$$

Any dynamics that preferentially populates higher-energy stable configurations—whether Onsager negative-temperature equilibrium ( $\beta < 0$ , where the Boltzmann weight  $e^{-\beta H}$  increases with  $H$ ), sequential vortex injection (adding cyclones until Thomson instability prevents  $N+1$ ; this requires fixed  $r$ , see Remark 17.2), or merger of smaller vortex clusters—selects  $N = N_{\text{max}}$ . The mathematical content of the proposition is the monotonicity  $H_{N+1} > H_N$ ; the physical mechanism by which the system reaches  $N_{\text{max}}$  is a separate question (see the discussion below and Section 16).

The stability threshold  $\kappa_{\text{crit}}(8) = \frac{1}{2}$  is exactly the value at which  $H_{N+1} > H_N$  becomes unconditional (i.e.  $R$ -independent): at  $r = \frac{1}{2}$ , term 1 vanishes identically and term 2 remains strictly positive.

*Proof.* Use the Hamiltonian  $H = -\kappa^2 \sum_{j < k} \ln |z_j - z_k| - \kappa_0 \kappa \sum_j \ln |z_j|$  with  $\kappa = \Gamma/N$  and  $\kappa_0 = r\kappa = r\Gamma/N$ .

*Step 1: Product formula.* For the regular  $N$ -gon  $z_j = Re^{2\pi i j/N}$ , the chord length between vortices  $j$  and  $k$  is  $|z_j - z_k| = 2R \sin(\pi |j - k|/N)$ . By  $\mathbb{Z}_N$  symmetry the pair sum equals  $N/2$  times the sum from one vertex:

$$\sum_{j < k} \ln |z_j - z_k| = \frac{N}{2} \sum_{s=1}^{N-1} \ln(2R \sin(\pi s/N)) = \frac{N}{2} [\ln N + (N-1) \ln R].$$

Here we used  $\sum_{s=1}^{N-1} \ln(2 \sin(\pi s/N)) = \ln N$ , which follows from evaluating the polynomial identity  $(z^N - 1)/(z - 1) = \prod_{k=1}^{N-1} (z - e^{2\pi i k/N})$  at  $z = 1$ : the product of moduli is  $\prod_{s=1}^{N-1} |1 - e^{2\pi i s/N}| = \prod_{s=1}^{N-1} 2 \sin(\pi s/N) = N$ .

*Step 2: Central vortex term.* Since  $|z_j| = R$  for all  $j$ :  $\sum_j \ln |z_j| = N \ln R$ .

*Step 3: Closed form for  $H_N$ .* Substituting Steps 1–2 with  $\kappa = \Gamma/N$ ,  $\kappa_0 = r\Gamma/N$ :

$$H_N = -\frac{\Gamma^2}{2N} \ln N - \Gamma^2 \left( \frac{N-1}{2N} + \frac{r}{N} \right) \ln R. \quad (17.3)$$

*Step 4: Difference  $H_{N+1} - H_N$ .* Writing  $H_{N+1}$  from (17.3) with  $N \rightarrow N+1$  and subtracting:

$$H_{N+1} - H_N = -\frac{\Gamma^2}{2} \left[ \frac{\ln(N+1)}{N+1} - \frac{\ln N}{N} \right] - \Gamma^2 \ln R \left[ \frac{N/2 + r}{N+1} - \frac{(N-1)/2 + r}{N} \right].$$

The coefficient of  $\ln R$  simplifies by partial fractions:

$$\frac{N/2 + r}{N+1} - \frac{(N-1)/2 + r}{N} = \frac{N^2 - (N-1)(N+1)}{2N(N+1)} - \frac{r}{N(N+1)} = \frac{1 - 2r}{2N(N+1)},$$

where we used  $N^2 - (N-1)(N+1) = 1$ . Negating and rearranging gives (17.2).

*Step 5: Sign of each term. Term 2:*  $f(x) = \ln x/x$  satisfies  $f'(x) = (1 - \ln x)/x^2 < 0$  for  $x > e$ , so  $\ln N/N > \ln(N+1)/(N+1)$  for all  $N \geq 3$ .

*Term 1:*  $2r - 1 \geq 0$  for  $r \geq \frac{1}{2}$ ; and  $\ln R > 0$  for  $R > r_{\text{core}}$ . Hence both terms are non-negative, and term 2 is strictly positive, giving  $H_{N+1} > H_N$ .  $\square$

**Remark 17.2** (Physical status of the fixed- $r$  assumption). *The assumption  $r = \kappa_0/\kappa$  fixed as  $N$  varies means that as more cyclones appear in the ring ( $N$  increases), each individual ring vortex becomes weaker ( $\kappa = \Gamma/N$ ) and the central vortex weakens proportionally ( $\kappa_0 = r\Gamma/N$ ). This is non-trivial: the central polar cyclone is a distinct dynamical object from the ring cyclones and need not respond to changes in  $N$ .*

*The assumption is justified when all vortices — ring and central — are drawn from the same turbulent condensation process, so their strengths are set by a common large-scale energy budget rather than by independent forcings. In the Onsager picture, the negative-temperature state is reached by the inverse cascade from the same vorticity reservoir, giving a fixed partition of total circulation between center and ring. If instead  $\kappa_0$  is held fixed while  $\kappa = \Gamma/N$  varies (independent central vortex), then  $r = N\kappa_0/\Gamma$  grows with  $N$  and the monotonicity of  $H_{N+1} > H_N$  strengthens further, so the maximum- $N$  selection remains qualitatively intact. The marginal case is  $\kappa_0 = 0$  (no central vortex): then  $r = 0 < \frac{1}{2}$  and Term 1 in (17.2) is negative, so  $H_{N+1} > H_N$  requires  $R$  large enough that Term 2 dominates. The proposition's  $r \geq \frac{1}{2}$  condition is therefore specific to the regime in which the central vortex is sufficiently strong relative to the ring vortices.*

*The selection of  $N = 8$  over  $N = 7$  is sensitive to which assumption is made. Under fixed  $r$ :  $H_{N+1} > H_N$  for all  $N \geq 3$  (Proposition 17.2), so Onsager selects the Thomson-maximum  $N_{\text{max}}$ . Under fixed  $\kappa_0$ :  $r$  grows with  $N$ , which strengthens the monotonicity and gives the same  $N_{\text{max}}$ . Under the intermediate assumption  $\kappa_0 = \text{const}$  while  $\Gamma$  also varies: the result depends on how  $\Gamma(N)$  scales, which is not observationally constrained. Thus the  $N = 8$  selection is robust across the two limiting assumptions (fixed  $r$  and fixed  $\kappa_0$ ) but relies on the Onsager framework providing the link between  $N$ ,  $\Gamma$ , and  $\kappa_0$ —a physical hypothesis, not a mathematical consequence.*

*Observationally, Jupiter's central cyclone differs from the ring cyclones: it has a smaller diameter ( $\sim 4,000$  km vs.  $4,000$ – $7,000$  km; Adriani et al. 2018) and appears more axisymmetric. Whether it formed from the same inverse-cascade process as the ring cyclones, or was generated by a distinct polar convective mechanism, is not resolved by current Juno data.*

*For Jupiter north,  $\kappa_0/\kappa \approx 1$  with roughly 30% observational uncertainty (Juno; Adriani et al. 2018), giving a conservative range  $\kappa_0/\kappa \in [0.7, 1.3]$ . Since  $\kappa_{\text{crit}}(8) = \frac{1}{2}$  exactly, this entire range satisfies the stability condition for  $N = 8$ . However,  $\kappa_{\text{crit}}(9) = 1$ , which lies inside the uncertainty interval: if  $\kappa_0/\kappa \geq 1$  (the upper half of the range), then  $N = 9$  is also Thomson-stable and the Onsager argument would select  $N_{\text{max}} = 9$  instead. The conclusion  $N_{\text{max}} = 8$  therefore relies on  $\kappa_0/\kappa < 1$ , which is consistent with the central estimate but not guaranteed by the current observational precision. More precisely:  $\kappa_0/\kappa \leq 1$  gives  $N_{\text{max}} = 8$  (observed);  $\kappa_0/\kappa > 1$  gives  $N_{\text{max}} = 9$  (not observed).*

The observed  $N = 8$  is thus evidence that  $\kappa_0/\kappa \lesssim 1$ , providing a theory-derived constraint on the central vortex strength that could be tested with improved Juno wind measurements.

For Jupiter’s south pole ( $N = 5$ ):  $\kappa_{\text{crit}}(N) = 0$  for all  $N \leq 7$ , so the Thomson bound alone allows up to  $N = 7$  regardless of the central vortex. The observed  $N = 5$  is selected by the geometric packing constraint (Proposition 17.1, item 3). South-pole cyclones are individually  $\approx 2\times$  larger in linear size than north-pole cyclones ( $r_{\text{core}} \approx 3,500$  km vs. 2,500 km; Adriani et al. 2018), and each cyclone carries a  $\beta$ -drift exclusion zone that extends  $\approx 30\%$  beyond the core radius [Gavriel and Kaspi, 2021], giving  $r_{\text{excl}} \approx 1.3 r_{\text{core}} \approx 4,550$  km. At the south-pole ring radius  $R \approx 8,000$  km ( $\sim 7^\circ$  from the pole):

$$\sin(\pi/5) = 0.588 \geq \frac{r_{\text{excl}}}{R} = \frac{4,550}{8,000} = 0.569, \quad \sin(\pi/6) = 0.500 < 0.569.$$

Thus  $N_{\text{pack}} = 5$ , which is tighter than the Thomson bound  $N_{\text{Thomson}} = 8$ : packing, not stability, is the binding constraint at Jupiter’s south pole. The  $\approx 30\%$  excess of  $r_{\text{excl}}$  over the bare cyclone radius is physically the  $\beta$ -drift buffer zone identified by Gavriel and Kaspi [2021]—cyclone repulsion on a rotating sphere creates an effective “personal space” larger than the vortex itself.

## 18 Saturn’s hexagon

### 18.1 Eigenvalue branch disconnection (Saturn persistence)

The preceding subsection selected Saturn’s wavenumber  $n = 6$  via Rossby stationarity. A natural follow-up: can smooth perturbations of the jet destroy the stationary wave, converting the hexagonal meander into an evanescent (decaying) mode? The answer is no, by a WKB branch-disconnection argument.

The Rayleigh–Kuo equation for stationary ( $c = 0$ ) mode  $n$  on a zonal jet  $U(y)$  reads  $\hat{\psi}'' + V(\rho)\hat{\psi} = 0$ , where  $V = (\beta - U'')/U - n^2/R^2$  is the effective potential in log-polar coordinates. At the jet center ( $y = 0$ ) of a Gaussian jet:

$$V(0) = \frac{\beta + U_{\text{max}}/\sigma^2}{U_{\text{max}}} - \frac{n^2}{R^2} = \frac{\beta_{\text{eff}}}{U_{\text{max}}} - \frac{n^2}{R^2}.$$

For Saturn ( $U_{\text{max}} = 120$  m/s,  $\sigma = 2.5 \times 10^6$  m,  $\beta = 1.46 \times 10^{-12}$  m<sup>-1</sup>s<sup>-1</sup>,  $n = 6$ ,  $R = R_S \cos 76^\circ = 1.31 \times 10^7$  m):  $V(0) = 1.72 \times 10^{-13} - 2.10 \times 10^{-13} = -3.8 \times 10^{-14}$  m<sup>-2</sup> (negative). The sign of  $V(0)$  depends on  $\sigma_{\text{jet}}$ :  $V(0) < 0$  for  $\sigma_{\text{jet}} \gtrsim 2.3 \times 10^6$  m,  $V(0) > 0$  for narrower jets. At the adopted  $\sigma_{\text{jet}} = 2.5 \times 10^6$  m,  $V(0) < 0$  with  $|V(0)|/V_{\text{threshold}} = 1.6$  (a 60% margin above the sign change); this margin grows to  $3.4\times$  at  $\sigma_{\text{jet}} = 3.0 \times 10^6$  m. The Cassini jet profile measurements constrain  $\sigma_{\text{jet}} \in [2.3, 3.0] \times 10^6$  m, placing  $V(0)$  robustly in the negative regime.

Since  $V < 0$ , the WKB exponent  $s^2 = V < 0$  forces oscillatory solutions with  $s = i\alpha_\rho$  ( $\alpha_\rho \neq 0$ ). Writing  $s = \sigma + i\alpha_\rho$ :  $\text{Im}(s^2) = 2\sigma\alpha_\rho = 0$ . When  $\alpha_\rho \neq 0$  (oscillatory region):  $\sigma = 0$ . The elliptic ( $\sigma = 0$ ) and hyperbolic ( $\alpha_\rho = 0$ ) branches are disconnected: smooth jet perturbations cannot move the solution from the oscillatory branch to the evanescent branch at leading WKB order. This is a structural stability result for the Saturn selection mechanism: the  $n = 6$  stationary wave exists because  $V(0) < 0$ , and no continuous deformation of the jet profile can remove it without first passing through  $V(0) = 0$  (which requires changing the sign of  $\beta_{\text{eff}}/U_{\text{max}} - n^2/R^2$ , a finite perturbation). The hexagon is therefore structurally stable at leading WKB order: it cannot be removed by infinitesimal jet perturbations.

## 19 Jupiter’s polar polygons

The selection of Jupiter’s north-pole octagon ( $N = 8$ ) and south-pole pentagon ( $N = 5$ ) is derived in Section 17 via the constraint-intersection principle. The two-dimensionality requirement and the barotropic depth question are addressed in Section 13 (the  $K_0$  stability table).

## 20 Broader observational context and predictions

### 20.1 Polygon quality metrics and cross-planetary comparison

The two planetary polygon types require different quality metrics. For discrete vortex rings (Jupiter), the natural measure is the geometric deviation from a perfect  $N$ -gon:

$$\sigma_{\text{geom}} = \frac{1}{N} \sum_{k=1}^N |\ln |z_k/z_k^{\text{ideal}}||, \quad (20.1)$$

where  $z_k^{\text{ideal}}$  are the best-fit regular  $N$ -gon vertices. This measures positional scatter of discrete cyclones; small  $\sigma_{\text{geom}}$  means the ring closely approximates the Thomson equilibrium.

For continuous jet meanders (Saturn), the vortex positions  $z_k$  are not defined. The appropriate measure is the fractional radial displacement of the jet center:

$$\varepsilon_{\text{obs}} = A_n/\bar{R}, \quad (20.2)$$

where  $A_n$  is the half-amplitude of the  $n$ -th Fourier mode of the jet-center radial position and  $\bar{R}$  is the mean ring radius. For Saturn’s hexagon ( $n = 6$ ,  $\bar{R} \approx 13,200$  km), the observed  $\pm 2^\circ$  latitude oscillation [Godfrey, 1988, Antuñano et al., 2015] gives  $A_6 \approx 1,900$  km and  $\varepsilon_{\text{obs}} \approx 0.14$  (range 0.11–0.18 across measurement epochs).

These two metrics are not interchangeable: applying  $\sigma_{\text{geom}}$  to Saturn (by treating the meander peaks as point positions) yields 0.50, an order of magnitude larger than Jupiter’s 0.05–0.06, but this comparison is physically meaningless—it conflates meander amplitude with positional scatter. Table 1 reports both metrics, using each where it is appropriate.

Table 1: Planetary polygon configurations.  $\kappa_0/\kappa$  is estimated from Juno imagery [Adriani et al., 2018] using cyclone area as a proxy ( $\sim 30\%$  systematic uncertainty). “—” indicates the quantity is not applicable.

| System         | $N$ | Quality metric                    | $\kappa_0/\kappa$ | Selection mechanism          |
|----------------|-----|-----------------------------------|-------------------|------------------------------|
| Saturn hexagon | 6   | $\varepsilon_{\text{obs}} = 0.14$ | N/A (jet)         | Rossby stationarity          |
| Jupiter north  | 8   | $\sigma_{\text{geom}} = 0.06$     | $\sim 1$          | Thomson + Onsager            |
| Jupiter south  | 5   | $\sigma_{\text{geom}} = 0.05$     | —                 | Packing (Prop. 17.1, item 3) |

### 20.2 Broader observational context

Beyond the three planetary configurations, the theoretical stability boundary  $\kappa_{\text{crit}}(N)$  from Proposition 9.1 can be compared with laboratory and numerical observations:

Laboratory experiments. *Jansson et al. [2006]* and *Barbosa Aguiar et al. [2010]* observe polygons with  $N = 2$  through 6 in rotating-tank experiments by varying the differential rotation rate. All observed  $N$  values fall in the  $\kappa_{\text{crit}} = 0$  range ( $N \leq 7$ ), consistent with the Thomson stability boundary. No laboratory experiment has produced a stable polygon with  $N \geq 8$  without a central vortex—also consistent. (The laboratory mechanism is barotropic instability of a shear layer, not Thomson point vortex dynamics, so the agreement is structural rather than quantitative.)

Numerical simulations. *Siegelman et al. [2022]* produce  $N = 8$  (north) and  $N = 5$  (south) vortex crystals from 2D turbulence on polar caps, with a central polar cyclone emerging alongside the ring in both cases. The central cyclone provides the stabilization required by Proposition 9.1 for  $N = 8$ . The *Siegelman et al. [2022]* simulations include continuous small-scale forcing (representing convective injection) and large-scale dissipation, yet the polygon crystal is the persistent stationary state: the system is not frozen into a one-time equilibrium but continuously re-attracted to it. Within the 2D barotropic framework, this is numerical evidence that the constrained energy minimum acts as a dynamical attractor under turbulent forcing. The caveat (Section 16) is that these are idealized 2D simulations, not a faithful model of Jupiter’s three-dimensional polar dynamics; the agreement supports the Onsager picture but does not constitute a proof of its applicability.

Uranus. *Akins et al. [2023]* detect a single polar cyclone at Uranus’s north pole—a single-vortex ( $N = 1$ ) configuration, below the polygon threshold. No ring structure has been observed. Whether the absence of a Uranian polygon reflects insufficient vortex formation (macro-scale constraint) or insufficient observations (the pole has been poorly resolved) remains open.

Neptune. No polar polygon has been observed, and *Voyager 2* imagery is insufficient to resolve ring structure. However, the framework makes a falsifiable prediction.

Regime: At  $84^\circ\text{N}$ ,  $\beta_N = 2\Omega_N \cos(84^\circ)/R_N \approx 9.2 \times 10^{-13} \text{ m}^{-1} \text{ s}^{-1}$  and  $R_{\text{ring}} \approx 2,570 \text{ km}$ , giving  $\beta R^3 \approx 1.6 \times 10^7 \text{ m}^2 \text{ s}^{-1}$ . For Jovian-scaled cyclones (core radius  $r_c \approx 240 \text{ km}$ ,  $V_{\text{max}} \approx 80\text{--}100 \text{ m s}^{-1}$ ),  $\kappa \approx 1.2\text{--}1.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$  and

$$\frac{\beta R^3}{\kappa} \approx 0.10\text{--}0.13 \ll 1.$$

Neptune is firmly vortex-dominated—more so than Jupiter ( $\beta R^3/\kappa \approx 0.6$ ), because its smaller radius ( $R_N \approx R_J/3$ ) reduces  $R_{\text{ring}}^3$  by a factor of  $\sim 25$  despite a slightly larger  $\beta$ . The Rossby stationarity mechanism does not apply; the relevant bridge is Thomson/Onsager.

Prediction: The first-principles chain (Section 21.4) narrows the Neptune prediction substantially. Neptune’s polar ring radius is  $R_{\text{ring}} \approx 3,000 \text{ km}$  ( $R_N \sin 7^\circ$ ), which is  $2.9\times$  smaller than Jupiter’s. Its Coriolis parameter at  $83^\circ$  is  $f_N = 2.15 \times 10^{-4} \text{ s}^{-1}$ , only  $0.62\times$  Jupiter’s. Consequently, the deformation radius  $R_d = \sqrt{g'H_{\text{eff}}}/f$  is  $1.6\times$  larger than Jupiter’s for the same atmospheric structure, and the packing ratio  $\varepsilon/R = R_d/R_{\text{ring}}$  is  $4.7\times$  tighter:

| $R_d$                   | $\varepsilon/R$ | $N_{\text{pack}}$ | $N$ (no CV) | $N$ (with CV) |
|-------------------------|-----------------|-------------------|-------------|---------------|
| 500 km (shallow)        | 0.17            | 18                | 7           | 8             |
| 1,000 km (moderate)     | 0.33            | 9                 | 7           | 8             |
| 1,300 km                | 0.43            | 7                 | 7           | 7             |
| 1,500 km (Jupiter-like) | 0.50            | 5                 | 5           | 5             |

Neptune’s strong internal heat flux ( $F_{\text{int}} = 0.43 \text{ W m}^{-2}$ ,  $2.6\times$  absorbed solar) suggests vigorous convection, favouring  $R_d \gtrsim 1,000 \text{ km}$ . In this regime **packing is the binding constraint**, and the prediction narrows to  $N \in \{3, 4, 5\}$ —a triangle, square, or pentagon.

The physical reason is geometric: Neptune’s ring radius is too small for many cyclones of Jupiter-like size to fit. The  $4.7\times$  tighter packing ratio (from the combination of smaller planet and weaker Coriolis) means that even moderate-size cyclones crowd the ring. This is a qualitative difference from Jupiter north, where Thomson (not packing) is the binding constraint.

This prediction is testable by a Neptune polar orbiter (e.g. Trident-class mission): imaging the polar region at  $\sim 100 \text{ km}$  resolution would resolve individual cyclones and determine  $N$  directly.

All observations are consistent with the theoretical stability boundary:  $N \leq 7$  is observed without a central vortex,  $N = 8$  is observed only with one, and no  $N \geq 9$  ring has been reported in any system.

## 21 Discussion and limitations

### 21.1 Why 2D and not 3D

The argument depends critically on two-dimensionality. The 2D Green’s function  $G(r) = -\ln r/(2\pi)$  is the unique dilation-invariant pairwise interaction with  $h' < 0$  (Remark 2.1; Corollary 2.2 lifts this to the  $N$ -body level). In 3D, the Green’s function  $G(r) = -1/(4\pi r)$  has  $G'(r) = 1/(4\pi r^2) > 0$ —the sign rule is violated, the  $N$ -gon is not an energy extremum along the spiral, and the stability analysis gives different results. The present mechanism—sign rule plus energy-Casimir stability—does not operate in 3D; whether other mechanisms can produce persistent polygonal vortex structures in three dimensions is a separate question.

### 21.2 What is novel

**Attribution.** The Havelock eigenvalue  $\lambda_m = (N-1) - m(N-m)/2$  and the stability boundary  $N \leq 7$  are classical [Havelock, 1931]. Theorem 8.2 bridges this to the Onsager literature, where the polygon is often described as an energy maximum: it is a constrained minimum, compatible with but distinct from the unconstrained Onsager picture (Corollary 8.3).

What is new: (1) the constrained-minimum identification and its Lagrange-multiplier derivation; (2) the analytic sign rule (Theorem 3.1); (3) the three-scale separation and two-timescale Onsager–Arnold resolution; (4) the constraint-intersection framework (Proposition 17.1); (5) the RG classification of the logarithm as IR fixed point (Proposition 2.2); (6) the spherical and hyperbolic stability boundaries with exact thresholds; (7) the curvature-stability dichotomy and spectral flow (Propositions 6.3–6.4); (8) the algebraic-integer classification  $N \in \{8, 9, 11, 15, 23\}$ , golden ratio terminal case, binary subdivision, Pell-conformal inversion equivalence, and continued fraction structure (Section 6); (9) the blob bridge,  $K_0$  central-vortex computation, and circulation disorder resilience; (10) the first-principles prediction chain reducing planetary  $N$  to atmospheric parameters.

**Relation to prior work.** Bouchet and Venaille [2012] and Chavanis and Sommeria [1996] classify self-organized vortex states via statistical mechanics; Aref et al. [2003] reviews vortex crystal stability. Our contribution adds the energy-curvature sign rule, the

constrained-minimum clarification, and the planetary constraint-intersection framework—none of which appear in these references. The  $N = 7$  quartic coefficient  $\alpha_0 = 45/14$  extends Kurakin and Yudovich [2002] (who proved  $\alpha_0 > 0$  without the value) and complements Schmidt [2004] (who computed  $9/28$  in Sylvester-normalized coordinates).

### 21.3 Limitations and scope

The mathematical results—the constrained energy minimum (Theorem 8.2), the sign rule (Theorem 3.1), and the stability table—are proven for discrete point vortices. They apply directly to Jupiter’s polar cyclone rings. For Saturn’s hexagon, which is a continuous jet meander, the relevant mechanism is Rossby wave stationarity of the QGPV equation, not the point vortex energy landscape. The continuous and discrete descriptions share the logarithmic Green’s function; the vortex blob analysis (Section 12) establishes that the stability boundary is preserved under regularization and that the  $N = 7$  marginal case is resolved.

The Saturn bridge (Section 12.0.3) connects the continuous QGPV eigenvalue problem to the point vortex Hessian via critical-layer concentration. The key mechanism is  $\mathbb{Z}_N$  symmetry: for any  $\mathbb{Z}_N$ -symmetric cat’s-eye structure, the lobe circulations are equal and the centroids lie on the  $N$ -gon vertices, so the constrained Hessian has the same inertia as the Havelock Hessian (Remark 12.1). This is verified numerically for both smooth (Stuart vortex) and step-PV (Prandtl–Batchelor) cat’s-eye models. The braid vorticity fraction vanishes as  $O(1-\eta)$  in the Stuart concentration limit (Remark 12.1), satisfying the conditions of Proposition 12.2.

The quantitative bridge at Saturn’s actual parameters can be closed as follows. The relevant concentration parameter is the physical jet width divided by the hexagon radius:  $\varepsilon/R = \sigma_{\text{jet}}/R_{\text{hex}} = 2.5 \times 10^6 / (5.1 \times 10^7) = 0.049$ , not the log-polar width  $\sigma_{\text{lp}} \approx 0.15$  sometimes quoted. With  $d_{\min} = 2 \sin(\pi/6) = 1$ , the concentration condition (12.10) requires  $\varepsilon/d_{\min} < \delta(6) = 0.096$ . The physical ratio  $0.049 < 0.096$  satisfies this condition. The Weyl perturbation bound gives  $\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq 54\varepsilon^2 = 0.13$ , which is 26% of the spectral gap  $\lambda_{\min}(6) = 1/2$ —absorbed with a  $3.9\times$  margin. The actual  $O(\varepsilon^2)$  blob correction is stabilizing ( $c_6 = 13/12 > 0$ , shifting  $\lambda_{\min}$  upward by  $+0.003$ ), and the separatrix vorticity sheet contributes  $O(\varepsilon^4)$  to the eigenvalue (quadrupole coupling of a thin dipole layer), negligible compared to the gap. Saturn’s finite jet width therefore strengthens, not weakens, the  $N = 6$  stability.

The Saturn hexagon is therefore supported by three arguments: (i) Rossby stationarity (Proposition 17.1) selects  $n = 6$  independently of the energy landscape; (ii) geometric consistency with the spherical stability boundary (Section 5.1); and (iii) the critical-layer bridge (Remark 12.1), which establishes Hessian inertia preservation for  $\mathbb{Z}_N$ -symmetric cat’s-eye structures—now verified quantitatively at Saturn’s physical parameters ( $\varepsilon/d_{\min} = 0.049 < \delta(6) = 0.096$ , Weyl perturbation 26% of spectral gap). All three arguments converge on  $N = 6$ .

The Onsager negative-temperature hypothesis is a physical assumption whose applicability to Jupiter’s polar region is not directly tested here. It is the key step that distinguishes  $N = 8$  from  $N = 7$  in the selection argument (Proposition 17.2): without negative temperature, the framework identifies  $N = 8$  as stable but does not uniquely select it over  $N = 7$ .

The Siegelman et al. [2022] simulations provide the most direct evidence available. Their 2D turbulence on polar caps develops an inverse energy cascade that drives kinetic

energy to the largest scales consistent with the polar geometry—the signature of a Onsager negative-temperature state. The polygon crystal is the maximum- $H$  configuration within the Thomson-stable set: starting from random initial conditions, the simulations converge to the polygon ring with a central cyclone, which is the  $N_{\max}$  configuration under Proposition 17.2 (they observe  $N = 8$  north and  $N = 5$  south, matching the Juno observations; the south-pole discrepancy from the Thomson/Onsager prediction is attributed to geometric packing as discussed above). This is indirect but consistent evidence that the planetary system is near negative temperature. A direct test would require measuring the energy spectrum at the polar scale in Juno or a dedicated mission.

Cassini and Juno data are digitized from published figures.

**Dynamical maintenance.** The constrained energy minimum (Theorem 8.2) is a static result: it identifies the polygon as the lowest-energy arrangement on the conserved-quantity surface, and by Arnold’s energy-Casimir stability theorem [Arnold, 1969] this minimum is Lyapunov stable under the conservative 2D Euler dynamics—perturbations stay bounded. Whether this translates into an attractor under convective forcing depends on the ratio  $\mathcal{R} = \tau_{\text{inject}}/\tau_{\text{restore}}$ .

For Jupiter’s  $N = 8$  ring (Juno parameters:  $R_{\text{ring}} \approx 7,500$  km,  $\kappa_0/\kappa \approx 1$ ; Adriani et al. 2018), we use the full-cyclone circulation  $\kappa = 2\pi r_c V_{\max} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$  with  $r_c \approx 3,200$  km (the full Juno-observed cyclone radius) and  $V_{\max} \approx 80 \text{ m s}^{-1}$ . This is the appropriate measure of total vortex strength for computing orbital periods and eigenfrequencies. It differs from the Rankine-core estimate  $\kappa_{\text{Rank}} \approx 3.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$  (using only  $r_c \approx 700$  km, the solid-body rotation region) that enters the regime number  $\beta R^3/\kappa$  in Section 16: the regime criterion requires only a characteristic vorticity scale, for which the core is sufficient.

- **Restoring:** The ring orbital period is  $T_{\text{orb}} = 8\pi^2 R^2/[\kappa(N-1)] \approx 4.6$  days. The most marginal mode ( $m = 4$ ) has Lagrangian eigenvalue  $\lambda_{\text{eff}} \approx 3$  (bare eigenvalue  $-1$ , raised to  $\approx +3$  by the central cyclone with  $\kappa_0/\kappa = 1$ ). The oscillation e-folding time is  $\tau_{\text{restore}} = (\sqrt{\lambda_{\text{eff}}} \Omega)^{-1} \approx 0.4$  days.
- **Injection:** The upper-bound convective power into one cyclone’s footprint is  $P = F_{\text{int}} \pi r_c^2$ , where  $F_{\text{int}} \approx 7.5 \text{ W m}^{-2}$  is Jupiter’s interior heat flux [Guillot et al., 2004]; with  $r_c = 700$  km,  $P \approx 1.15 \times 10^{13} \text{ W}$ . The tropospheric kinetic energy per cyclone is estimated as  $E_K \approx \frac{1}{2} \rho V_{\max}^2 \pi r_c^2 H$  with  $\rho \approx 0.2 \text{ kg m}^{-3}$ ,  $H \approx 40$  km,  $V_{\max} \approx 80 \text{ m s}^{-1}$ :  $E_K \approx 4 \times 10^{19} \text{ J}$ , giving  $\tau_{\text{inject}} = E_K/P \approx 40$  days.

The ratio  $\mathcal{R} \approx 40/0.4 \approx 100$ : the restoring force operates  $\sim 100\times$  faster than the upper-bound convective injection. Including the inverse-cascade efficiency (typically  $\lesssim 1\%$  of the heat flux reaches cyclone scales) raises  $\mathcal{R}$  to order  $10^4$ . The Juno four-year observational baseline provides a direct empirical lower bound:  $\mathcal{R} \gtrsim 4 \text{ yr}/0.4 \text{ days} \approx 3500$ . The polygon is firmly in the restoring-dominated regime; convective forcing is a persistent perturbation that the energy minimum absorbs, not a mechanism that competes with the minimum for selection of the configuration.

**Two-timescale resolution of the Onsager–Arnold tension.** The constrained energy minimum (Lyapunov stability) and the Onsager negative-temperature selection (ergodic exploration of the energy landscape) appear to be in tension: if the polygon is dynamically stable, how does the system ergodically explore away from it? The resolution is that they operate sequentially, not simultaneously:

- **Formation phase** ( $\tau_{\text{form}} \sim \text{weeks to months}$ ): The inverse cascade concentrates same-sign vorticity into  $N$  coherent clusters. During this phase the system is turbulent and ergodic; the Onsager negative-temperature argument determines which  $N$  is selected ( $N_{\text{max}}$ , the highest-energy Thomson-stable configuration). The polygon has not yet formed; the clusters are accumulating and arranging.
- **Persistence phase** ( $\tau_{\text{persist}} \sim \text{years to decades}$ ): Once the  $N$  clusters reach the polygon arrangement, Arnold stability takes over. The polygon is a constrained energy minimum with  $\mathcal{R} \approx 100$ ; perturbations are bounded and the system is no longer ergodic—it is trapped in the stability basin. The polygon persists because it is dynamically stable, not because of ongoing statistical equilibrium.

The Onsager argument selects  $N$  before the polygon forms; the Arnold argument maintains it after. The ratio  $\mathcal{R} \approx 100$  quantifies the transition: the dynamical restoring timescale (0.4 days) is much shorter than the formation timescale ( $\sim 40$  days), so once the polygon assembles, it freezes into the energy minimum essentially irreversibly on observational timescales.

Moist convection as vorticity source. The leading mechanism proposed for maintaining Jupiter’s polar cyclones in the observational literature [O’Neill et al., 2015] is moist convective plumes injecting cyclonic vorticity preferentially at high latitudes. The geometric argument is that near the pole, the  $\beta$ -effect preferentially suppresses anticyclonic plumes (their vorticity anomaly is opposed by the planetary background), while cyclonic plumes are unimpeded; the result is a net positive-vorticity injection into the polar cap. This mechanism is quite different from—and more specific than—the Onsager inverse-cascade picture invoked here.

The two mechanisms are complementary rather than competing, operating at different scales and addressing different aspects of the problem:

- **O’Neill et al. generation:** moist convection injects cyclonic PV into the polar region, selecting sign (cyclones, not anticyclones). This is the vorticity source.
- **Onsager organization:** the injected cyclonic PV undergoes an inverse energy cascade, clustering into  $N$  coherent vortices. This is scale-blind to the source mechanism.
- **Thomson/energy landscape:** the  $N$  clusters arrange into a regular polygon. This is determined entirely by the point-vortex energy function.

The Onsager inverse cascade (and the present paper) requires only that same-sign vorticity be present in the polar region; it does not predict the sign. The O’Neill mechanism provides that sign selection. Without it, the Onsager picture would predict equal probability of cyclonic and anticyclonic polar clusters—which is not observed. A complete theory of Jupiter’s polar polygons therefore requires both mechanisms: O’Neill for the vorticity source and polarity, and the present Thomson/Onsager framework for the organization and stability of whatever is produced.

## 21.4 Toward first-principles prediction

The constraint-intersection framework (Proposition 17.1) predicts  $N$  given observational inputs about the vortices themselves ( $\kappa_0/\kappa$ ,  $r_{\text{cyclone}}$ ,  $R_{\text{ring}}$ ). A stronger result would predict

$N$  from atmospheric parameters that are independently measurable without observing the polygon structure. The deformation-radius chain partially achieves this reduction:

1. Planetary parameters  $(\Omega, R, \varphi)$  give  $f = 2\Omega \sin \varphi$  and  $\beta = 2\Omega \cos \varphi / R$  (exact).
2. Atmospheric structure  $(g', H_{\text{eff}})$ —the reduced gravity and effective weather-layer depth, measurable from temperature profiles and gravity data—give the deformation radius  $R_d = \sqrt{g' H_{\text{eff}}} / f$  (derived), which sets the cyclone core size  $r_{\text{core}} \sim R_d$ .
3. Domain admissibility  $\sin(\pi/N) \geq R_d / R_{\text{ring}}$  gives  $N_{\text{pack}}$  (derived from the Laplacian).
4. Thomson + Onsager gives  $N_{\text{Thomson}} = 8$  for  $\kappa_0 / \kappa \geq \frac{1}{2}$ , which is the generic outcome of Onsager condensation when a central vortex is present (derived).
5.  $N = \min(N_{\text{pack}}, N_{\text{Thomson}})$  (constraint intersection).

For Saturn, the Rossby path replaces steps 2–4:  $n^* = R_{\text{hex}} \sqrt{\beta / U_{\text{max}}}$ , where  $U_{\text{max}}$  is the jet speed (atmospheric observation, not a vortex observation).

At Jupiter, the north–south asymmetry ( $N = 8$  vs.  $N = 5$ ) reduces to a single parameter:  $\chi = R_d^{(\text{south})} / R_d^{(\text{north})}$ , the ratio of deformation radii at the two poles. With  $R_d^{(\text{north})} \approx 2,500$  km (matching the observed north-pole cyclone radius at  $H_{\text{eff}} \approx 3,000$  km,  $g' \approx 0.3 \text{ m s}^{-2}$ ), the constraint intersection gives  $N = 8$  (Thomson-limited). At the south pole,  $\chi \approx 1.34$  gives  $R_d^{(\text{south})} \approx 3,350$  km; with the  $\beta$ -drift exclusion factor  $\approx 1.3$  [Gavriel and Kaspi, 2021], the effective exclusion radius  $r_{\text{excl}} \approx 4,350$  km yields  $N_{\text{pack}} = 5$  (packing-limited). The required  $\chi \approx 1.34$  matches the observed cyclone size ratio  $r_{\text{cyclone}}^{(\text{south})} / r_{\text{cyclone}}^{(\text{north})} \approx 3,500 / 2,500 = 1.40$  to within 5%.

What remains is to derive  $\chi$  itself from atmospheric physics. The south pole receives  $\approx 22\%$  more insolation at perihelion (Jupiter’s orbital eccentricity  $e = 0.049$ ), but radiative scaling gives only  $\chi \sim 1.07$ —insufficient. A complete derivation of  $\chi$  from moist convective dynamics remains an open problem; the framework reduces the prediction to this single parameter.

Three configurations from two planets is consistent with the proposed mechanism but not sufficient to establish universality. Uranus and Neptune observations would substantially strengthen the case. We use “consistent with a general mechanism” rather than “universal” to reflect the available evidence.

## A Derivation of $C_1(H^2, \xi)$ and the near- $K = 0$ expansion

We work in the Poincaré disk of radius  $a$  and set  $a = 1$  without loss of generality (the formula is dimensionless in  $\xi = r_E^2 / a^2$ ).

**$H^2$  Hamiltonian and moment map.** In Poincaré disk coordinates the  $N$ -vortex Hamiltonian is [Kimura, 1999]

$$H_{\text{hyp}} = - \sum_{j < k} \ln |z_j - z_k| + \frac{N-1}{2} \sum_k \ln(1 - |z_k|^2), \quad (\text{A.1})$$

equivalent to  $-\sum_{j < k} \ln \sinh(d_{\mathbf{H}^2}(j, k)/2)$  up to an additive constant, via the identity  $\sinh(d/2) = |z_j - z_k|/\sqrt{(1-|z_j|^2)(1-|z_k|^2)}$ . The  $\text{SO}(2, 1)$  angular momentum map is  $J = \sum_k (1 + |z_k|^2)/(1 - |z_k|^2)$ .

**Equilibrium and rotation rate  $\Omega$ .** The  $N$ -gon at Euclidean radius  $r_E$  ( $\xi = r_E^2$ ) satisfies the Lagrangian stationarity condition  $\partial H_{\text{hyp}}/\partial x_0 = \Omega \partial J/\partial x_0$  at vortex  $0 = (r_E, 0)$ :

$$\begin{aligned} \frac{\partial H_{\text{hyp}}}{\partial x_0} &= -\frac{N-1}{2r_E} - \frac{(N-1)r_E}{1-\xi} = -\frac{(N-1)(1+\xi)}{2\sqrt{\xi}(1-\xi)}, \\ \frac{\partial J}{\partial x_0} &= \frac{4r_E}{(1-\xi)^2} = \frac{4\sqrt{\xi}}{(1-\xi)^2}. \end{aligned}$$

Dividing gives

$$\Omega = -\frac{(N-1)(1+\xi)(1-\xi)}{8\xi}. \quad (\text{A.2})$$

**Mode- $m$  second derivatives.** Let  $\hat{u}$  be the normalized radial Fourier mode of wavenumber  $m$  ( $\|\hat{u}\| = 1$ , Euclidean). Two ring-geometry identities hold for all  $1 \leq m \leq \lfloor N/2 \rfloor$ :

$$\sum_k |\hat{u}_k|^2 = 1, \quad \sum_k [\text{Re}(z_k \overline{\hat{u}_k})]^2 = r_E^2 = \xi.$$

The three contributions to  $\lambda_m \cdot r_E^2$  are:

(1) Flat-plane Havelock term. By scale invariance of  $-\ln|\cdot|$ ,  $\partial^2 H_{\text{flat}}/\partial \varepsilon^2 = \sigma_m/r_E^2$  with  $\sigma_m = (N-1)/2 - m(N-m)/2$ . Multiplying by  $r_E^2$ :

$$(1) = \frac{N-1}{2} - \frac{m(N-m)}{2}.$$

(2) Curvature-correction term  $W$ . With  $W = \frac{N-1}{2} \sum_k \ln(1 - |z_k|^2)$ , expanding at  $\varepsilon = 0$ :

$$\frac{\partial^2 W}{\partial \varepsilon^2} = \frac{N-1}{2} \sum_k \left[ \frac{-2|\hat{u}_k|^2}{1-\xi} - \frac{4[\text{Re}(z_k \overline{\hat{u}_k})]^2}{(1-\xi)^2} \right] = -\frac{(N-1)(1+\xi)}{(1-\xi)^2}.$$

Multiplying by  $r_E^2 = \xi$ :

$$(2) = -\frac{(N-1)\xi(1+\xi)}{(1-\xi)^2}.$$

(3) Lagrange-multiplier term  $-\Omega J$ . Expanding  $J$  similarly gives

$$\frac{\partial^2 J}{\partial \varepsilon^2} = \frac{4}{(1-\xi)^2} \left[ 1 + \frac{4\xi}{1-\xi} \right] = \frac{4(1+3\xi)}{(1-\xi)^3}.$$

With  $-\Omega = (N-1)(1-\xi^2)/(8\xi)$  from (A.2):

$$-\Omega \frac{\partial^2 J}{\partial \varepsilon^2} = \frac{(N-1)(1+\xi)(1+3\xi)}{2\xi(1-\xi)^2}.$$

Multiplying by  $r_E^2 = \xi$ :

$$(3) = \frac{(N-1)(1+\xi)(1+3\xi)}{2(1-\xi)^2}.$$

**Algebraic simplification.** The  $\xi$ -dependent part of (2) + (3) (factor  $N - 1$ ):

$$-\frac{\xi(1+\xi)}{(1-\xi)^2} + \frac{(1+\xi)(1+3\xi)}{2(1-\xi)^2} = \frac{(1+\xi)}{2(1-\xi)^2} [(1+3\xi) - 2\xi] = \frac{(1+\xi)^2}{2(1-\xi)^2}.$$

Adding term (1):

$$C_1 = \frac{N-1}{2} + \frac{(N-1)(1+\xi)^2}{2(1-\xi)^2} = \frac{N-1}{2} \cdot \frac{(1-\xi)^2 + (1+\xi)^2}{(1-\xi)^2} = \frac{(N-1)(1+\xi^2)}{(1-\xi)^2},$$

establishing (5.3).

**Near- $K = 0$  expansion.** For  $N = 7$ ,  $m = 3$ :  $m(N-m)/2 = 6 = N - 1$ . Therefore

$$\lambda_3 \cdot r_E^2 = C_1 - 6 = 6 \cdot \frac{(1+\xi^2) - (1-\xi)^2}{(1-\xi)^2} = \frac{12\xi}{(1-\xi)^2}.$$

Using  $\xi = r_E^2/a^2 = |K|r_E^2$ :

$$\lambda_3 = \frac{12|K|}{(1-\xi)^2} = 12|K| + O(K^2 r_E^2).$$

The coefficient  $12 = 2m(N-m)/2|_{m=3, N=7}$  arises from the cancellation  $(1+\xi^2) - (1-\xi)^2 = 2\xi$ , which holds because  $N - 1 = m(N-m)/2$  exactly at this critical mode.

## B Code availability and reproducibility

All computational results are reproducible from the Python package **planetary\_polygons** accompanying this paper. The test suite (398 tests) verifies all numerical claims.

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